

AN ALMA ARCHIVAL SEARCH FOR SPECTRAL TECHNOSIGNATURES TOWARD STARS WITHIN 40 PC: METHODOLOGY AND FIRST SURVEY RELEASE

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ABSTRACT

We search the ALMA Science Archive for technosignatures from nearby stars: every star within 40 pc that usable public archival observations cover (archival bycatch). This nearest-first release is the development sample of the survey – the detection statistic and line mask were developed on its data – and covers 88 stars in 82 systems through 431 windows over 93.1 GHz of unique frequency, in which each stellar position is compared with 512 identically treated controls: 118 fine-resolution windows are searched for drift-resolved spectral carriers, and the 313 coarse windows are sensitive only to unresolved spectral excess. Four windows raise a statistical screening flag, not a candidate detection: two toward β Pic and one toward HD 48370 are consistent with, and very likely caused by, known circumstellar or foreground CO emission; one, toward CP–72 2713, is unclassified and not significant at the survey level. No credible technosignature is found. The searched subsample over-represents F and G stars ($\times 3.1$ – 3.4) and under-represents M dwarfs ($\times 0.6$ – 0.7) relative to the 40 pc stellar census. Trigger thresholds – the transmitter powers at which a signal first exceeds the screening test, not completeness-corrected limits – span 1.6×10^{13} to 2.8×10^{17} W equivalent isotropic radiated power at native channel resolution, deep enough, toward the nearest systems, for an unresolved carrier matching the total power of an Arecibo-like planetary radar (a power scale, not a matched transmitter model). Completeness is measured on a single fine-channel configuration and transferred by assumption: no completeness number in this release is validated across more than one instrumental configuration, the transfer error is unbounded, and the coarse class has no completeness curve at all. The survey constrains nothing outside its searched windows and archival epochs; the statistic is frozen before the remaining, pre-registered confirmatory datasets are searched.

Subject headings: extraterrestrial intelligence, radio lines: stars, submillimetre: stars, surveys, astrobiology

1. INTRODUCTION

A technological civilisation, signalling deliberately or simply going about its affairs, may leave a detectable imprint on the electromagnetic spectrum: a *technosignature* (Sheikh 2020; Vidal et al. 2026). Six decades of radio searches now cover many thousands of stars yet, as Wright et al. (2018) quantified, sample only an infinitesimal fraction of the multi-dimensional space of transmitter frequency, bandwidth, power and duty cycle – 6×10^{-18} of the haystack’s volume. Three gaps persist: the millimetre and submillimetre regime above ~ 50 GHz is essentially unsearched; most surveys target interest-selected classes of star, not the local stellar population; and few systematically reprocess *existing* interferometric archives.

Status of statistical inference in this release. (i) *The 107 datasets processed here are the development sample of the 208-dataset survey.* (ii) *The symmetric detection statistic and the ± 50 km s^{−1} molecular-line mask were both defined, and the statistic once revised, using these data; the revision was decided on its algebraic merits before any flagged window was re-examined (§5.4, Appendix P).* (iii) *Every rank probability and false-alarm figure in this release is therefore retrospective and descriptive of the development sample, not an independent validation of the pipeline false-positive rate; “confirmed calibration” is reserved for the confirmatory stage.* (iv) *The remaining datasets form the frozen, held-out confirmatory sample, to be searched under the frozen statistic, mask and thresholds (§5.3).*

This paper addresses all three, asking how strongly existing ALMA observations can constrain unresolved artificial spectral emission from nearby stellar systems. In plain terms, what follows characterises the stars ALMA has happened to observe, not the 40 pc population as a whole: the searched subsample over-represents F and G stars and under-represents M dwarfs (§3), and every limit below inherits that bias. We take an ALMA archive-complete sample: every star within 40 pc of the Sun with public ALMA archival coverage, irrespective of spectral type, disc-hosting status or known planets, and mine it for technosignatures in unresolved spectral carriers – channel-confined excess at native

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ALMA resolution, with “narrowband” reserved for the assumed intrinsically narrow transmitter, not for the resolving power of the data (§4) – and for anomalous millimetre continuum, correcting for proper motion and Doppler drift across the adopted search domain. The processing sets the search class, and §4.5 measures it: a continuous narrowband beacon is the best case, short-duty and fast-drifting emitters the worst; coarse-channel windows retain sensitivity to persistent unresolved excess power only (§4). Four archival-specific analyses accompany the classical search – closure-phase vetting, a continuum anomaly screen, a chirp and periodicity re-analysis, a cross-target frequency-occupancy check – of which only the continuum screen enters dispositions; the others are capability demonstrations (§4.4, Table 15). One worked case runs through the paper: a crossing towards β Pictoris, the release’s largest star-versus-control contrast, attributed to the disc’s known circumstellar CO across two transitions and three epochs (§5.3) – the machinery exists to make that attribution routine. Per-target tables are in Appendix H; dispositions, false-alarm accounting, injection completeness and the occurrence machinery consolidate in Tables 5 and 7 with Appendices N and D. We publish this first release at 51 per cent completion because the citable content is methodological – the archive-conditioned archive-complete sample definition, the empirically estimated local false-alarm machinery, the measured completeness. A first set of per-target limits follows, every number tied to a frozen snapshot that later releases supersede row by row. “The survey” is the full 208-dataset programme; “this release” its 107 processed datasets.

2. BACKGROUND

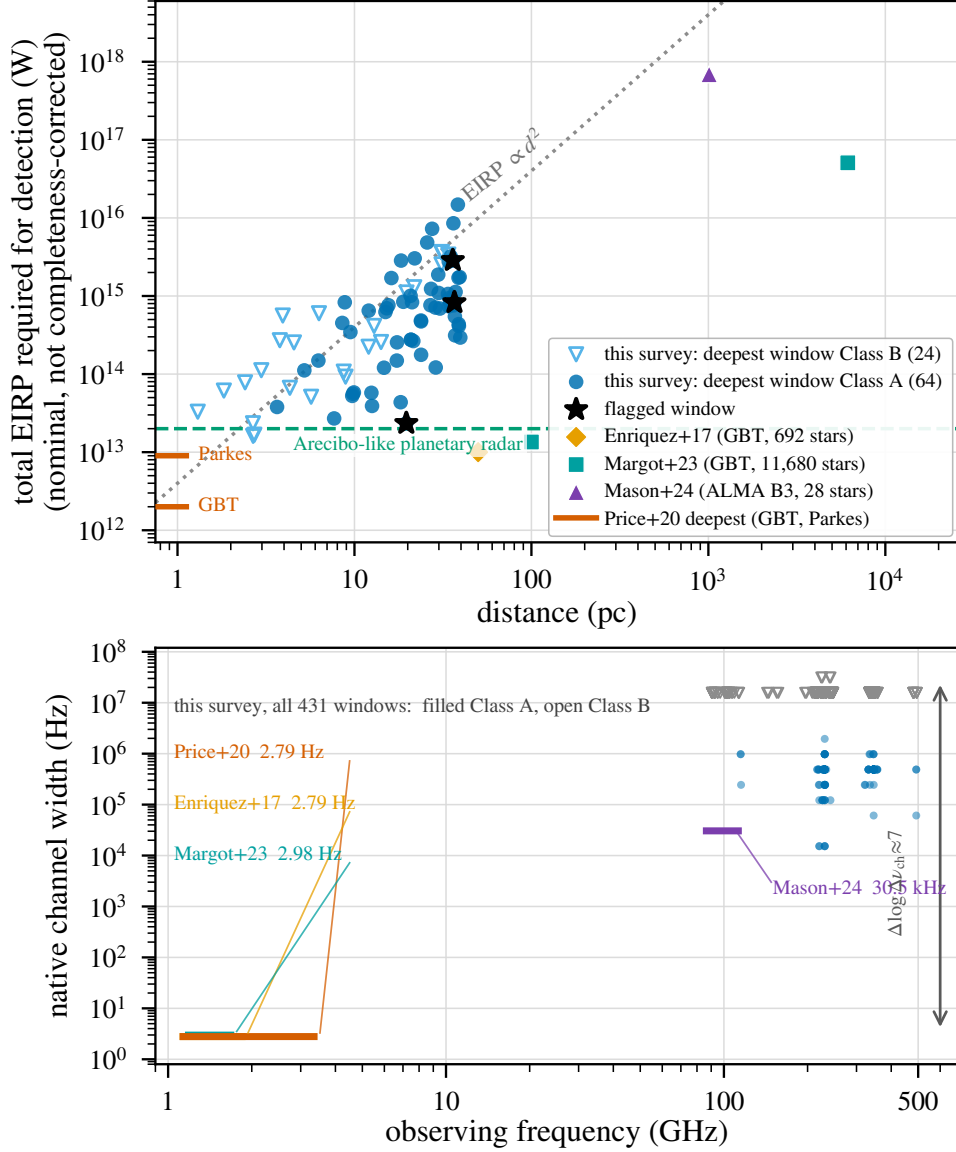
These six decades – from Cocconi & Morrison (1959) and Drake (1960), through the targeted programmes reviewed by Tarter (2001) and Project Phoenix (Backus & Project Phoenix Team 2002) – divide into wide-field commensal surveys and targeted programmes, despite physical reasons to go higher: interstellar scattering and dispersion fall steeply with frequency, and a given aperture delivers a tighter beam per watt of EIRP. Nor is artificial mm-wave emission physically exotic – directional communication, radar and beamed power transmission are all engineered coherent emission at millimetre wavelengths already, on scales from spacecraft links to laboratory sources – so plausibility needs no assumption that anyone operates Arecibo-scale transmitters at hundreds of gigahertz. (Zhang et al. 2020; Czech et al. 2021; Morrison et al. 2023; Li et al. 2026; Johnson et al. 2023; Valente et al. 2025; Tremblay et al. 2024); machine-learning RFI rejection is now central (Ma et al. 2023; Choza et al. 2024; Jacobson-Bell et al. 2025; Poznanski et al. 2025; Garrett et al. 2026) – a machine-learning capability our classical threshold search lacks. As Margot et al. (2023) note – a separate, statistical point – a non-detection bounds the transmitter fraction only if the sample represents the underlying population, which interest-selected samples do not. Essentially all this activity lies below ~ 10 GHz; above it the mm/submm regime has been searched once, also with ALMA (Mason et al. 2024), despite physical reasons to go higher: interstellar scattering and dispersion fall steeply with frequency, and a given aperture delivers a tighter beam per watt of EIRP. Nor is artificial mm-wave emission physically exotic – directional communication, radar and beamed power transmission are all engineered coherent emission at millimetre wavelengths already, on scales from spacecraft links to laboratory sources – so plausibility needs no assumption that anyone operates Arecibo-scale transmitters at hundreds of gigahertz. The countervailing costs are real – atmospheric transmission windows and receiver noise worsen, a carrier costs more photons per watt per hertz, diffraction-limited pointing and phase stability tighten correspondingly, and no demonstrated transmitter technology delivers mm-wave radar powers – they price the search, not forbid it.

Mason et al. (2024) searched archival Band 3 windows toward 28 “stellar bycatch” stars – stars observed for other science that a search can use for free – placing $\text{EIRP}_{\min} > 7 \times 10^{17}$ W for the nearest and identifying high-frequency work’s challenges: drift rates scale with frequency, smearing erodes sensitivity without a drift search, and the molecular line forest confuses. Our pilot (White 2026; under review, citation provisional, manuscript available from the authors on editorial request) extended that methodology to TRAPPIST-1 across Bands 3, 6 and 7 ($\text{EIRP}_{\min} \sim 5\text{--}10 \times 10^{13}$ W, no detection) and built the ranked 100-star list this survey grew from; its pipeline is that design applied to one target, without the archive-complete sample selection, control-ring calibration or completeness machinery introduced here. New here are the archive-complete sample selection (§3), the 512-position control-ring local false-alarm estimation, closure-phase vetting, the spectral-type-normalised continuum screen, the chirp and periodicity extension and the cross-target frequency-occupancy check – each new, to our knowledge, *in an ALMA archival technosignature survey*, though we claim no wider priority over their centimetre-wave counterparts. No execution block is analysed in both papers: the pilot’s 100-star search products are superseded, not re-used, here, and every window here was re-reduced and re-searched from the raw archive data with the pipeline of §4. Where a star is treated in both, the numbers to cite are this release’s frozen products; any discrepancy resolves in their favour.

Figure 1 places this release’s nominal EIRP thresholds among published searches in *total power*; the per-unit-bandwidth convention of a pre-freeze draft is withdrawn and logged with the other validation changes (§4.6). Read with care (the in-panel note carries the same warning): vertical position is not technosignature merit – this survey’s competitive EIRP toward the nearest systems is predominantly the d^2 factor (d spans a factor $\sim 10^3$ in d^2), not sensitivity to Hz-wide carriers, which a 15.625-MHz channel and a 2.8-Hz SETI channel respond to differently (§4).

A non-detection must become a quantitative exclusion statement. Wright et al. (2018) formalised the “Cosmic Haystack” search volume, Margot et al. (2023) its conversion into an upper bound on the transmitter-hosting fraction, and Sheikh et al. (2025a) asked where Earth’s technosignatures would be detectable – a calibration point for any survey’s sensitivity.

Hence the case for an archive-complete sample over interest-selected lists. Most surveys select by proximity, spectral type, planets or Earth Transit Zone membership; the ALMA-specific form, disc selection, skews the sample young and dynamically active. We instead take every star within 40 pc in the usable field of at least one public ALMA observation (§3): a “bycatch” sample – stars in the archive because another programme observed their field, not because anyone



Total power for an unresolved carrier at each survey's native resolution; not a spectral-power density.
Native resolutions span 7 decades across the surveys plotted -- the subject of panel (b).

FIG. 1.— Literature context of this release's 5σ trigger thresholds, in total power (§2). (a) EIRP against distance. Blue: this survey, deepest window per star (88 stars, 1.30–39.63 pc; filled: that window is Class A, drift-resolved; open: Class B, unresolved excess); orange: Enriquez et al. (2017); aqua: Margot et al. (2023); violet: Mason et al. (2024); left-edge ticks: deepest per-target limits of Price et al. (2020). Grey dotted: $EIRP \propto d^2$ at constant received flux, a guide not a fit. Green dashed: an Arecibo-like planetary radar, a technological power scale, not a model. (b) Native channel width against observing frequency for every searched window of this survey (filled Class A, open Class B) and for the four literature programmes (coloured segments): native resolutions differ by $\sim 10^7$ between the Hz-resolution radio surveys and ALMA's correlator channels. All values are nominal trigger thresholds, and conventions differ among these surveys: context, not a ranking (in-panel note).

selected them for a search – of the ALMA-observed subset of the local stellar population in a fixed volume, not of that population itself. This removes technosignature-specific but not archival astrophysical selection; our 168 stars are ~ 1 per cent of the catalogued population within 40 pc, and that residual coverage selection is characterised as a completeness function (§3, Table 5).

3. SAMPLE SELECTION

Our sample is every star within 40 pc of the Sun (Gaia DR3 parallaxes, distances direct inversions $d = 1/\pi$; every entry $\pi \geq 25$ mas with $\sigma_\pi/\pi \leq 0.8$ per cent) whose proper-motion-corrected position at the observation epoch falls in the field of view of at least one public ALMA execution block, not necessarily at the pointing centre: mosaics and wide pointings serendipitously cover more than one catalogued star. Disc, planet and spectral type are descriptive metadata here, not selection criteria. Throughout, a *star* is one catalogue entry and a *system* the set of entries bound to it; a *target/band* dataset is one star in one ALMA band; an *execution block* (EB) one archival observation; and a *window* is one spectral window of one execution block. This gives 168 unique target stars, processed nearest-first;

because trigger thresholds scale with d^2 at fixed noise, the order is sensitivity-ranked, not an unbiased draw, and the prevalence statements of §6 are conditional on it.

Composition follows from the archival selection, and is tabulated against the reference census in Table 16 rather than asserted here: of the searched stars carrying a Gaia temperature, 41 per cent are M dwarfs against 69 per cent of the reference census, while F and G stars are over-represented by factors of 3.4 and 3.1. The archive prefers earlier types and nearby systems. The conditional transmitter-fraction limits of §6 therefore apply to this archivally selected mix, which under-represents the M dwarfs dominating the local volume – a mismatch of astrobiological priority, not statistical validity: the non-detection bears on the ALMA-observed, F/G-enriched mix of Table 16, not preferentially on habitable-zone or M-dwarf prevalence. Eighteen of the 88 processed stars are confirmed exoplanet hosts (§6); the debris-disc hosts (β Pic, AU Mic, ϵ Eri, τ Cet) contribute dusty continua, kept from masquerading as anomalies by the continuum lane’s spectral-type-normalised screen (§4). Window counts are likewise concentrated: β Pic (15 windows), AU Mic (12) and TRAPPIST-1 (12) alone carry 39 of the 431 searched windows (9.0 per cent) from 3 systems, so window-weighted occurrence statements lean on a few heavily observed young, active or disc-hosting systems.

The 168 stars are complete with respect to, and conditional only on, the 40-parsec reference catalogue intersected with the public-archive snapshot (2026 September 8) defining this release. Against that Gaia DR3 solar-neighbourhood catalogue (17 566 stars within 40 pc passing the parallax-quality cut of Table 16) they are $\sim 1\%$; that list is itself incomplete at both ends, so the denominator is a convention and the fraction says only that coverage is small. Table 16 leaves the selection function inspectable. Age and disc-hosting flags are not uniformly available and are documented qualitatively; a machine-readable selection-function table ships with the release.

Stellar counts throughout are catalogue-entry counts: the 168 sample entries comprise 82 independent systems among those searched – seven designation-linked component pairs (SIMBAD designations; per-entry rows, distances and unresolved-pair notes in Appendix H). Since the 44 spectrally covered entries include the unresolved UV Ceti and AT Mic pairs, the 107 star/band datasets, 431 windows and 102 execution blocks derive from 82 systems; the paired rows are bookkeeping, not independent trials.

“Usable public ALMA coverage” is a five-criterion decision tree, frozen for this release absent an erratum: (i) at least one public execution block (EB) whose QA2-calibrated visibilities – QA2 being the observatory pipeline’s calibrated, quality-scored data products – cover the star’s epoch-propagated position in the primary beam, with no minimum integration, sensitivity, antenna count, mosaic-tile count or array configuration beyond QA2 itself; (ii) that position within one primary-beam FWHM of the field centre (largest accepted offset ~ 0.7 FWHMs, Sirius B⁴); (iii) each spectral window admitted individually, on the pipeline’s quality vetting alone; (iv) no programme-category restriction, science, calibration and observatory projects entering on identical terms; (v) a fixed archive snapshot date (2026 September 8), re-evaluated only at release boundaries. No star or window was excluded by an unlisted criterion, none for short integration (the shortest retained are 1.0 and 1.5 min).

ALMA project metadata quantify pointed versus serendipitous coverage: almost every spectral-window-covered star is the pointed science target in at least one execution block, the exceptions a few blocks on another programme’s field or calibrator-only, leaving Wolf 219 the sole genuinely serendipitous star. An archive-complete sample’s residual selection function is the union of ALMA target lists – a contrast with interest-selected samples of degree, but of *quantified* degree: a bycatch fraction of 1 star in 44, 1 EB in 57.

4. DATA AND METHODOLOGY

Survey vocabulary – target/band dataset, spectral window, execution block, control ring, candidate, molecular-line exclusion mask – is defined in Table 13, Appendix B.

Figure 3 summarises the design: one shared archive-ingestion stage feeds two independent search lanes. For each target we mine, within a bounded budget, all calibrated (or locally recalibrated) archival execution blocks covering the star’s Gaia-propagated position, and search their visibilities in (i) the *primary* lane, a Doppler-drift-corrected spectral-carrier search following Mason et al. (2024) and Enriquez et al. (2017), converting each non-detection into an equivalent isotropic radiated power limit $\text{EIRP}_{5\sigma} = 4\pi d^2 S_{\min} \Delta\nu$ ($S_{\min}$ five times the searched window’s per-channel rms, primary-beam corrected; $\Delta\nu$ its native channel width; d the target distance; a fully worked numerical example in Appendix C). The quoted thresholds are nominal: their systematic budget – the ALMA absolute flux scale (5–10 per cent, band- and epoch-dependent), the primary-beam model at the stellar offset (bounded in §5.3), and visibility calibration – multiplies $S_{\min}$ and hence $\text{EIRP}_{5\sigma}$ directly, while *Gaia* distances contribute negligibly at these distances. The same visibilities feed (ii) an *ancillary* line-excluded continuum lane (Appendix K). “ 5σ ” throughout is the local instrumental *trigger* threshold of Table 1, separately and empirically estimated (§5.3). “Narrowband” is reserved for the assumed intrinsic signal morphology, never for the spectral resolving power of the archival observations: in the majority of windows an intrinsically Hz-wide carrier is unresolved within a 15.625 MHz channel, and the data could not demonstrate that a one-channel excess is intrinsically narrow. The search is accordingly reported as two formal survey classes, distinguished by the dimensionless drift parameter

$$\eta_{\text{drift}} = |\dot{\nu}_{\text{max}}| T / \Delta\nu_{\text{ch}}, \quad (1)$$

the channel-width fraction the *maximum* searched drift $|\dot{\nu}_{\text{max}}|$ traverses over an integration T at native channel width $\Delta\nu_{\text{ch}}$: a window is *drift-resolving* when $\eta_{\text{drift}} \gtrsim 1$, only then carrying a carrier across enough channels for its motion to

⁴ Retained with bounded 3.6–16 per cent primary-beam corrections; its EIRP values therefore carry a systematic the holography-beam recomputation, not yet in the released snapshot (Data Availability), will settle (§4.2).

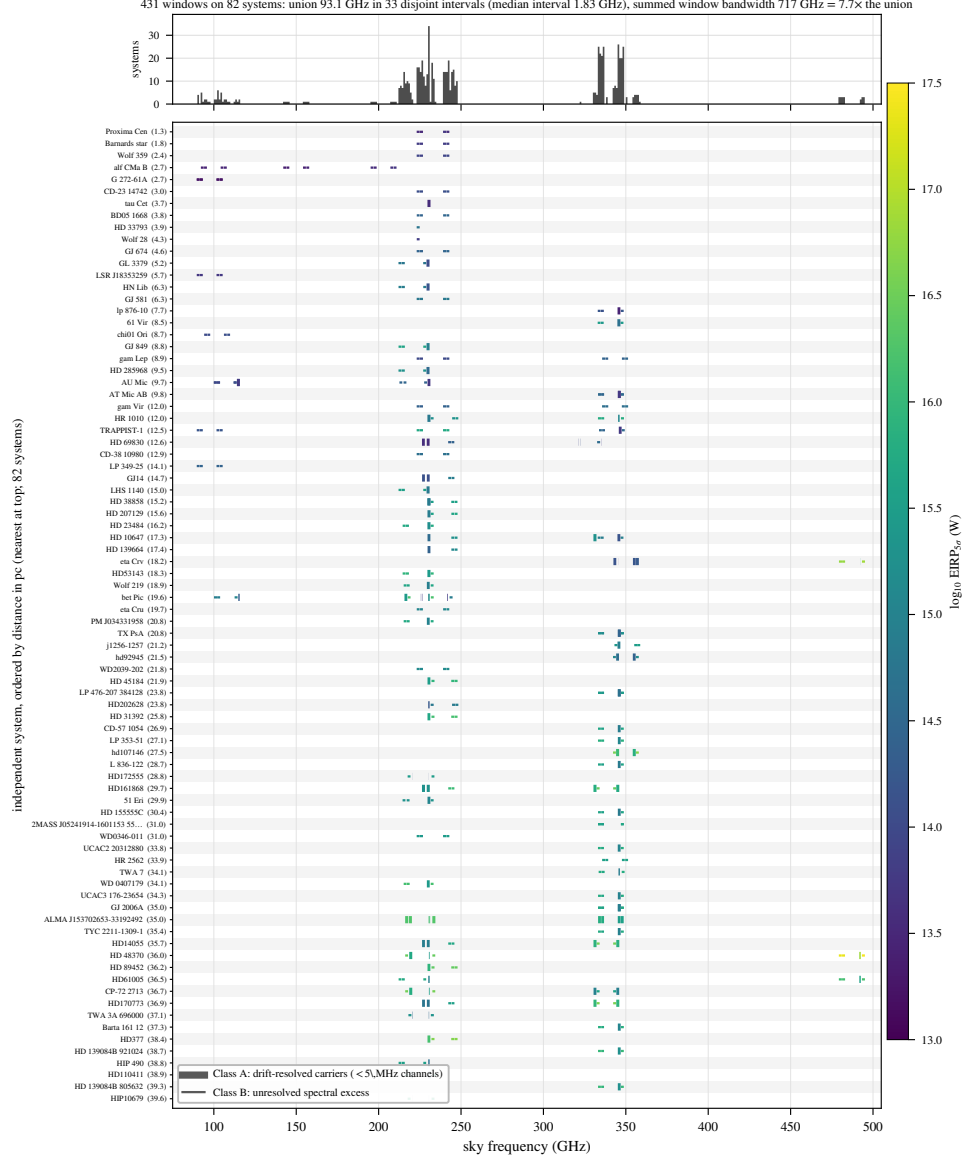


FIG. 2.— What this survey sampled in frequency. Each searched window is a horizontal segment on its host system’s row, rows ordered by distance, coloured by nominal EIRP_{5σ}, with line weight distinguishing the drift-resolved (Class A) from the unresolved-excess (Class B) channelisation; the upper panel counts systems per gigahertz of sky frequency. The narrow, clustered, repeatedly revisited coverage this shows – 33 islands concentrated near 230 and 345 GHz, summed window bandwidth 7.6× the union – is what the in-band transmitter fractions of §6 condition on (Scope of the constraints, §4). Thresholds are nominal in the sense of §4.

be measured rather than assumed. The drift ceiling $|\dot{\nu}_{\max}|$ inside η_{drift} (Eq. 1) is a boundary of the search grid – the widest trial drift the stack evaluates – not a physical prior on transmitter drift rates: carriers drifting faster are simply missed, and the resulting selection function is made explicit only in §6.1 (Table 5, Fig. 13). *Class A*, the drift-resolved spectral-carrier search: the 118 windows of fine channelisation, native widths 15.259–1953.125 kHz (median 488.3 kHz; 62 per cent at the 488.3-kHz mode), spanning $\eta_{\text{drift}} = 0.34$ –422 (median 12.8) – the only class with measured end-to-end recovery of a drifting carrier. *Class B*, the unresolved spectral-excess search: the 313 coarse windows (73 per cent of the sample), native widths 15.625–31.25 MHz (309 windows at the 15.625-MHz mode), spanning $\eta_{\text{drift}} = 0.004$ –1.35 (median 0.36) – sensitive to persistent near-stationary excess power, with no survey-threshold completeness curve and no drift discrimination. Class membership is one pipeline variable, the native channelisation, and it is bimodal: no window falls between 1953.125 kHz and 15.625 MHz, an empty gap of factor 8.0 (one 1953.125-kHz window sits on the Class A side), so any boundary inside the gap assigns identical membership – figures label the classes by name, never by a boundary frequency. The physical criterion behind the split is η_{drift} : only 12 long-integration Class B windows reach $\eta_{\text{drift}} \geq 1$, and then only marginally (maximum 1.35), where one-to-two channel traverses still support no discriminative drift fit after baseline removal; conversely 6 short Class A windows fall below unity and are, in drift terms, conservatively treated as Class A anyway. Results are given per class first and combined only where statistically appropriate (§5.3). Every limit, disposition, false-alarm calibration and completeness statement derives from the spectral-carrier lane; the chirp/periodicity extension and molecular-line census are in Appendices F and L.

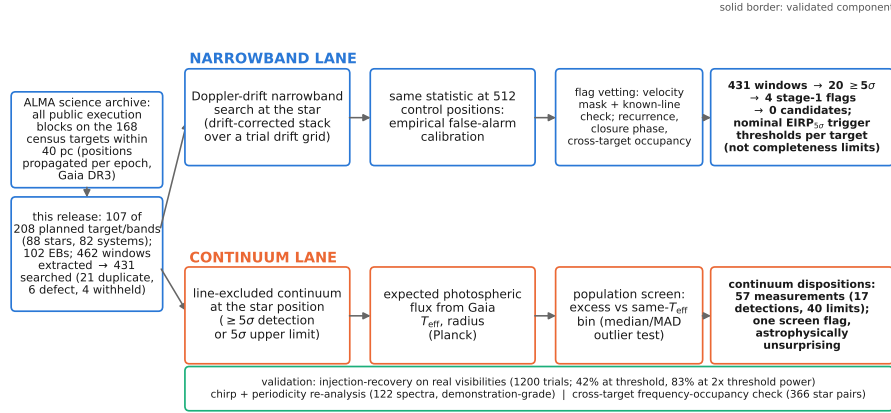


FIG. 3.— Design of the survey. All public execution blocks on the 168 targets are mined from the ALMA archive, positions propagated to each epoch from Gaia DR3, and searched along the two independent lanes described in the text. Solid borders mark validated components; the validation rail summarises the injection-recovery (1200 trials, Appendix D), chirp/periodicity and frequency-occupancy analyses.

Both lanes exclude catalogued molecular transitions at tolerances suited to their statistics. The continuum lane masks the eight common species (CO , ^{13}CO , C^{18}O , HCN , HCO^+ , CS , CN , SiO) at native resolution before integrating; the carrier lane applies a $\pm 50 \text{ km s}^{-1}$ stellar-frame velocity exclusion mask (§§4.2, 5.3) to any threshold crossing. The mask is *excluded search space*: the unmasked survey covers 93.1 GHz of unique sky frequency, the effective default quoted throughout is the $\approx 88.2 \text{ GHz}$ surviving the mask, and no constraint is placed on carriers inside the excluded velocity regions (§5.3). Masking reduces the frequency space a candidate can occupy, not the per-channel noise, so no threshold in the per-target file changes. All flux densities and EIRP thresholds are total intensity (Stokes I), formed from the two parallel-hand cross-correlations the archive delivers; no cross-hand ($V/Q/U$) extraction or circularity test was performed. The construction is nonetheless polarisation-blind in amplitude: I is the summed parallel-hand power, so for fixed total transmitter power the response to a strongly circular or linear artificial signal equals that to an unpolarised one, to within instrumental cross-polarisation leakage (a few per cent); what full Stokes analysis would add is discrimination against weakly polarised astrophysical backgrounds, not depth (§6.4).

How to read every limit in this paper. *Every $\text{EIRP}_{5\sigma}$ value here is a nominal, local instrumental detection threshold – the trigger power P_{trig} : five times the per-channel rms of the searched window, at that window’s native channel width, for a transmitter at the star’s position matching the searched spectral morphology, with a measured recovery that rises with the fraction of the track the carrier occupies its channel and is highest for a continuous one (§4.4, Appendix D). It is not a calibrated completeness limit: completeness is the power P_x at which recovery reaches x per cent. $P_{50} = 1.10 P_{\text{trig}}$ and $P_{90} > 2 P_{\text{trig}}$ for the calibration configuration’s drifting class; for every other configuration P_{50} and P_{90} are not determined – the single measured curve is transferred by assumption (§6.2) – so P_{trig} is quoted throughout rather than any completeness level. It is a total-power threshold set by ALMA’s coarse native channelisation, not a Hz-resolution narrowband sensitivity. The same reading applies to every subsequent “ 5σ ” and to the per-target file (Appendix H); it is stated once here and not repeated at each occurrence.*

In brief, the three detectable morphologies, quantified with real windows in §6.1: a continuous frequency-stationary carrier (best case, essentially ideal gain toward the nearest systems); a continuously drifting carrier in fine channels (42 per cent recovered at threshold, 75–83 per cent at twice it); an intermittent duty-1/4 carrier (91 per cent fine, 100 per cent coarse channels).

4.1. The search statistic, and the words used for its output

The primary statistic is formed on extracted spectra, not images or visibilities. The extraction operates on the QA2-calibrated measurement set as delivered: per integration and per native channel, with the archive’s QA2 flagging carried through untouched and no spectral regridding or interpolation at any stage (Table 2, step 3); it returns the Stokes I intensity at the propagated sky position $\mathbf{x}$, formed from the two parallel-hand correlations (§4). Baseline removal is applied *after* extraction (step 4), and the primary-beam correction enters the quoted thresholds and fluxes (§4.2) rather than the extracted amplitudes the statistic ranks. The estimator’s visibility-domain weighting is a fixed property of the released code (Data Availability), which itemises the quantities consumed and emitted per window and is the reproducibility reference for steps 1–3. For a sky position $\mathbf{x}$, each per-integration spectrum is baseline-subtracted by a running median along the *frequency* axis – there is no temporal median at any stage (§4.4) – the residual dynamic spectrum is stacked along each trial linear drift rate $\dot{\nu}$, and the statistic is the largest resulting single-channel amplitude

TABLE 1
NOMENCLATURE FOR SEARCH OUTPUT AND THE TWO SENSES IN WHICH A PROBABILITY IS QUOTED.

Term	Meaning
crossing	one channel $\times$ drift cell at the stellar position with $T \geq 5$: a threshold crossing (state 1)
stage-1 statistical flag	a window whose T_* exceeds all 512 control statistics (“star-exceeds-ring”) – a screening device, not a candidate significance (state 2)
candidate	a stage-1 statistical flag surviving every veto; none in this release (state 3)
5σ trigger threshold (P_{trig})	local instrumental trigger level, <i>not</i> a completeness limit
completeness (P_x)	the EIRP at which recovery of the searched class reaches x per cent; P_{50} , P_{90} not determined outside the calibration configuration
$p_{\text{rank}, \text{min}}$	$1/(N_{\text{ctrl}} + 1) = 1/513$, the rank <i>resolution</i> of the control ensemble
$P(\text{false flag})$	the survey false-positive probability after thresholding, correlation and non-Gaussianity; not the same quantity
crosswalk	“stage-1 statistical flag” (star-exceeds-ring) $\approx$ the “hit”/“event” of the wider literature (Enriquez et al. 2017; Margot et al. 2023) after their spatial-veto stage; the SP statistic $\approx$ their peak signal-to-noise ratio, with the false-alarm probability calibrated empirically by the control ring rather than a Gaussian tail
Class A / Class B	drift-resolved spectral-carrier search (fine channels) / unresolved excess-power coarse search (§4)
f_{dwell}	intra-track dwell fraction: the fraction of one continuous observation a carrier occupies its channel
p_{epoch}	epoch activity probability: the chance a transmitter is on during a given analysed epoch (§6.2)

in units of the window’s noise,

$$T(\mathbf{x}) = \max_{\nu, \nu} \frac{S(\mathbf{x}, \nu, \dot{\nu})}{\sigma(\mathbf{x})}, \quad (2)$$

where S is the drift-stacked flux density in one native channel and $\sigma(\mathbf{x})$ a single robust scale per position and window, estimated as $1.4826 \times$ the median absolute deviation of that position’s baseline-subtracted spectrum over all channels: local in position, global in frequency. The baseline median’s width W is likewise a single fixed channel count for all 431 windows, independent of native channel width, so the linewidth ceiling it imposes scales in frequency with each window’s channelisation (Appendix C). The crossing channel is *not* excluded from the scale – standard for a median-based estimator, whose 50-per-cent breakdown point makes a single-channel feature immaterial over a window’s hundreds of channels; the per-window local recomputation below excludes it explicitly. $T_* \equiv T(\mathbf{x}_*)$ at the proper-motion-propagated stellar position; the same operation, drift grid, channelisation and noise estimator give $\{T_i\}$ at all 512 control positions, which lie on a single ring centred on the target, evenly spaced in azimuth ($360/512 = 0.70$ degrees apart), at a radius well inside the primary beam but many synthesised beams from the target. Neighbouring controls are correlated below the synthesised-beam scale, so a window’s ring carries $N_{\text{eff}} \simeq 30\text{--}43$ independent spatial trials, as the symmetric reprocessing runs measure directly (Appendix N). Ranking T_* against 512 values fixes the smallest per-window p -value the design can express, $1/513$: the ring estimates the local false-alarm rate but cannot authenticate an individual candidate, and the survey-wide Bonferroni reference 1.2×10^{-4} lies two orders of magnitude below the floor (component v, §5.3). Table 2 sets out the data path in order, so the statistic can be reimplemented without the appendices.

Being global in frequency, that scale can misstate the noise *local* to a crossing wherever a window carries bandpass-edge structure, atmospheric features or a line forest. The retained products keep the per-window scale but not the per-channel spectra, so no local estimate around a crossing can be recomputed for this release survey-wide – but for the four flags we have now measured it. Re-extracting their retained per-integration spectra and recomputing the scale locally, within a band around the feature and excluding the feature itself, gives local-to-global ratios of 0.97 (CP–72 2713), 1.29 (HD 48370), 0.93 (β Pic B3) and 1.02 (β Pic B6). The global scale is thus accurate to a few per cent except towards HD 48370, where the local value sits 29 per cent higher because that field is filled with CO: rescaling both the statistic and its ring by the local value leaves it flagged ($T_* = 21.0$ against a corrected ring maximum of 20.8) – the disposition already assigned to it. For CP–72 2713 the local scale is slightly *lower*, so its excess is marginally understated rather than inflated, and no disposition moves. The retained products keep no drift-stacked cube, so this tests the noise scale and not T_* itself; a per-crossing local scale inside the search is a next-release item (§6.4). The same estimator runs at the star and at all 512 control positions, so a badly misstated σ would displace the control distribution, and the pseudo-star ranks are uniform in every band and at both channelisations (§5.3) – the one departing stratum is the fine-channel one, and that departure is the four flagged windows themselves.

The output vocabulary nests strictly (Table 1); no object of this release is a candidate.

The searched drift-rate range is the larger of (a) a generic frequency-scaled default ($12\text{--}13 \text{ Hz s}^{-1} \text{ GHz}^{-1}$), the Mason et al. (2024) fiducial with a threefold safety margin – a design boundary, not a claim that faster drift is implausible – and (b) for confirmed exoplanet hosts, the maximum line-of-sight orbital acceleration of the innermost known planet, so a transmitter with a short-period planet cannot drift out of the window. Drift rate maps onto line-of-sight acceleration

TABLE 2

THE STATISTICAL DATA PATH, IN ORDER. STEPS 1–8 RUN IDENTICALLY AT THE STELLAR POSITION AND ALL 512 CONTROLS; ONLY STEP 9 ONWARD DISTINGUISHES THEM.

1	calibrated measurement set from the ALMA archive
2	stellar position propagated to the observation epoch (Gaia DR3)
3	spectral extraction at that position, native channelisation
4	per-integration baseline removal: running median along frequency
5	stack along each trial linear drift rate
6	robust per-position noise scale $\sigma(\mathbf{x})$ (MAD)
7	$T(\mathbf{x})$ from Eq. 2
8	repeat 3–7 at 512 control positions on a ring
9	threshold: $T_* \geq 5$ (a crossing)
10	rank test: $T_* > \max_i T_i$ (stage-1 statistical flag)
11	molecular-line mask, $\pm 50 \text{ km s}^{-1}$ stellar frame
12	vetting: recurrence, closure phase, cross-target occupancy

TABLE 3

LINE-OF-SIGHT ACCELERATION AND DRIFT RATE AT 93.1 GHz FOR CIRCULAR ORBITS OF SHORT-PERIOD PLANETS (PERIODS AND SEMIMAJOR AXES FROM THE NASA EXOPLANET ARCHIVE), AGAINST THE ADOPTED DRIFT-RATE SEARCH DOMAIN OF $12\text{--}13 \text{ Hz s}^{-1} \text{ GHz}^{-1}$ ($|a_{\text{los}}| = 3.6\text{--}3.9 \text{ m s}^{-2}$): TRANSMITTERS ON THE INNERMOST TRAPPIST-1 PLANETS SIT AT OR BEYOND IT. ECCENTRICITY RAISES THE PEAK LINE-OF-SIGHT ACCELERATION OVER THE CIRCULAR VALUE BY UP TO $(1+e)/(1-e)^2$ AT PERIASTRON: $1.4\times$ AT $e = 0.1$, $1.9\times$ AT $e = 0.2$, SO PLANETS WITHIN A FACTOR ~ 2 OF THE CEILING – THE BORDERLINE ROWS – MOVE OUTSIDE THE SEARCHED DOMAIN AT MODEST ECCENTRICITY; THE CIRCULAR-ORBIT ENTRIES SHOULD BE READ WITH THIS MARGIN.

System	P (d)	a (au)	a_{los} (m s^{-2})	$\dot{\nu}$ (Hz s^{-1})
Earth analogue	365.25	1.000	0.006	1.84
Proxima b	11.19	0.0485	0.31	95
TRAPPIST-1 d	4.05	0.0223	1.07	334
TRAPPIST-1 c	2.42	0.0158	2.14	665
TRAPPIST-1 b	1.51	0.0115	4.00	1243

through the first-order Doppler relation

$$a_{\text{los}} = c \frac{\dot{\nu}}{\nu}, \quad (3)$$

so the ceiling is most compactly quoted in cross-band-comparable form $\dot{\nu}/\nu$: $12\text{--}13 \text{ Hz s}^{-1} \text{ GHz}^{-1}$ is $\dot{\nu}/\nu = (1.2\text{--}1.3) \times 10^{-8} \text{ s}^{-1}$, i.e. $|a_{\text{los}}| = 3.6\text{--}3.9 \text{ m s}^{-2}$ (Eq. 3), independent of observing frequency – the acceleration at $\sim 0.04 \text{ au}$ around a solar-type star. Earth-motion contributions ($\lesssim 0.13 \text{ Hz s}^{-1} \text{ GHz}^{-1}$) lie two orders of magnitude below, subsumed by the grid. The grid itself is uniform in $\dot{\nu}$, spanning the searched range in n_{drift} steps (2–1944 across windows, recorded per window in the released file), so the worst-case mismatch – a carrier midway between trial rates – leaves a residual drift that traverses at most 0.26 of one native channel over the longest fine-class track (0.29 over the coarse class): the mismatched carrier stays within a single channel of its trial track throughout, and the stacked-amplitude loss is bounded by that sub-channel smearing alone, an order below the recovery slope measured between 5σ and 10σ . The injection campaign injected on grid points; continuous randomisation between grid points is pre-registered with the stratified campaign of §6.4. Table 3 grounds the ceiling physically: a transmitter co-orbiting an Earth analogue trails it by three orders of magnitude, Proxima b sits well inside, and the innermost TRAPPIST-1 planets sit at or beyond the adopted domain – very short-period systems are its sacrificial edge. The domain is a search definition, not a physical prior: an artificial emitter need not be tied to a known planet at all (a free-flying platform, rotating structure or deliberately chirped carrier can accelerate as it chooses), and such emitters are unconstrained, not excluded. These are *apparent* topocentric drift rates, including deliberate transmitted chirp and acceleration-induced drift, which the survey does not distinguish; our constraints concern linear apparent drift within the searched range only.

Across the 431 windows maximum searched drift rates span $1.1\text{--}5.9 \text{ kHz s}^{-1}$, native channel widths $15.3 \text{ kHz--}31.25 \text{ MHz}$ (median 15.625 MHz), trial drift rates per window 2–1944 (median 4). The threefold margin matters: at $1\times$ the release would have been blind to two of its threshold-crossing windows, including the AU Mic crossing; at $5\times$ no disposition changes, transmitters drifting faster than $\sim 13 \text{ Hz s}^{-1} \text{ GHz}^{-1}$ remaining unconstrained. Intra-integration smearing is negligible almost everywhere: at most 1.10 channels, median 0.001; only three windows have $\eta_{\text{smear}} < 0.99$ (marked * in Appendix H; Table 14), the worst the β Pic Band 6 fine window (effective threshold $1.74\times$ nominal). Channel widths are effective resolutions, verified against the measurement sets’ spectral window tables (Appendix C).

4.2. Frequency and velocity reference frames

Every frequency axis here – searched spectra, tabulated ranges, candidate frequencies – is *topocentric*, as delivered by the ALMA correlator, no velocity-frame shift applied to any data product at any stage. Velocities enter only in the

vetting stage’s stellar-frame line-exclusion mask, whose conversion chain is in Appendix C. The stacking is coherent in frequency and in the drift-rate trial but preserves no electric-field phase coherence between integrations.

The native channel width is also the linewidth convention for every EIRP threshold here – the commonest confusion when comparing technosignature limits across surveys. The derivation is explicit: for a transmitter intrinsically narrower than the channel ($\delta\nu_{\text{tx}} \ll \Delta\nu_{\text{ch}}$), channel dilution spreads its total line flux F_{line} across the channel, $S_{\nu,\text{ch}} = F_{\text{line}}/\Delta\nu_{\text{ch}}$; triggering requires $S_{\nu,\text{ch}} \geq S_{\text{min}}$, so the minimum detectable total power is $4\pi d^2 S_{\nu,\text{ch}} \Delta\nu_{\text{ch}}$ – independent of $\delta\nu_{\text{tx}}$ below the channel width. $\text{EIRP}_{5\sigma}$ is thus the minimum *total* power of a channel-confined carrier, not a spectral power density; dividing by $\Delta\nu_{\text{ch}}$ converts to the minimum detectable spectral power for wider transmitters (a 10^{15} W threshold at 15.625-MHz channelisation is $6.4 \times 10^7 \text{ W Hz}^{-1}$, and the same division applies wherever the two quantities appear together below). The 1-Hz and 3-Hz limits of Hz-resolution surveys are total powers in the same sense, each the best case of its own channelisation, which is what makes the axes of Fig. 1 commensurable at all. Channelisation is the dominant price the archive exacts, but we deliberately do not project an Hz-resolution sensitivity: a radiometer-equation scaling $P_{\text{min}} \propto \sqrt{\Delta\nu}$ across six orders of magnitude in spectral resolution is a thought experiment, not an achievable ALMA sensitivity, and quoting it invites the misreading this paper avoids. The worked example (UV Ceti, Band 3), sub-channel qualifications, smearing table and channel-width verification are in Appendix C.

The spectral-carrier search covers every spectral window of an observation, capped at 8 per target to bound compute cost; the cap bound exactly one target/band (61 Vir Band 7: four of eight windows searched), whose frequency-space selection function is therefore tabulated. Of the 107 star/band datasets, 53 have every configured window searched; the remaining 3 (τ Ceti, Kapteyn’s Star, Van Maanen’s Star) are represented by their deepest single window, marked in Appendix H, full multi-window coverage awaiting the next release. Each searched window yields its own $\text{EIRP}_{5\sigma}$ row; Figure 9 plots only the deepest per target/band.

Velocity coincidence, not drift, is the operative discriminant when a crossing is checked against the eight masked species (§4): circumstellar emission sits at rest offset by the gas’s kinematics, whereas a technosignature may appear at any frequency and drift rate. The searches as executed used a narrower one-channel-width implementation, agreeing with the velocity-space form on every disposition here; its justification and the bandwidth removed are in §5.3.

Both carrier-lane thresholds and continuum fluxes are corrected for ALMA’s primary-beam response at the star’s offset from the phase centre, negligibly for most targets (median factor 1.0001, 90th percentile 1.012, none above 2%). Two sit at large offset. ϵ Eri Band 6 lies at ~ 0.7 primary-beam FWHM, where the correction reaches $2.9\text{--}3.4\times$ and the Gaussian beam form fails; rather than publish provisional values we *withhold* those four windows, which enter no number here but are released, flagged, in the per-target file and return once the holography beam model is applied (that recomputation is not yet in the released snapshot, Data Availability). Sirius B is retained and quoted as such: corrections $3.6\text{--}16$ per cent across bands, a bounded systematic rather than an invalid model. Excluding it too would change neither endpoint of the quoted EIRP range, nor the median ($1.3 \rightarrow 1.4 \times 10^{15} \text{ W}$), nor any conclusion.

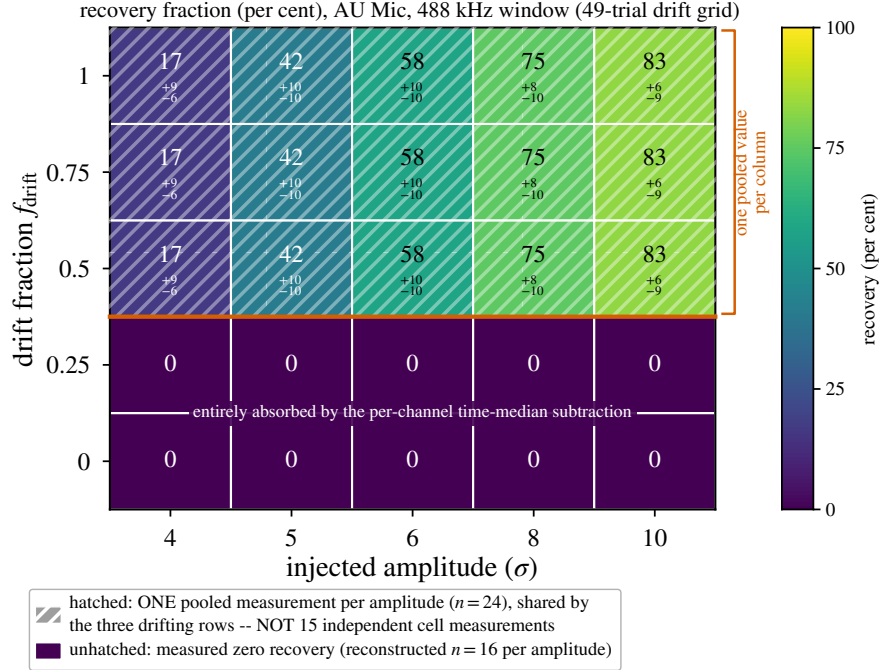
4.3. Reference transmitter benchmarks

The EIRP thresholds of §5 are compared throughout to two human-transmitter benchmarks, both from Sheikh et al. (2025a): an Arecibo-like planetary radar (Drake 1974) (S-band $\text{EIRP } 2 \times 10^{13} \text{ W}$ – the most powerful directed transmitter an Earth-equivalent civilisation could point at us, quoted purely as a technological-power benchmark), and unintentional Earth mobile-network leakage, no signalling intent (EIRP of order $4 \times 10^9 \text{ W}$).

4.4. Ancillary and validation analyses

Four analyses beyond the spectral-carrier search are executed here, each specified in full in Table 15 and summarised only as far as the main text depends on it.

Injection-recovery validation (Appendix D). Synthetic tones of known amplitude were injected into real visibilities and recovered with the unmodified pipeline (1200 trials across six window configurations, three bands and both resolution classes; trial-level records were not retained for that campaign, and the dwell campaign of §4.5 now supplies them). The pipeline performs no temporal subtraction at any stage: its baseline step is a per-integration running median along *frequency* (Table 2, step 4), which by construction cannot remove a feature confined to one or two channels, and its de-drift step is an inverse-variance-weighted sum over integrations, which accumulates a persistent carrier coherently. One validation-history item must be stated plainly: the campaign’s original report of 0 of 500 coarse-window recoveries was an artefact of its recovery *criterion*, not of the search pipeline – a drift-matching clause that could not be satisfied on coarse windows by construction. It was caught by a quantitative bound, not by inspection: the all-zero return arrived at injected amplitudes of $20\text{--}200\sigma$, where the machinery-only lane’s measured amplitude fidelity of $-9/+14$ per cent guarantees a detection, so no working criterion can return zero there; that impossibility isolated the fault to the criterion. The identification is logged as a pre-freeze validation change (Table 21, Appendix P), its wall-clock record the same versioned audit trail as the statistic revision of §5.4. On code paths, stated plainly: the campaign and the machinery-only lane share one pipeline and one trial list; the faulty clause sat in the criterion applied to that pipeline’s output, so the lane is the same path with the clause’s decision step bypassed – an internal control isolating the clause, not an independent implementation. The corrected criterion has not been re-derived by an independently written implementation even on a subset of trials; it is retained as a regression fixture, and the cross-check is recorded as an open item (§6.4). The independent evidence is the dwell campaign of §4.5, whose post-baseline detection criterion is separately derived and which measures persistent, near-static carriers as *recovered*, not suppressed (Table 4) – the reading the artefact had reversed. The primary search statistic has no automated injection test on every pipeline



Trial-level records were not retained; errors are binomial Wilson 68 per cent intervals on the pooled class.

FIG. 4.— What the pipeline recovers, for the best-calibrated fine-channel configuration, against injected amplitude and drift fraction f_{drift} (injected drift as a fraction of the searched ceiling). Carriers with $f_{\text{drift}} \leq 0.25$ show zero through-pipeline recovery at every amplitude – a measured property of the through-pipeline path, distinct from the coarse-window criterion artefact withdrawn in §4.4: post-baseline injection (§4.5) recovers near-static carriers, so the loss originates in the full path, not in the search statistic, and the searched-class completeness quotes below apply to the drifting class only. Carriers with $f_{\text{drift}} \geq 0.5$ recover progressively, from 42–83 per cent between 5σ and 10σ . Trial-level records were not retained for this campaign, so surviving fractions pool grid phase, sub-channel placement and the three drifting rows: the figure is drawn as discrete cells, not an interpolated surface, each column’s hatched rows sharing one pooled measurement ($n = 24$ per amplitude) rather than fifteen independent ones. Intervals are binomial Wilson 68 per cent; absorbed rows are measured zeros.

change; three quantified checks bound undetected pipeline faults instead: the machinery-only lane (fidelity $-9/+14$ per cent – the bound that caught this artefact), the AU Mic re-analyses of Table 12 (three treatments, recomputation agreeing with the released products), and the pseudo-star rank uniformity (§5.3: 220 672 pooled ranks uniform, mean 0.5010 against 0.5, KS $p = 0.37$), which any defect of the statistic itself would displace. For the drifting class on the fine window recovery is 17, 42, 58, 75 and 83 per cent at 4, 5, 6, 8 and 10σ – measured on the single calibration configuration (one star, one Band 6 window) and transferred by assumption to all 118 fine windows, an error the bracketing of §6.2 treats as a $0.5\text{--}2\times$ rescaling – giving $\text{EIRP}_{50} = 1.10 \text{EIRP}_{5\sigma}$ and, as a bound, $\text{EIRP}_{90} > 2.0 \text{EIRP}_{5\sigma}$ (Fig. 12, Table 18). Tables here stay in nominal $\text{EIRP}_{5\sigma}$; the conditional in-band transmitter fraction of §6 is the one place measured completeness is folded in, Band 8 completeness being transferred by shared construction, not measured.

4.5. Dwell-fraction completeness

The searched class is defined by how long a carrier occupies one native channel, so we measure recovery against that directly. A carrier was injected into the retained per-integration spectra of 12 real windows, six coarse and six fine, at amplitudes of 2–20 times the per-integration noise and dwell fractions $f_{\text{dwell}} = 0.10\text{--}1.00$, six realisations each: 3888 trials, released with one row per injection (Table 4). Its detection criterion – recovered when the logged per-integration S/N reaches five – was independently re-derived from the released rows: over 3888 trials and 12 windows the rule and the logged flag agree 3888 times (0 mismatches; highest unrecovered S/N 5.00, lowest recovered S/N 5.04 – a clean separating gap). Recovery *rises* monotonically with dwell. Pooling amplitudes at or above 4σ , coarse windows recover 100 per cent of continuous carriers, 100 per cent at $f_{\text{dwell}} = 0.25$ and 83 per cent at 0.10; fine windows give 100, 91 and 71 per cent.

An end-to-end check confirms it through the unmodified pipeline rather than a re-implementation. Injecting a zero-drift, always-on tone into the calibrated visibilities of a coarse Band 7 window at 1, 3 and 10 times the per-integration rms and running the released extraction and search returns $T_\star = 20.5, 65.3$ and 217.7, recovered within one channel of the injected frequency and flagged as detections in every case. For 639 integrations the ideal coherent gain is $\sqrt{639} = 25$, so a carrier at the per-integration noise level should reach $\sim 25\sigma$: the measured 20.5 is that, less weighting and flagging losses. Nothing is being suppressed.

Three distinctions discipline how these numbers are read. *Injection point*: the dwell campaign injects into retained per-integration spectra, after the baseline step of Table 2; the end-to-end check injects into visibilities, before it. Both recover static and near-static carriers, so coarse-window amplitude is characterised by the dwell experiment –

TABLE 4

RECOVERY OF HIGH PER-INTEGRATION-S/N INJECTED CARRIERS AS A FUNCTION OF DWELL FRACTION, POOLING INJECTED AMPLITUDES $\geq 4\sigma$ PER INTEGRATION, FROM 3888 TRIALS OVER 12 REAL WINDOWS. RECOVERY RISES WITH DWELL: A CONTINUOUSLY TRANSMITTING CARRIER IS THE EASIEST CASE, NOT THE HARDEST. THIS EXPERIMENT ESTABLISHES THE DEPENDENCE ON TEMPORAL OCCUPANCY, NOT THE ABSOLUTE COMPLETENESS AT THE SURVEY DETECTION THRESHOLD (THAT IS THE END-TO-END CAMPAIGN OF FIG. 4).

f_{dwell}	coarse windows	fine windows
1.00	100%	100%
0.50	100%	99%
0.25	100%	91%
0.10	83%	71%

83–100 per cent recovery at $\geq 4\sigma$ per-integration amplitude (Table 4) – which is not a survey-threshold completeness measurement. *Drift discrimination*: on coarse windows every in-grid drift is sub-channel over the track, so the search recovers the carrier’s amplitude but carries no information on its true drift rate – the coarse majority (313 of 431 windows) is drift-undiscriminating, and no drift-rate claim is made for it. *The one unmeasured cell*: through-pipeline recovery of near-static carriers on *fine* windows. The 1200-trial campaign failed them under a criterion that mixes amplitude and drift matching and retained no records to separate the two (Appendix D); no end-to-end zero-drift injection has been run on a fine window. Table 18 marks that cell unmeasured. The earlier claim that the pipeline suppresses persistent carriers is withdrawn (§4.4, §4.6).

Drift-ceiling caveat. The fixed $12 \text{ Hz s}^{-1} \text{ GHz}^{-1}$ searched ceiling, line-of-sight $|a| = 3.6 \text{ m s}^{-2}$, is uniform across every window: two orders of magnitude clear of Earth-rotation and Earth-orbit drifts, but 11 per cent *below* the 4.0 m s^{-2} of a close-in planet on a TRAPPIST-1 b-like orbit. The limits of §6 are therefore conditional on it (Figure 13).

Closure-phase vetting (Appendix E). A one-sided point-source test – evidence against celestial point-source behaviour, never an authentication – unavailable to single-dish and beamformed surveys, needing multiple independent baselines. Implemented and simulation-validated, it has at this survey’s flux densities *no discriminating power*: per-baseline S/N never exceeds 5.1 in the positive-control fields, so no flagged window could have failed it whatever its nature. It is a proof-of-principle diagnostic for deeper releases, not an operative filter here, and no window is described as having “survived” it.

Extended and cross-target screens (Appendices F and G). One interference class deserves its own statement: satellite constellations grew across the 2013–2025 archival baseline. No known ordinary downlink allocation maps directly into the searched frequencies, but that excludes neither harmonics, intermodulation products nor unconventional emitters, and no exclusion claim rests on allocation alone: the primary defences are the control ring and the time-domain tests. A satellite enters star and ring extractions alike within an integration; the channel-width mismatch and the drift-stacked statistic dilute transient passes; a pass confined to one block and one target would still evade the cross-target occupancy check by construction. A near-field emitter, moreover, need not share a celestial source’s far-field interferometric response – among the reasons the next release adds a dedicated per-integration RFI screen and visibility-domain checks. A chirped (quadratic-drift) extension and per-channel periodicity search ran on the 122 retained spectra as a demonstration, not a survey-grade screen; null behaviour is as expected (star-exceeds-ensemble rates 10.7 and 13.1 per cent against an 11-per-cent chance rate), no instance is a candidate, and no limit derives from them. Separately, crossings towards different targets were cross-matched for recurrence at the same sky frequency – the archival substitute for the ON/OFF cadence this single-pointing survey lacks. Of the 366 star pairs with overlapping searched ranges, none falls within the matching kernel, subject to one bookkeeping caveat (64 crossings of one β Pic window untested) in the appendix.

4.6. Validation changes made before the frozen analysis

The pipeline as released differs from earlier working versions in five substantive ways – threshold convention, search statistic, baseline-operator description, line-exclusion width and occurrence framing – each settled before the analysis was frozen and logged once, with its effect on released products, in Table 21 (Appendix P) rather than narrated where it arose; asymmetric-derived quantities are historical there and enter no operative number. None changes any released data product except where stated there. The boxed *development/confirmation split* of §1 governs all of it: performance estimated on this release is retrospective validation; the remaining survey sample is the first independent confirmatory dataset.

Scope of the constraints. *What was searched*: 33 frequency islands totalling 93.1 GHz (88.2 GHz after molecular-line masking) at 15.3 kHz–31.25 MHz channelisation, during the analysed archival intervals, for carriers within the searched drift ceiling and one native channel in width, at nominal trigger thresholds whose measured recovery is 42 per cent (single configuration, transferred by assumption). Inside that box the non-detection is a constraint. *Nothing in this paper bounds the prevalence of*: (i) drift-rate structure on the 313 coarse windows, which carry no drift discrimination and no survey-threshold completeness curve (§4.5); (ii) transmitters emitting *outside the searched spectral windows* – those 33 islands, not ALMA’s frequency range (Fig. 2); (iii) transmitters active *outside the archival epochs* – the limits assume near-continuous emission and are unconstrained below epoch activity ~ 0.1 (Table 23); (iv) drift rates *above* the searched ceiling $|a| = 3.6 \text{ m s}^{-2}$, excluding the most

TABLE 5

THE SURVEY AS FROZEN FOR THIS RELEASE, AND THE PAPER’S HEADLINE NUMBERS IN ONE PLACE. EVERY COUNT QUOTED IN THIS PAPER RECONCILES WITH THIS TABLE; SUBSET NUMBERS ARE IDENTIFIED AS SUCH WHERE USED. THRESHOLDS ARE NOMINAL (BOXED RULE, §4). THE FINE- AND COARSE-CLASS RECOVERY ENTRIES ARE NOT PARALLEL MEASUREMENTS: THE FIRST IS A THRESHOLD-COMPLETENESS EXPERIMENT, THE SECOND A DWELL EXPERIMENT AT HIGHER PER-INTEGRATION AMPLITUDE (§4.5). THE OCCURRENCE-RATE FORMALISM IS A METHODOLOGICAL DEMONSTRATION (§6.2, APPENDIX Q) AND IS DELIBERATELY ABSENT FROM THESE HEADLINE NUMBERS.

Quantity	Value
Census stars within 40 pc	168
Stars searched; independent systems	88; 82
Star/band datasets	107 of 208 planned (51%)
Spectral windows searched	431
extracted; repeats, defective, withheld	462; 21, 6, 4
Class A drift-resolved / Class B unresolved-excess	118 / 313
Execution blocks	102
Distance range	1.30–39.63 pc
Unique frequency union	93.1 GHz (33 intervals, 89.6–495.1 GHz)
Windows by ALMA band (3/4/5/6/7/8)	40/4/4/216/155/12
Effective after line-exclusion mask	≈88.2 GHz
Native channel widths	15.3 kHz–31.25 MHz; median 15.625 MHz
Drift-rate ceilings	1.1–5.9 kHz s ^{−1} (12.0–13.3 Hz s ^{−1} GHz ^{−1})
On-source time, exposure axes	Table 10
Nominal $S_{\min}$	0.48 mJy–1.30 Jy (median 6.7 mJy)
Nominal trigger power P_{trig}	1.6×10^{13} – 2.8×10^{17} W (median 1.4×10^{15})
Recovery at threshold, fine/drift class	42^{+16}_{-14} % end-to-end (118 windows)
coarse class: dwell experiment, $\geq 4\sigma$ per integration	83–100% (313; not threshold completeness)
Threshold crossings ($\geq 5\sigma$ at the stellar position)	20 windows (all Class A)
Star-exceeds-ring windows ($\geq 5\sigma$)	4 windows (all Class A)
circumstellar or foreground CO	3
unclassified	1
Credible technosignatures	0

close-in orbits (Fig. 13); (v) carriers placed *inside* the $\pm 50 \text{ km s}^{-1}$ molecular-line mask – excluded search space, not searched-and-empty; (vi) emission *broad*er than one native channel, for which the thresholds must be rescaled; (vii) anything below the nominal threshold, where measured recovery is 42^{+16}_{-14} per cent (Appendix D); (viii) any transmitter-frequency prior spanning ALMA’s tuning range, under which the conditional bound vanishes (§6.2).

5. RESULTS

The results are presented in three tiers of decreasing evidential weight: the search itself – the non-detection, per-window thresholds and their measured completeness (§5.2, §5.3); the four flagged windows and their dispositions (§5.3, §5.4); and ancillary by-products – continuum, line catalogue, audits – that gate nothing in the primary lane (§5.5). One robustness statement belongs with the headline itself: the non-detection is stated under two late-stage corrections, and both made the reported picture more conservative, not less. The mid-analysis statistic revision (§5.4, Table 11) only ever removed flags – seven windows before, four after, no window in which the star dominated its own controls – and the withdrawal of the coarse-window 0-of-500 recovery artefact (§4.4) restored a sensitivity claim (persistent near-static carriers are recovered, not suppressed) rather than removing one. The underlying evidence for both consolidates in Appendix P.

Strict volume ordering (nearest star first; §3) weights these results to the very nearest stars, spanning 1.30–39.63 pc. Per-target values, including each band’s searched frequency range, are tabulated in Appendix H; the released machine-readable file (`per_target_results_v3.32.csv`) additionally gives the one-line-per-star view – deepest threshold per star – and marks the two continuum-only stars.

5.1. *Barnard’s Star and Wolf 359*

Barnard’s Star and Wolf 359 – the second- and fourth-nearest stellar systems – have not, to our knowledge, previously been searched for technosignatures at ALMA frequencies. Both are non-detections in public Band 6 archival data: nominal EIRP_{5 σ} trigger thresholds of 6.2 – 6.9×10^{13} W (Barnard’s Star, 4 windows) and 7.8 – 9.1×10^{13} W (Wolf 359), with continuum upper limits of 0.575 and 0.453 mJy.

5.2. *Data-quality exclusions and open items*

Of the 208 planned target/bands, 107 are searched to completion and reported here, the rest queued for later releases bar a few failures excluded from all counts. Four classes within the 107 are absent from the released per-target file (Appendix H) and itemised with forensics in Appendix P: (i) four process-level search failures, each still carrying a continuum non-detection; (ii) one pair excluded for data quality; (iii) windows partially withheld for a noise-handling defect; and (iv) one relabelled entry.

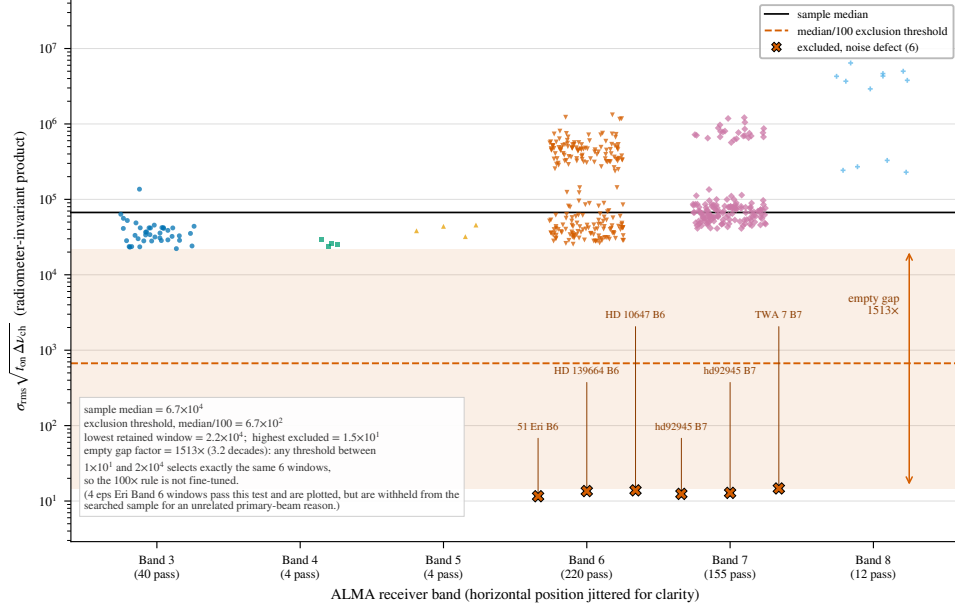


FIG. 5.— Quality control on the noise estimate (§5.2): the radiometer-invariant product $\sigma\sqrt{t_{\text{on}}\Delta\nu_{\text{ch}}}$, the six excluded windows a factor 1.5×10^3 below the lowest retained across an empty gap.

THE COARSE-NOISE DEFECT, AND THE BOUND ON ITS REACH

A reproducible defect produces implausibly small noise in a minority of coarse-channel windows; naming the two affected did not generalise as the sample grew, so item (iii) is now caught by a physical test. The radiometer relation $\sigma \simeq \text{SEFD}/\sqrt{N_{\text{bl}}\Delta\nu_{\text{ch}}t_{\text{on}}}$ makes $\sigma\sqrt{t_{\text{on}}\Delta\nu_{\text{ch}}} \simeq \text{SEFD}/\sqrt{N_{\text{bl}}}$ a prediction, not a convention: across a heterogeneous archive it varies by factors of a few, so a window $100\times$ below the sample median is a bookkeeping defect, not a fluctuation; we exclude such windows. The failure mode itself is not yet identified – reproducible, confined to coarse-channel windows, and to be named in the completed survey’s pipeline release, whose first change promotes this physical test from a release-time screen to a pre-count regression gate on every window – but two positive facts bound it. First, the retained population’s radiometric quantity is continuous from $0.33\times$ to $\sim 4\times$ the sample median with no secondary mode: a tenth-strength version of the defect would sit inside the same empty gap – between $0.1\times$ and $0.33\times$ the median – where no retained window lies. Second, as a window-level bookkeeping defect it deflates σ at the stellar and control positions alike, so it can only over-state sensitivity, never manufacture a star-exceeds-ring window; the six sit three decades below the retained population across an empty gap and are absent from the per-target file and every count. Could the defect recur undetected among the retained windows? The test screens each window on its own measured quantity, so a recurrence with the observed signature cannot pass it; a recurrence an order of magnitude weaker would pass, but would then sit inside the continuous retained population, where its only effect is the same one-directional over-statement of sensitivity. Six of 431 windows fail, across five stars, a factor 1.5×10^3 below the lowest retained across an empty gap of 3.2 decades (Figure 5); the threshold is not fine-tuned, any value between 1.5×10^1 and 2.2×10^4 selecting the same six. Their nominal thresholds, $2.0\text{--}7.8 \times 10^{12}$ W, would otherwise have set this paper’s headline depth; excluding them, the best is 1.6×10^{13} W. 21 of the 462 extracted rows are duplicate catalogue entries for the same physical window – the same (star, execution block, frequency range, channel width) key appearing twice in the archive query. All are repeated archive products or alternative calibrations of the same execution block, never independent observations, so one row per key is the correct statistical unit. The retention rule is deterministic and quantity-independent: the first-listed row of the key in the archive query’s own identifier order, a rule that consults no measured value and so cannot prefer a crossing or a deeper threshold; it is frozen unchanged for the confirmatory sample. Most repeats are identical, but two are not – two reductions of the same τ Ceti Band 6 window differ by 44 per cent in threshold, and the two HR 1010 Band 6 rows straddle the 5σ threshold (5.10 and 4.95) – so the crossing count is rule-dependent at the level of one window. The four stage-1 statistical flag windows are not: no discarded row is both a crossing and star-exceeds-ring, and every row, kept or discarded, ships with its status, making the choice auditable row by row. The audit is otherwise clean apart from four windows of elevated system temperature, and every sample identifier re-resolves independently in SIMBAD and *Gaia* (Appendix P).

Item (iv) is a cautionary example for archival work: the catalogue entry “ γ Lupi” resolved to the F4V debris-disc host HD 139664, holder of the historical identifier “* g Lup” – a name-lookup substitution caught by inverting the pipeline distance and recovering the *Gaia* DR3 parallax (Appendix P).

5.3. Technosignature search

All 107 star/band datasets yielded at least one carrier-lane result (process-level failures, §5.2, uncounted), spanning 431 spectral-window measurements after the catalogue audit removed 21 duplicate rows (Appendix H); by ALMA band, B6/216 and B7/155 dominate (Table 19 gives the per-band breakdown). EIRP $_{5\sigma}$ trigger thresholds span 1.6×10^{13} –

TABLE 6

THE SPECTRAL-CARRIER DECISION CASCADE OF THIS RELEASE, FOR THE 431 WINDOWS SEARCHED AFTER QUALITY CONTROL (§5.2); VOCABULARY AS IN TABLE 1.

Stage	Count
Channel $\times$ drift trial cells (all windows)	4.60×10^7
Control positions compared per window	512
Windows searched after quality control	431
Windows containing ≥ 1 crossing ($\geq 5\sigma$ at the stellar position)	20
Windows containing a stage-1 statistical flag	4
attributed to circumstellar or foreground CO	3
indistinguishable from the control ensemble	1
Candidates (surviving every veto)	0

TABLE 7

THE FOUR WINDOWS CONTAINING A STAGE-1 STATISTICAL FLAG AND THE EVIDENCE DISPOSITIONING EACH. T_* IS THE SEARCH STATISTIC AT THE STELLAR POSITION (Eq. 2), ‘RING MAX’ THE LARGEST OF THAT WINDOW’S 512 CONTROL STATISTICS. $\Delta\nu$ AND Δv ARE THE SAME OFFSET OF THE CROSSING FROM THE NEAREST OF THE EIGHT MASKED SPECIES’ CATALOGUED TRANSITIONS, IN FREQUENCY AND IN VELOCITY, TOPOCENTRIC; THE $\pm 50 \text{ km s}^{-1}$ MASK APPLIES IN THE STELLAR FRAME AFTER THE CONVERSION CHAIN OF APPENDIX C. ITS APPLICATION TO THESE FOUR WINDOWS IS A *developmental* DISPOSITION – THE WIDTH WAS FIXED AFTER INSPECTING THE β PIC CROSSINGS IT MUST EXCLUDE – WHEREAS THE SAME RULE, NOW FROZEN, IS THE *confirmatory* CRITERION FOR THE REMAINING DATASETS (§4.6). NO DISPOSITION RESTS ON A SINGLE TEST; THE HD 48370 ROW IS THE LIMITING CASE – A NEAR-TIE OF RANK (27.10 AGAINST 26.83) DISPOSITIONED BY VELOCITY COINCIDENCE WITH INDEPENDENTLY PUBLISHED EMISSION (§5.3).

Window	ν (GHz)	T_*	ring max	$\Delta\nu$ (MHz)	Δv (km s^{-1})	Disposition
β Pic B3	115.2605	14.65	7.27	−10.36	−26.9 (CO 1→0)	circumstellar CO
β Pic B6	230.5164	11.68	9.12	−21.87	−28.4 (CO 2→1)	circumstellar CO
HD 48370 B6	230.7305	27.10	26.83	−26.33	−34.2 (CO 2→1)	foreground CO
CP−72 2713 B7	345.1152	5.81	5.68	−1526.2	−1326 (CO 3→2)	unclassified; n.s. at survey level

$2.8 \times 10^{17} \text{ W}$, median $1.4 \times 10^{15} \text{ W}$ (Figures 9, 2, 1); as expected for a volume-ordered survey the deepest are the nearest systems’ ($1.6 \times 10^{13} \text{ W}$ for the UV Ceti pair at 2.7 pc), the shallowest η Corvi, the first Band 8 target searched. Against the benchmarks of §4.3, only the two resolved components of G 272-61 at 2.7 pc reach below the Arecibo-like planetary-radar EIRP.

Presented in two stages, the statistics strictly before the astrophysics. *Before astrophysical masking*: of the 431 searched windows, 20 contain a $\geq 5\sigma$ on-source crossing, and four of those carry a *stage-1 statistical flag* – the stellar statistic exceeding all 512 ring controls, a screening device rather than a candidate significance. *Astrophysical interpretation*: three of the four flags have independent CO evidence – matched transitions, velocities and epochs at β Pictoris, independently published foreground emission at HD 48370 – and one, towards CP−72 2713, is unclassified; none survives vetting, so this release reports *zero candidates*. The $\pm 50 \text{ km s}^{-1}$ mask is not the instrument of rejection in this developmental sample – the astrophysical evidence is; the frozen rule applies prospectively to the confirmatory sample (§4.6). Table 6 follows the cascade, Table 7 each flagged window and the evidence dispositioning it.

Do the 4 stage-1 statistical flags exceed this survey’s own calibration, or is any distinguishable from the empirical background or known astrophysical emission? No to both.

SURVEY-LEVEL FALSE-ALARM CALIBRATION

Each window-level test compares the peak signal-to-noise ratio at the star’s position with the same statistic at 512 controls; the drift grid and thousands of channels apply identically to both, so the per-window trials factor is absorbed empirically. The statistic is *symmetric*: the on-source quantity is the single propagated stellar position, as each control is. Every number in this account derives from that frozen statistic. The retired asymmetric variant that maximised over a small region at the star only – and every diagnostic it produced, from its 15.1-per-cent star-exceeds-ring rate to the 4–19 per cent bracket it induced on $P(F \geq 3)$ – is development history, labelled as such and quarantined in Appendix P; it contributes nothing to the operative false-alarm probability. That probability has five components, in the order the procedure applies them.

(i) *The empirical spatial rank probability*. The per-window significance quantity of this survey is not a Gaussian-tail p-value but an *empirical spatial rank probability under the control-position null*: under exchangeability one designated position of 513 ranks first with probability exactly $1/513$ – a rank statement independent of whether the noise is Gaussian or heavy-tailed, so heavy tails cannot explain an excess. The survey expectation is $431/513 = 0.84$ windows, and $P(\geq 5 | 0.84) = 1.7 \times 10^{-3}$ under an independent-Poisson approximation. That is a real excess over the noise null, for an astrophysical rather than statistical reason: in 3 of the 5, genuine celestial line emission sits at the stellar position (Table 7), making the star rank first by construction rather than by chance, so those 3 are not draws from the noise null at all. Removing them leaves 2 rank-first windows against 0.84 expected, $P(\geq 2 | 0.84) = 0.21$ – entirely unremarkable: the CP−72 2713 window of §5.3, and one towards Barnard’s Star whose stellar peak (4.40 against a ring maximum of 4.38) does not reach the trigger threshold and so is not a flagged window at all.

Three levels stay distinct throughout: an *instrument/noise* null, calibrated by the control ring; an *astrophysical-source* null, addressed by line identification and kinematics; and only then a technosignature interpretation. The

chance budget must account for flags unattributed after the first two, not every window in which the star happens to be brightest. Two statistics appear: the *SP statistic* of Eq. 2 – the single stellar position against 512 single-position controls – is operative over all 431 windows and the source of every count in this paper; the *SR statistic* additionally maximises over the source region at star *and* controls and, needing raw-visibility reprocessing, exists for seven windows only, as a consistency check.

(ii) *The $\geq 5\sigma$ amplitude requirement.* The joint criterion gating a candidate is dominated by its threshold leg: only 20 of 431 windows contain any $\geq 5\sigma$ crossing at the stellar position, each contributing at most the exchangeable rate $1/513$ – the 0.84 expectation of component (i). The rank, not the trial count, is the calibrated quantity: a window’s channel $\times$ drift grid holds up to 1.1×10^5 effective trials (Appendix N), more than 512 controls could resolve individually, so the control maximum absorbs whatever effective trial count the noise realisation carries and the per-window probability is expressed as a rank with resolution $1/513$ rather than a Gaussian tail at some N_{eff} . Four flagged windows exceed the budget ($P(\geq 4 | 0.84) = 0.9$ per cent) with no tension: excluding the three astrophysical, it need explain one unattributed crossing against 0.78 expected across the 402 non- β Pictoris windows. The estimate absorbs execution-block systematics into the fixed block margins, assuming only that crossing and star-exceeds-ring status are exchangeable within a block.

(iii) *The line mask.* The velocity-space mask (Appendix I) removes, from every crossing, the velocity space of known molecular emission before that crossing can count as unattributed. It is an analysis choice, not a measurement: it never lowers the noise-null budget above, and no disposition in this paper rests on the mask alone – the three CO attributions are kinematic (Table 7). Statistic, mask and thresholds were frozen and deposited before the confirmatory sample is processed (the repository tag named in the Data Availability statement carries the frozen code and its timestamp); what that deposit protected against is quantified as development history in Appendix P.

(iv) *The family-level pseudo-target Monte Carlo.* A Monte-Carlo calibration applies the full selection, not the rank alone, to pseudo-targets: each trial draws one control per window uniformly from its ring as the pseudo-star and passes it through the operative gates ($T \geq 5\sigma$, then the velocity mask). The expected count of pseudo-target windows flagged by the full procedure is 0.20 (82 per cent of trials flag none, 18 per cent at least one). This is the *prespecified family-level procedure* for the release – one expectation carrying the threshold, the mask and the ring geometry simultaneously, on real data; the Poisson expectation of component (i) and the Bonferroni reference of component (v) are checks against it, not independent claims (Appendix N). Worked: 0.84 expected flags from the rank test alone; the amplitude gate and the mask together remove about three-quarters of that budget; and the Monte-Carlo procedure – which carries threshold, mask and ring geometry simultaneously – lands at 0.20 expected flagged windows. On the 384 windows of stars outside the development set (§1) the same procedure expects 0.17 and the control ranks remain uniform ($p = 0.73$) – the held-out behaviour the confirmatory sample will test.

(v) *The control-ensemble floor.* The controls can show a crossing unremarkable but cannot certify one significant survey-wide: the ring’s smallest expressible per-window probability is $1/513$, two orders above the Bonferroni reference 1.2×10^{-4} over 431 windows – and that reference is itself conservative, windows sharing execution blocks, so it is a threshold to compare against, not the calibrated family-wise probability (§4; box below). Authentication therefore falls to the localisation, recurrence and astrophysical attribution that dispositions each flag below. Stated in advance for later releases: N windows contribute $\sim N/513$ chance flags, so the full ~ 2000 -window programme expects ~ 4 , and one unattributed marginal crossing per release is the *expected* outcome, not an accumulating anomaly. We pre-commit that no crossing in this programme will be advanced as a candidate on rank statistics alone: promotion needs evidence the ensemble cannot supply – recurrence at an independent epoch, or a ring enlarged until the rank probability resolves survey-wide significance ($\gtrsim 10^4$ positions for a Bonferroni threshold over the completed survey; §6.4) – and until then every such crossing is reported with its disposition stated explicitly, and released with its products. No flagged window is promoted by any number in this account: the three CO dispositions rest on kinematic attribution to emission resolved, mapped or independently published, and the CP–72 2713 standing is a rank statement, invariant to the bookkeeping.

The trials chain, start to finish. The release evaluates 4.60×10^7 channel $\times$ drift cells; the measured lag-one correlation of the drift-tracked statistic ($\rho_{\text{lag1}} = 0.989\text{--}0.991$) collapses these to $8.5 \times 10^4\text{--}1.1 \times 10^5$ effectively independent cells per window. Lag one is sufficient for the use this number is put to: the effective- N feeds only the Bonferroni robustness figure below, while every operative calibration – the ring ranks and the pseudo-target Monte-Carlo above – is empirical, reproducing the observed flag counts from real control positions with no effective- N input at all. The control ring calibrates the resulting maximum empirically – no Gaussian tail is taken at any N_{eff} – leaving per-window resolution $1/513$. Across 431 windows: Bonferroni reference 1.2×10^{-4} (conservative); rank-budget expectation 0.84 flags; family-level Monte-Carlo expectation 0.20. Each number’s provenance is tabulated in Table 17.

Is the ring exchangeable with the star? Exchangeability is the central assumption of the statistic, not a technical footnote: the $1/513$ expectation holds only if the star and its 512 controls are exchangeable under the null, which spatially varying noise, sidelobes, correlated pixels or phase-centre-localised calibration residuals could spoil; we measure it by stratum (Fig. 6). The median per-window control maximum is flat across bands – 4.46, 4.59, 4.42, 4.52, 4.63 and 4.69 in Bands 3 to 8 – and a Kolmogorov–Smirnov test of the star’s rank among its controls against $U(0,1)$ is consistent in every band ($p = 0.20\text{--}0.83$) and the coarse-channel stratum ($p = 0.65$). Fine-channel windows sit higher

(control maximum 5.76 against 4.42), as expected from their far larger channel $\times$ drift trial count; not a violation, the test being conditional on the window.

Those tests are conditional on the stellar rank; for the stronger question – exchangeability of the ensemble itself – we ranked each of a window’s 512 controls against the other 512-1 by the statistic and add-one rule applied to the star, pooled over the release. The 220 672 pseudo-star ranks are uniform to the resolution the ensemble can express (mean 0.5010 against 0.5; the Kolmogorov–Smirnov $D = 1/512$ is the discreteness floor of a 512-point rank statistic, $p = 0.37$), in every band and at both channelisations, and a two-sample comparison of the 431 real stellar ranks with that pooled distribution gives $p = 0.38$: the stellar position behaves like one more position in its own ring – the null calibration the control-ring argument needs, measured not assumed (Fig. 11 makes the same comparison at the amplitude level).

One stratum does depart, reported rather than pooled away: the fine-channel windows fail the same $U(0,1)$ test at $p = 0.015$ ($n = 118$), containing four windows whose on-star peak exceeds their own control maximum against 0.22 expected – precisely the four flagged windows of Table 7. That is consistent with the presence of the astrophysical line features identified below, not a misbehaving ring; but the departure is genuine, so fine-window rank probabilities are slightly optimistic. It is also localised, not global: excluding the 20 fine windows that contain any $\geq 5\sigma$ stellar crossing restores uniformity ($p = 0.39$, $n = 98$), and excluding the development-set stars entirely leaves $p = 0.73$ over the remaining 384 windows ($p = 0.23$ in their fine stratum) – the departure is carried by the flagged and crossing windows, not by a miscalibrated ring. Two stratifications the retained products cannot support, storing no per-control radial position or primary-beam gain – so the per-control gain evaluation below is analytic, not measured. The control ring shares the star’s radius from the phase centre exactly (positions enter the extraction as exact phase centres, not pointing estimates), which makes the exchangeability violation a property of the beam’s *azimuthal* structure at fixed radius: for a circularly symmetric beam it is identically zero, at any offset. The radially symmetric component is bounded per band from the Gaussian form $G = \exp[-2.77(r/\theta_{\text{PB}})^2]$ with $\theta_{\text{PB}} = 1.13\lambda/D$: the 12-m primary beams span 51–64 arcsec in Band 3, 24–27 in Band 6, 16–18 in Band 7 and 12–12 in Band 8, so at the median stellar offset (0.1 arcsec) the star-versus-ring gain differential is at most 0.019 per cent even in the highest-frequency band – the effect lives entirely in the beam’s departure from circularity, not in the offset itself. At the largest retained offset (Sirius B, $r \simeq 0.23\theta_{\text{PB}}$) the Gaussian gain changes by 128 per cent per $0.01\theta_{\text{PB}}$ of effective radius, so percent-level holography or ellipticity terms at that radius map to percent-level differentials; the withheld ϵ Eri windows (§4.2) sit beyond the regime entirely. The sign of any true differential is one-sided and conservative: the primary beam attenuates an unresolved carrier’s extracted amplitude by $G(r)$ while the visibility-domain thermal noise it is measured against carries no such factor, so a suppressed control can only *inflate* star-exceeds-ring outcomes under the null – the safe direction for every disposition drawn here. The empirical rank distributions themselves (Fig. 6, and the pseudo-target calibration of §5.3) bound the net effect: expected flags 0.20 against a rank-only budget of 0.84, with no excess in any stratum. Constructing controls at *equal gain* – sampling ring azimuth at fixed holography-model gain rather than fixed radius – is pre-registered as the default for the completed survey, which removes the term by construction. Remaining robustness checks – trials-count reconciliation under channel adjacency, control-ring geometry – are in Appendix N.

THE FULLY SYMMETRIC REGION-MAX CHECK

The SR statistic applies identical source-region, drift-rate and frequency maximisation at the star’s position and at each of the 512 controls under one search rule. The frozen release products cannot meet that, so its domain is the seven locally reprocessed windows – six validation, plus the AU Mic window’s first-epoch block – all giving $T_\star = 3.7$ –4.9 against control maxima of 6.7–7.7, $p = 0.81$ –1.00: the star is exchangeable with its ring wherever the test runs at full symmetry. The single-position (SP) calibration above is therefore the operative release-wide criterion over all 431 windows, the SR statistic a validation subset; the two are never interchanged. The machinery ships with the release; the full statement, with the family-wise trials factors of Table 17, is in Appendix N.

ASTROPHYSICAL END-TO-END VALIDATION: β PICTORIS CO

Two of the four flagged windows are towards β Pictoris, a debris disc with reported CO emission (Matrà et al. 2017): one in Band 3 at 115.2606 GHz, 10.56 MHz below the CO($J=1\rightarrow0$) rest frequency, the other in Band 6 at 230.5161 GHz, 21.87 MHz below CO($J=2\rightarrow1$). Neither coincidence is exact, so the automated check missed them. Through the frame chain (Table 9; Fig. 8) the offsets are one kinematic component – -2.7 and -3.5 km s^{-1} from stellar systemic, 0.74 km s^{-1} apart – in the two lowest CO rotational transitions of a system with detected circumstellar CO; the unflagged coarse Band 6 window, from a third epoch six years earlier, independently shows CO at -3.6 km s^{-1} in the same frame – all three epochs and both transitions within 0.9 km s^{-1} of each other, deep inside the $\pm 50\text{ km s}^{-1}$ mask. We attribute both crossings to this kinematically offset CO. The pair is the survey’s astrophysical end-to-end validation – a known signal run through the same search that the null results run through – and what it recovers is the check list: the frequencies, to the frame chain’s stated tolerance; the velocity concordance across epochs and transitions; the localisation, the Band 3 stellar peak exceeding all 431 control maxima and the Band 6 peak all but its own CO-filled ring; and the drift class, zero to within the grid. Not quantified is flux against calibrated imaging (Matrà et al. 2017): line fluxes are not among the retained carrier products. A known line remains the more convincing end-to-end control precisely because nothing about it was tuned. A three-panel figure – the Band 3 and Band 6 features and both aligned in stellar velocity – would complete the record, but the channel-level spectral products of those execution blocks are not carried by the frozen release, and re-extracting them is the same declined visibility-level operation as

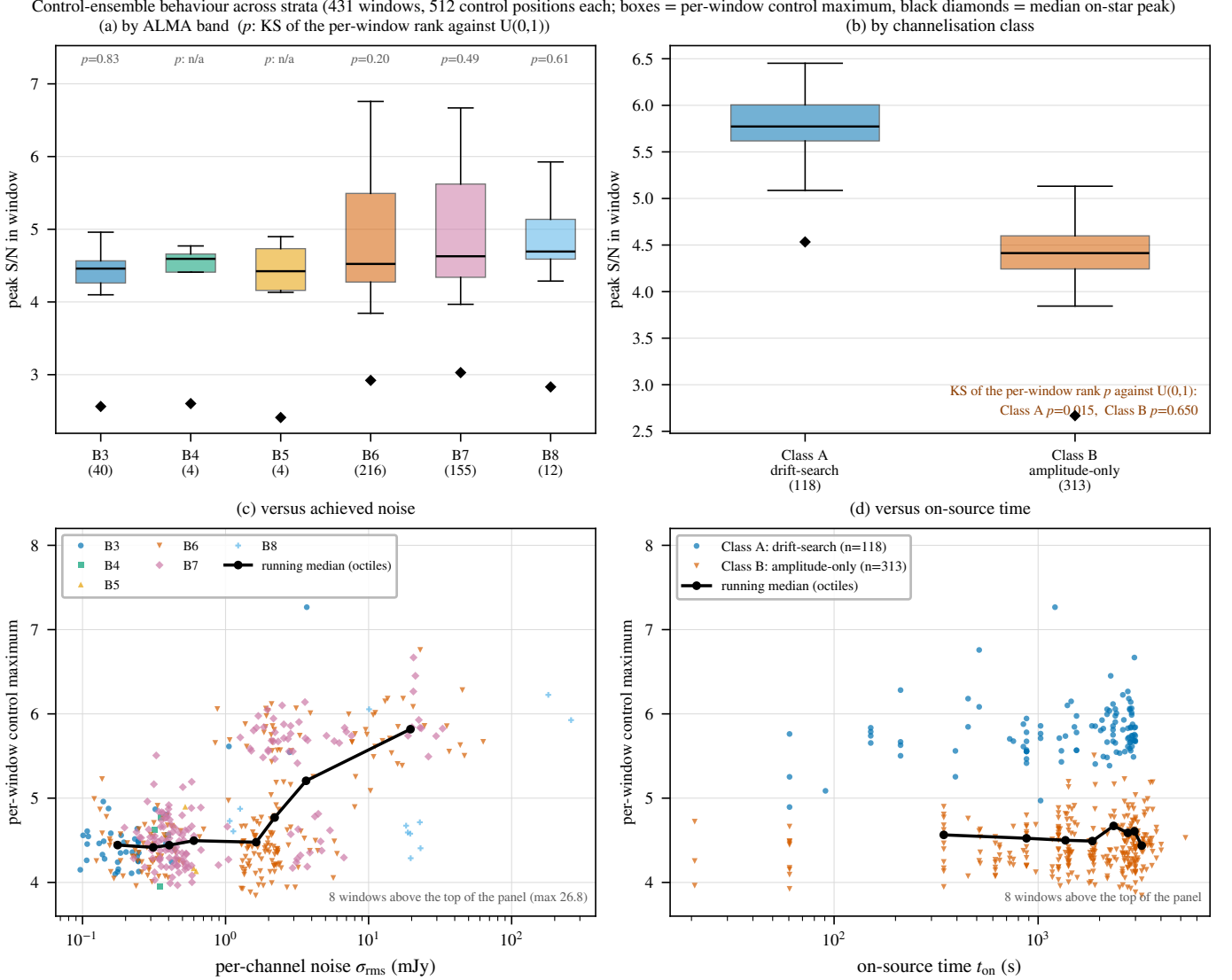


FIG. 6.— Exchangeability of the 512-position control ring by stratum. Boxes: per-window control maximum; diamonds: median on-star peak; p : the Kolmogorov–Smirnov test of the text. The ensemble tracks the on-star statistic in every band and in both channelisation classes; the one departure is the drift-search class ($p = 0.015$), driven by the four flagged windows of Table 7.

the CP–72 2713 tests (Table 8); Fig. 8 carries the velocity alignment, and the validation itself rests on the numbers above.

Their empirical significance under the release background is nonetheless extreme, so the attribution is no dismissal: the Band 3 region peak of 20.3 exceeds all 431 per-window control maxima (add-one $p < 0.0044$), the Band 6 peak of 12.8 all but one – that window’s own CO-filled control ring, reaching 15.2, by far the release maximum ($p \approx 0.0088$). In robust units – the stellar statistic minus the ring median, over $1.4826 \times$ the ring median absolute deviation – the standardised excesses are $\zeta = 36.0$ and 26.8 , the yardsticks against which the remaining flagged windows should be read. The alternative needs a transmitter coincidentally placed at matched offsets from two CO transitions in the same star.

DISPOSITIONS OF THE REMAINING FLAGGED WINDOWS

The third, towards HD 48370 in Band 6, is the release’s largest excess in absolute and smallest in relative terms: $T_\star = 27.10$ against a control-ring maximum of 26.83, the star exceeding its ring by one per cent of the statistic – a standardised excess of $\zeta = 4.2$, near the CP–72 value below, though its amplitude and velocity evidence differ. Star and ring rising together is what emission filling the primary beam looks like, and the window sits -34.2 km s^{-1} topocentric from CO($J=2 \rightarrow 1$). The attribution is independently published: Cataldi et al. (2023) report prominent CO($2 \rightarrow 1$) and $^{13}\text{CO}(2 \rightarrow 1)$ emission towards HD 48370 at $v_{\text{bary}} \approx 41 \text{ km s}^{-1}$, well separated from its 23.6 km s^{-1} systemic velocity and attributed by them to foreground cloud contamination; our window recovers it, and the crossing is consistent with, and very likely caused by, that previously reported foreground CO emission. The coincidence is with their quoted v_{bary} , not an independent frame-chain re-derivation on our side. It is also a stated limitation of the spatial test: this window is identified by the velocity coincidence and near-equality of star and ring, not the rank statistic.

TABLE 8

THE UNIDENTIFIED CP–72 2713 BAND 7 FEATURE: EVERY QUANTITY THE FROZEN RELEASE PRODUCTS CARRY, FOR PLANNING THE TARGETED RE-OBSERVATION. ROWS MARKED † REQUIRE VISIBILITY-LEVEL RE-EXTRACTION OR TARGETED RE-OBSERVATION AND ARE NOT PERFORMABLE ON THE FROZEN PRODUCTS; THEY ARE THE OPEN ITEMS OF THE PROMOTE/RETIRE PROTOCOL IN THE TEXT.

Quantity	Value
Execution block	A002_Xff0235_X4a6d; epoch and proposal code in the machine-readable release (Appendix A)
Band; window centre	7; 345.1152 GHz (bandwidth 1.73 GHz)
Channel width	488 kHz (0.42 km s^{-1})
On-source time	50 min (2994 s); single execution block
Drift grid searched	129 trial drifts to $\pm 4.14 \text{ kHz s}^{-1}$
Map rms; 5σ threshold	2.10 mJy; 10.5 mJy
Stellar statistic T_*	5.81
Ring maximum (512 controls)	5.68 (margin 0.13: the 1/513 resolution floor)
Feature flux	12.2 mJy
Trigger power; feature EIRP	$8.26 \times 10^{14} \text{ W}$; $9.60 \times 10^{14} \text{ W}$
Stellar distance	36.7 pc
Nearest catalogued transition	masked set: CO($J=3 \rightarrow 2$) at -1326 km s^{-1} ; wider re-query: $\text{H}^{13}\text{CN}(J=4 \rightarrow 3)$ at -195 km s^{-1}
Polarisation hands; time splits; baseline/antenna subsets	†
Visibility-domain localisation; calibrator fields; atmospheric transmission at 345 GHz	†
Synthesised beam; position uncertainty	† (in the machine-readable release linkage; not re-derived here)

THE UNIDENTIFIED CP–72 2713 FEATURE

The fourth, towards CP–72 2713 in Band 7, is the only flagged window with no astrophysical attribution, and is marginal: $T_* = 5.81$ against a ring maximum of 5.68, a margin of 0.13: the stellar position ranks first among the 513 tested spatial positions, but the available control ensemble is too small to establish survey-level statistical significance – 0.13 is the smallest margin 512 controls can express, a resolution floor, not a measurement of significance; in the robust units above its standardised excess is $\zeta = 4.6$, two orders below the β Pictoris controls. That margin is smaller than the scatter across this star’s own windows, whose other fine-channel control maxima are 5.16, 5.41, 5.67 and 5.85: its rings reach 5.81 routinely where the star does not cross threshold. The nearest transition of the eight masked species lies 1326 km s^{-1} away; since that mask is an automated contamination cut, not a complete line catalogue, and a chemically rich or circumstellar environment can present many further ALMA-band transitions, we re-queried a wider catalogue, whose closest entry is $\text{H}^{13}\text{CN}(J=4 \rightarrow 3)$ at -195 km s^{-1} , still far outside any circumstellar tolerance. The defensible reading is “no plausible known transition identified”, a statement about the catalogues consulted, not a proof that none exists; and with a single execution block at this frequency no recurrence test is available. Atmospheric or calibration spectral structure near 345.115 GHz is controlled the same way everything here is: it enters star and ring positions alike within the one execution block, so it widens the ensemble rather than mimicking a stellar excess, and isolating any telluric contribution needs the calibration products beneath the frozen release. None of these facts is decisive alone. Figure 7 shows the crossing within its window’s full 512-position control distribution: T_* sits at the upper edge of an unremarkable ensemble, not beyond it – one draw of 513 against the release’s expected ring maxima (median 4.56, 90th percentile 5.85; a ring maximum at or above 5.68 occurs in 17 per cent of windows, at or above 5.81 in 13 per cent). We record it as an unclassified stage-1 flag – a statistical or astrophysical feature, not candidate evidence – with explicit promote/retire criteria: *promote* only on recurrence at an independent epoch – a targeted ALMA re-observation at 345.1152 GHz, specified in Table 8 but not yet scheduled, tests it directly – or on an enlarged control ensemble resolving the rank probability below the survey-wide Bonferroni threshold ($p < 1.2 \times 10^{-4}$, requiring of order 10^4 control positions against the present 1/513 floor); *retire* on a clean second epoch or on identification with catalogued emission. One further check is open rather than performed: a Gaia/2MASS/WISE cross-match for an unresolved background source. Its catalogue half could list sources in the synthesised beam, but without the visibility-domain localisation it could not say which, if any, the excess belongs to – a null match would not retire the flag, and a populated match could not promote it – so the check is scheduled with that localisation rather than run as a catalogue query that cannot move the disposition. Until either, it stays released and reported for what it is: unclassified, and statistically non-significant at the survey level. The enlargement and the visibility-level tests of Table 8 are not deferred for want of interest: each needs re-extraction beneath the frozen products, which they do not carry, while the re-observation at 345.1152 GHz needs no reprocessing and is the faster arbiter.

LINE EXCLUSION: DEFINITION, COST, AND LIMITS

The searches used an automated line check matching crossings to catalogued rest frequencies within one spectral channel at zero drift. The β Pictoris windows showed that inadequate: both fell outside its one-channel tolerance (by 10.6 and 21.9 MHz) while lying, after the frame chain, within 3.6 km s^{-1} of the stellar systemic of a resolved, mapped disc – exactly the regime a channel-width tolerance is blind to. Every crossing was therefore reclassified before publication with a velocity-space mask ($\pm 50 \text{ km s}^{-1}$ in the stellar frame around each catalogued transition), the pipeline’s line-exclusion definition from this release on. Its kinematic justification, empirical conservatism, that reclassification and the counter-argument that a communicator might deliberately choose these natural reference frequencies are in Appendix I. The mask’s application to this release is *retrospective* – a post-hoc classification criterion, not a detection threshold:

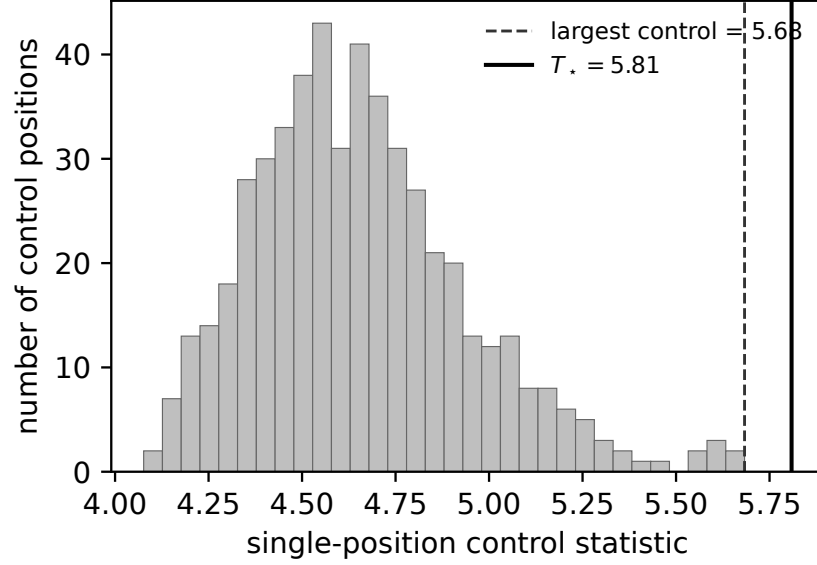


FIG. 7.— The CP-72 2713 Band 7 crossing inside its own control ensemble: histogram of the 512 single-position control statistics of that window, with $T_* = 5.81$ (solid) and the largest control value, 5.68 (dashed), marked.

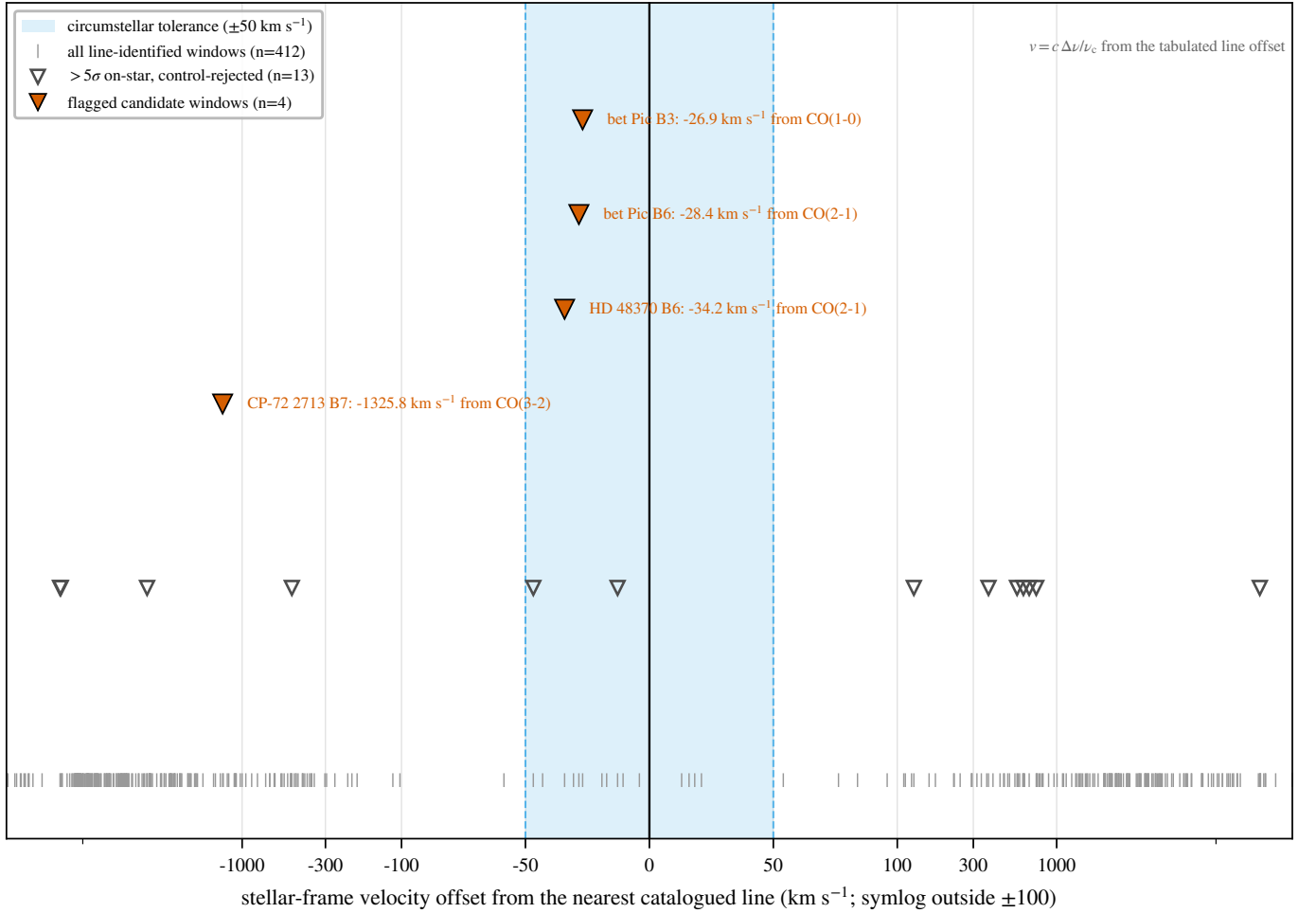


FIG. 8.— Stellar-frame velocity offsets of the flagged windows, through the frame chain (topocentric $\rightarrow$ barycentric $\rightarrow$ stellar; Table 9). Shaded band: the $\pm 50 \text{ km s}^{-1}$ molecular-line exclusion mask. *Top*: the two β Pictoris windows fall well inside the mask as one kinematic component, resolved in the inset. *Bottom*: the flagged window towards CP-72 2713 and the AU Mic crossing the symmetric statistic does not flag (§5.4) lie far outside any circumstellar tolerance, dispositioned by their spatial behaviour, not the mask.

TABLE 9

LINE-ATTRIBUTION AUDIT OF THE THREE LINE-ATTRIBUTED WINDOWS (FIRST THREE ROWS) AND THE COARSE β PICTORIS BAND 6 WINDOW CARRYING NO STAGE-1 FLAG YET SHOWING THE SAME CO COMPONENT, THROUGH THE FRAME CHAIN OF §4.2. TOPOCENTRIC OFFSETS ARE OBSERVED MINUS CATALOGUED REST FREQUENCY (CO $J=1\rightarrow0$ 115.271202 GHz, CO $J=2\rightarrow1$ 230.538000 GHz, PICKETT ET AL. 1998); THE BARYCENTRIC CORRECTION IS THE EARTH-EPHEMERIS TERM AT THE EXECUTION-BLOCK MID-TIME (BLOCK UIDS IN THE MACHINE-READABLE RELEASE); SYSTEMIC RADIAL VELOCITIES ARE CATALOGUED (SIMBAD; *Gaia* DR3 FOR β PICTORIS, GAIA COLLABORATION ET AL. 2022; DR2 FOR AU MIC, FOUQUÉ ET AL. 2018).

Flag	UTC start	f_{obs} (GHz)	Transition	Topo. offset (MHz; km s^{-1})	Bary. corr. (km s^{-1})	v_{sys} (km s^{-1})	Stellar-frame offset (km s^{-1})
β Pic B3	2022-03-02 23:14	115.260644	CO(1 $\rightarrow$ 0)	−10.56; −27.46	−7.90	+16.84	−2.72
β Pic B6	2019-03-12 00:14	230.516128	CO(2 $\rightarrow$ 1)	−21.87; −28.44	−8.15	+16.84	−3.45
AU Mic B6	2014-08-18 04:01	230.825230	CO(2 $\rightarrow$ 1)	+287.23; +373.51	−10.02	−4.71	+378.82
β Pic B6 [†]	2013-10-06 06:23	230.528134	CO(2 $\rightarrow$ 1)	−9.87; −12.83	+7.63	+16.84	−3.62

[†]Coarse Band 6 window without a stage-1 flag (Appendix H), shown because it independently shows the same CO component as that band’s flagged fine window.

TABLE 10

EXPOSURE OF THIS RELEASE ALONG ITS AXES, BY CHANNELISATION CLASS (CLASS A FINE, CLASS B COARSE). GROSS BANDWIDTH SUMS ALL 431 WINDOWS, OVERLAPS DOUBLE-COUNTED; UNION IS THE DISJOINT SKY-FREQUENCY COVERAGE; PER-STAR UNION SUMS EACH STAR’S OWN UNION OVER BANDS. $C(\nu)$ IS THE FREQUENCY-EXPOSURE CURVE OF THE TEXT, 2-GHz BINS.

Axis	Value
Windows (Class A / B)	431 (118 / 313)
On-source time per window	21–5274 s (median 2087 s)
Gross bandwidth	700.8 GHz (A 23%, B 77%)
Unique sky-frequency union	93.1 GHz (A: 29.4 GHz)
after line-exclusion mask	$\approx$ 88.2 GHz
Summed per-star union	371.9 GHz
On-source $\times$ bandwidth	1.4×10^6 s GHz (= 393 star-hour-GHz)
$C(\nu)$: covered-bin median / peak	2.4 / 18 star-hours

the half-width was adopted after the β Pictoris crossings had been seen, and it excludes exactly those two windows, changing no other disposition. A sensitivity test over half-widths: reclassifying all 20 crossings and recomputing masked coverage at ± 20 , ± 30 , ± 50 and $\pm 100 \text{ km s}^{-1}$ (nearest-catalogued-line approximation, one tube per window regardless of in-band position – an upper bound, exceeding the full-catalogue 3.2 per cent at ± 50) masks 2.1, 3.1, 5.2 and 10 per cent of gross bandwidth, leaves 19, 15, 13 and 13 of the 20 crossings and 4, 2, 1 and 1 of the 4 star-exceeds-ring windows unmasked. At every width the only unmasked unattributed flag is CP–72 2713 (at -1326 km s^{-1} , masked at no plausible width); the CO attributions rest on kinematics and published detections, not on the mask; no width admits a new unattributed flag. The half-width moves which CO crossings are masked, not any conclusion. We therefore pre-register the same $\pm 50 \text{ km s}^{-1}$ stellar-frame definition, unchanged, for all remaining planned datasets: the confirmatory sample tests a frozen rule, not a tuned one, and masked regions are reported throughout as *excluded search space*, never as searched-and-empty. Two standing policies accompany the pre-registration. First, the eight-species mask is the automated first-pass screen only: crossings outside it will in future releases be scored against a Splatalogue/CDMS/JPL query of all catalogued transitions in the searched interval, assigning each an astrophysical-line probability from source and environment rather than a binary species veto – the procedure CP–72 2713 would have needed had it fallen in a line-rich field. Second, the mask width and any other post-hoc-adjusted parameter will be applied and audited by a team member not involved in the original crossing identification.

The masking cost uses one definition: every JPL catalogued transition of the eight species (276), displaced by each system’s radial velocity, $\pm 50 \text{ km s}^{-1}$ stellar-frame half-width, intersected with each window and merged. Framed on the unique-frequency union: the release holds 93.1 GHz of archival spectral coverage, of which 88.2 GHz is included in the primary technosignature test and 4.96 GHz (5.3 per cent) is deliberately excluded because of molecular-line confusion (Figure 2) – a loss of possibly interesting parameter space accepted by design, not a contamination afterthought. On the window-summed gross bandwidth the same mask removes 22.20 GHz of 700.8 GHz (3.2 per cent; breakdown in Table 20; the three totals in circulation – 93.1 GHz unique union, 88.2 GHz searched, and the per-band unions of Table 19, dominated by Band 6 – are the same quantity before the mask, after it, and decomposed by band), the per-window usable fraction having median 0.960 (10th percentile 0.790). The full exposure accounting – gross and union bandwidth, the summed per-star union, on-source $\times$ bandwidth and the frequency-exposure curve $C(\nu) = \sum_i t_i(\nu)$ – is consolidated by class in Table 10: the archive’s tuning concentrations concentrate 30 per cent of all exposure in the five best-populated bins, which any frequency-prior calculation must weight accordingly (Fig. 2 shows where the coverage is, $C(\nu)$ how it is weighted). The mask governs dispositions, not the noise-derived thresholds, so the quoted EIRP_{5 σ} values are unchanged; no signal was found outside the excluded velocity regions in any window.

The mask is a design selection effect: a signal coincident with a molecular line is not thereby “false” but *excluded from the technosignature search domain*, the astrophysical prior there being overwhelming and this pipeline unable to separate the two. We place no constraint on transmitters in the excluded regions – 5.3 per cent of the unique-

TABLE 11

EVERY WINDOW FLAGGED BY THE ORIGINAL REGION-MAX STATISTIC, UNDER BOTH STATISTICS, SO THE REVISION CAN BE AUDITED ACROSS ALL OF THEM, NOT JUST THE AU MIC WINDOW. S_{reg} : MAXIMUM OF EQ. 2 OVER THE $n_{\text{src}} = 135$ -POSITION SOURCE REGION; ‘CTRL MAX’: THE LARGEST OF THE 512 SINGLE-POSITION CONTROLS IT WAS COMPARED WITH; T_* : THE SYMMETRIC SINGLE-POSITION STATISTIC AT THE PROPAGATED STELLAR POSITION; ‘RING MAX’: THAT WINDOW’S LARGEST CONTROL STATISTIC; ν : THE WINDOW CENTRE. FOR CP–72 2713 THE REGION MAXIMUM ALREADY LAY ON THE STAR, SO THE TWO STATISTICS COINCIDE EXACTLY.

Star	Band	ν (GHz)	region-max statistic		symmetric statistic		Symmetric outcome
			S_{reg}	ctrl max	T_*	ring max	
HD 48370	6	230.7305	27.95	26.83	27.10	26.83	flagged
β Pic	3	115.2605	20.31	7.27	14.65	7.27	flagged
β Pic	6	230.5164	12.78	9.12	11.68	9.12	flagged
AU Mic	6	230.5384	5.97	5.85	5.22	5.85	not flagged
ALMA J1537–3319	6	217.0183	5.83	5.63	5.09	5.63	not flagged
CP–72 2713	7	345.1152	5.81	5.68	5.81	5.68	flagged
HD 14055	7	345.1377	5.80	5.74	5.22	5.74	not flagged

TABLE 12

THE AU MIC CROSSING FREQUENCY (230.8252 GHz) UNDER THE THREE TREATMENTS THAT DECIDE IT: THE RELEASE’S FROZEN PRODUCTS (REGION-MAX STATISTIC), THE FULLY SYMMETRIC REPROCESSING OF THE SAME FIRST-EPOCH BLOCK (2014 AUGUST 18, OBSERVATORY CALIBRATION RECIPE VERBATIM) AND THE SECOND-EPOCH BLOCK OF THE SAME PROGRAMME (2014 JUNE 5, IDENTICAL SEARCH TREATMENT). THE PEAK DRIFT IS RECOMPUTED OVER A DIFFERENTLY WEIGHTED CUBE IN EACH COLUMN, SO THE ~ 10 –20 PER CENT SPREAD MEASURES THE DRIFT-LOCATION UNCERTAINTY OF A $\lesssim 2.5\sigma$ PER-CHANNEL FEATURE (APPENDIX D MEASURES EXACTLY THIS EFFECT), NOT PHYSICAL DRIFT EVOLUTION – AND A DRIFT-LOCATION ERROR OF THIS SIZE CANNOT CHANGE A DISPOSITION, WHICH RESTS ON AMPLITUDE AND RANK.

Statistic	Release (region-max)	Symmetric reprocess.	Second epoch (recurrence)
Star statistic	5.97	4.93	4.77
Control maximum (512)	5.85	7.70	6.78
Star rank p (add-one)	—	0.81	0.97
Single-position star statistic	5.22	—	—
Per-channel S/N (zero drift)	—	2.46	2.22
Peak drift (Hz s^{-1})	1418.5	1593	1281

frequency union – and a search for deliberately line-adjacent carriers would need a complementary strategy, pre-registered here: masked channels are retained, not discarded, and future releases will score them in a separate *line-coincident* technosignature catalogue, discriminating molecular from technological emission by the axes this survey already measures per window – spatial extent against the control ring, line ratios across transitions, recurrence across epochs, linewidth, and drift – rather than by molecular proximity alone. Nor is the exposure confined to the flagged windows: 131 of 431 windows (30 per cent) contain an in-band catalogued transition of the eight species, as do all four flagged windows, yet in none does a crossing fall within one native channel of a rest frequency – precisely the regime the velocity-space criterion covers.

5.4. Changing the detection statistic, and the AU Mic crossing

The initial region-max statistic maximised over an $n_{\text{src}} = 135$ -position region centred on the star against 512 *single-position* controls; under it a crossing towards AU Mic in Band 6 at 230.83 GHz was carried as a candidate. The defect belongs to the statistic – the asymmetry raises the star-exceeds-ring probability under a pure-noise null to 20.9 per cent, an algebraic argument holding for any window and derived before any crossing was re-examined (Appendix P) – so we replaced it with the symmetric single-position form of Eq. 2, applied identically to star and controls, re-ran the release and froze the criterion. The chronology is fixed and auditable: the AU Mic crossing was first computed, and carried, under the region-max statistic; the symmetric replacement was then derived algebraically, re-run across the whole release, and frozen; only afterwards was any flagged window – AU Mic’s included – inspected under the new criterion; the wall-clock record is the versioned audit trail of the frozen release (Appendix P; Data Availability). Table 11 audits the revision across every window the original statistic flagged: the symmetric criterion flags a strict *subset* (seven before, four now) – both β Pic windows and HD 48370 retain their flags under either statistic, while AU Mic, ALMA J1537–3319 and HD 14055 drop out, in all three because the stellar position never exceeded its own control ring – the one configuration a localised transmitter cannot produce. Because this release is also the sample on which the statistic was developed, we label its false-alarm calibration *retrospective* – developmental rather than confirmatory. The pre-registration binds only what follows: for the remaining 208–107 planned datasets the criterion is fixed before those data are searched, making them the confirmatory sample.

Under the symmetric criterion the AU Mic crossing is not flagged: 10 of its own ring’s 512 controls reach or exceed the stellar statistic, an add-one within-window rank of $p = 0.021$ – a different quantity from the release-wide rate at which any window’s ring maximum reaches 5.97 ($p = 0.081$, Appendix J) – and far from the $1/513$ floor either way. It also fails two pre-named tests, localisation and recurrence (Table 12).

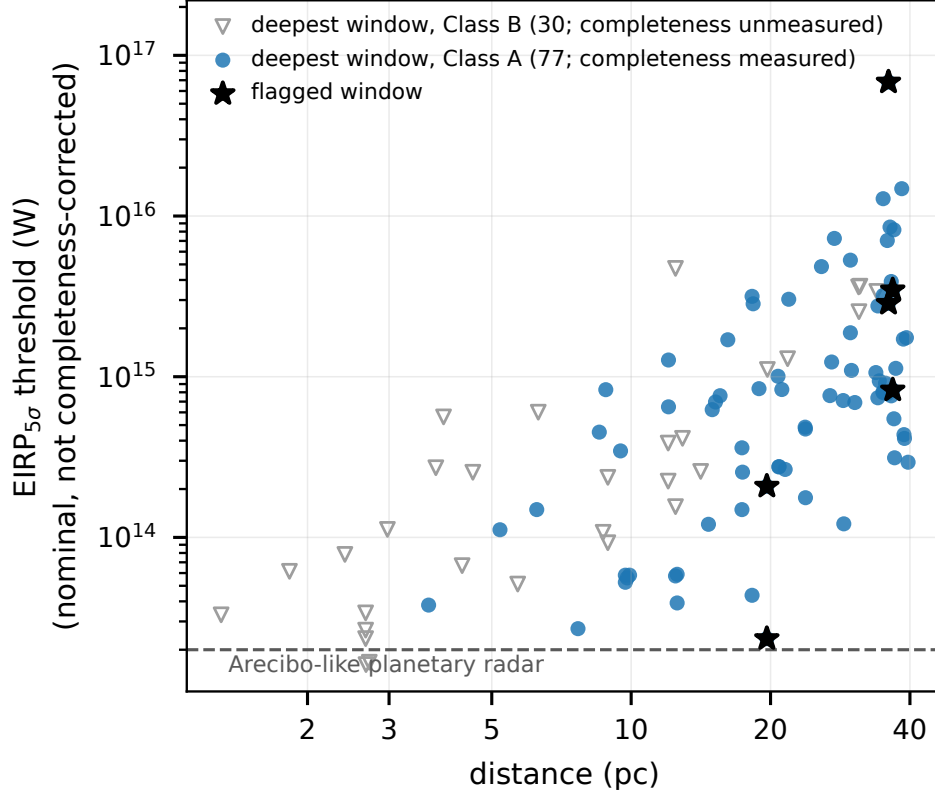


FIG. 9.— Nominal $\text{EIRP}_{5\sigma}$ trigger thresholds (boxed rule, §4) against distance, one deepest window per star and band (107 datasets, 88 stars). Symbols mark whether that window is Class A (drift-resolved), where the injection campaign measures end-to-end recovery of the drifting class, or Class B (unresolved excess), where detection completeness is measured post-baseline but the drift axis is indiscriminating (Table 18); survey-wide the split is 118 Class A against 313 Class B windows. Stars mark the four windows with a stage-1 statistical flag (Table 7). The dashed line is an Arecibo-like planetary radar, 2×10^{13} W, a scale for human transmitter power, not a claim that the survey reaches Earth-technology detectability (§4.3).

Localisation and recurrence are in Appendix J (Table 12): neither the reprocessed first epoch nor the one further public Band 6 block covering 230.83 GHz shows a localised excess, the second epoch a clean non-recurrence – rejecting a continuous or repeating source at this frequency and these epochs, though not intermittent transmission. Each star was observed in one to three execution blocks, searched independently, so the non-detection concerns the epochs observed, not permanent activity; §6 turns the epoch structure into an explicit epoch-activity constraint.

5.5. Ancillary results

Three ancillary results are reported in full in the appendices.

Integration-time audit (Appendix O). All 227 window-level on-source records were audited against the archive’s exposure accounting; two of the 58 star-execution-block groups over-count (AT Mic by 1.43, corrected to the archive’s 877 s; CD–38 10980 by 7 per cent), the rest are conservative, and no EIRP threshold depends on integration time, all deriving from measured per-channel rms: no disposition changes and the exposure-weighted coverage of §5.3 is at worst optimistic by under 1 per cent.

Continuum search (Appendix K). An exploratory by-product over the 57 target/bands processed far enough for a usable measurement: seventeen $\geq 5\sigma$ detections, forty non-detections reaching 0.036–3.6 mJy (median 0.52 mJy); every detection is astrophysically unsurprising, the auroral emitter LSR J1835+3259 meriting deeper follow-up. No technosignature disposition rests on it.

Serendipitous molecular-line catalogue (Appendix L). The line-exclusion step identifies every $\geq 5\sigma$ outlier channel as a by-product; across the subset scanned, 18 groups were found, six plausibly real, all reported openly.

6. DISCUSSION

6.1. What signals could ALMA archival data actually detect?

Three concrete cases, one per measured morphology, with real windows and the completeness entries of Table 18 (Fig. 10 maps every window).

(i) *A continuous, frequency-stationary carrier* in a coarse channel: the best case, not the worst. Toward the UV Ceti pair (2.7 pc, deepest window $\text{EIRP}_{5\sigma} = 1.6 \times 10^{13}$ W), a carrier of that power confined to its channel is recovered at essentially the ideal $\sqrt{N}$ gain (§4.5) – an unresolved, continuous carrier matching the *total* EIRP of an Arecibo-like planetary radar is detectable toward the nearest systems even at 15.625-MHz channel resolution (a power-scale comparison, not a simulated radar; §4.3).

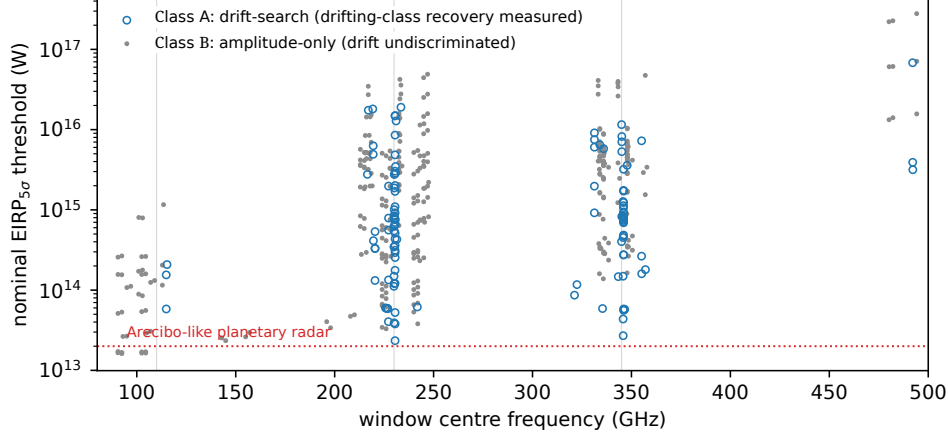


FIG. 10.— Two-dimensional sensitivity of this release: nominal EIRP trigger threshold (boxed rule, §4) against window centre frequency for all 431 searched windows, symbol by channelisation class – the axis along which completeness differs (Table 18). Class A windows (circles) have measured end-to-end recovery of the drifting class; Class B windows (grey) retain sensitivity to persistent unresolved excess power (dwell experiment, $\geq 4\sigma$ per integration; no survey-threshold completeness curve) and no drift discrimination. Grey lines mark the Bands 3, 6 and 7 concentrations; the dotted line is the Arecibo-like planetary-radar benchmark.

(ii) *A continuously drifting carrier* in a fine 488-kHz channel toward τ Ceti (3.65 pc, threshold 3.8×10^{13} W): recovery is 42 per cent at the threshold for $f_{\text{drift}} \geq 0.5$ and 75–83 per cent at $1.6\text{--}2\times$ threshold (Appendix D), so a drifting transmitter above $\sim 8 \times 10^{13}$ W – twice threshold – is recovered in 83 per cent of trials wherever its drift stays inside the searched ceiling.

(iii) *An intermittent carrier, on for a quarter of the track*, toward HD 10647 (17.4 pc, 1.5×10^{14} W): dwell recovery at $f_{\text{dwell}} = 0.25$ is 91 per cent (fine) and 100 per cent (coarse), so a duty-1/4 transmitter above $\sim 2 \times 10^{14}$ W is detected toward such mid-distance systems. Smaller dwell fractions degrade rapidly (Table 4); below $D \sim 0.1$ only the nearest windows carry information. The denominators, counting systems whose deepest relevant threshold lies below each case power: 1 of 82 at the case-(i) power (2 at twice it), 8 at the case-(ii) power and 24 at the case-(iii) power.

6.2. What does the null result mean?

This first stand-alone processed release is preliminary by construction: 107 of 208 planned target/bands (51 per cent) are complete, and that subset is sensitivity-ranked, not an unbiased draw; the full sample will reach larger distances and shallower faint-end limits. No credible technosignature has been found; the $\text{EIRP}_{5\sigma}$ values of Appendix H are the nominal powers at which a matching transmitter enters the detection pipeline (boxed rule, §4) – completeness-calibrated exclusion holds only where the measured recovery curve applies (Class A; the framework below). The flagged-window dispositions are established in §5.3 and not repeated here.

The searched space uses one definition throughout (§5.3, Figure 2). One exclusion in it deserves the converse emphasis: the line mask removes $\pm 50 \text{ km s}^{-1}$ stellar-frame tubes around eight common species, and those rest frequencies are exactly the ‘magic’ channels Cocconi & Morrison (1959) argued a communicator might choose as natural reference points. This survey is blind by construction to a transmitter parked there in the stellar frame – a false-positive control purchased with 3.2 per cent of gross bandwidth and one physically motivated class of signal, accepted knowingly and only partially narrowed by the pre-registered catalogue-probability screen of §5.3. Of the 41 planets known around the 18 planet-hosting targets only TRAPPIST-1 b reaches the drift ceiling, in its edge-on worst case at $\sim 13 \text{ Hz s}^{-1} \text{ GHz}^{-1}$; GJ 581 e, 61 Vir b, and TRAPPIST-1 c reach roughly half, the remaining 33 well inside the searched grid.⁵ Table 5 (§5) consolidates the axes on which a transmitter is or is not detectable; Table 19 decomposes the release by ALMA band. Coverage eligibility and search completeness differ. The archive of public usable ALMA data (§3) sets eligibility; every held execution block in scope was searched, so no EB-level selection lies behind the 102. Availability is measured directly, not inherited from release bookkeeping: a proper-motion-matched ALMA archive audit (2026 September 9; public rows, all bands, pointing within 30 arcsec of the propagated stellar position at the observation epoch) counts 107 unique public on-star EBs over the subset run so far, re-finding all but one searched EB – TRAPPIST-1’s Band 7 execution A002/Xcccc19/X5564, which `obscure` carries only through its calibrator-field row, an archive-representation artefact biasing that star’s availability low by one. Availability exceeds scope by the recurrence pool drawn on by the AU Mic second-epoch test (§5.3); the ledger `v323_star_coverage.json` gives per-star distance, availability, scope and searched frequency intervals, and the selection funnel itself is Table 6 with the selection function of Table 16.

Figure 13 shows the drift ceiling as an explicit selection function of planet period and host mass: the search covers the drift of a transmitter on any known planet in the sample bar the innermost TRAPPIST-1 planets, while a faster drift falls outside the searched grid, sacrificed by design (Appendix F recovers some non-linear morphologies). The host-mass dependence is weak ($a \propto M_{\star}^{1/3}$ at fixed period), so the ceiling is effectively a function of orbital period alone;

⁵ Star–planet relative accelerations $GM_{\star}/r^2$ from NASA Exoplanet Archive planetary-companion parameters (Mission Star/Planet table, accessed 2026 September), maximised over orbital phase and inclination.

a transmitter not co-located with a *known* planet is covered only if its drift falls in the searched grid (Scope box, §4).

The release sits against Wright et al. (2018)’s published haystack axis by axis. On frequency (10 MHz–115 GHz) only our Band 3 falls inside (20.6 of 115 GHz, 0.18 of the axis, and on it Hz-resolution surveys reach far more finely); the remaining 66.6 GHz (Bands 4–8, 125–495 GHz) lies wholly outside – the genuinely new territory. On transmitted bandwidth (0–20 MHz), a native-channel search probes a fraction set by channel width (window-mean 0.63, median 0.78). On sensitivity, our median threshold (7.3 mJy across a 15.625-MHz channel, $\phi \approx 1.1 \times 10^{-21} \text{ W m}^{-2}$) detects a 10^{13} W transmitter only within $\sim 0.9 \text{ pc}$. Against the sole prior ALMA-archival search (Mason et al. 2024) (28 bycatch targets at $\geq 1.01 \text{ kpc}$, deepest EIRP_{min} $6.91 \times 10^{17} \text{ W}$), this release’s 82 systems, 41 of them inside 20 pc, reach median per-target thresholds $\sim 1250\times$ deeper. These fractions are not multiplied into a combined coverage: the axes are strongly coupled (distance and sensitivity above all), so a product of marginals would misstate the overlap by orders of magnitude.

What the zero-detection result bounds is stated once here and carried everywhere below: fully conditioned, on each target’s own searched frequencies, epochs and morphology class. The occurrence formalism that will be applied once configuration-dependent completeness is established for the completed survey factorises the per-system detection probability into four terms,

$$C_i(P) = F_i R_i(P) D_i C_{\text{morph},i}, \quad (4)$$

F_i : *frequency occupancy* – the transmitter’s sky frequency lies in system i ’s searched intervals;

$R_i(P)$: *amplitude recovery* – a transmitter of EIRP P would have crossed system i ’s threshold had it emitted there;

D_i : *epoch activity* – it emitted during the analysed archival interval;

$C_{\text{morph},i}$: *morphology match* – the measured recovery of a carrier of the searched drift and temporal morphology (Appendix D).

The product treats the four factors as independent; for one strategy the assumption is strained (a continuously transmitting narrowband emitter trivially satisfies high D_i while carrying small F_i), for which the thresholded form, using R_i alone, is the honest one. The bound below sets $F_i = D_i = 1$ *by conditioning, not by measurement*. With that convention the probability of zero credible detections if a fraction f of systems hosts one is

$$P(N_{\text{det}} = 0 | f) = \prod_{i=1}^{N_{\text{sys}}} (1 - f C_i), \quad (5)$$

and the one-sided 95% upper limit f_{95} solves $P(N_{\text{det}} = 0 | f_{95}) = 0.05$, treating the 82 systems as independent. The formulation differs from the transmitter-fraction limits of Enriquez et al. (2017) and Margot et al. (2023): those surveys quote occurrence over samples with roughly uniform per-target frequency coverage, whereas this framework conditions on each target’s own archival frequency support.

Reading rule. *Among the ALMA-observed systems, fewer than $f_{95}(P)$ per cent host a transmitter of EIRP $\geq P$ emitting during the archival observations, at a sky frequency in the particular spectral intervals searched for that system, with drift and temporal morphology in the searched class.*

The numerical demonstrations – the f_{95} evaluation, its bracketing, duty-cycle variants and transfer-error analysis – are consolidated in Appendix Q, labelled *illustrative calculation only*: they state no survey result, appear among no headline quantities (Table 5), and should be cited, if at all, only with the $F_i=D_i=1$ conditioning and the single-configuration completeness transfer attached – never as a measured occurrence limit, and never as a statement about the 40 pc stellar population.

Because selection is on ALMA archival coverage alone, useful subsamples fall out without compromising the archive-complete framing, subject to the unresolved-pair caveat of §3. First, 18 of the 88 processed stars (41 known planets) are confirmed exoplanet hosts (Proxima Centauri, τ Ceti, ϵ Eridani, β Pictoris and the seven-planet TRAPPIST-1 system among them), extractable as a sub-survey from Appendix H; HN Lib and LHS 1140, each with a habitable-zone planet, now carry an ALMA technosignature non-detection beside their planet-search literature. Second, four searched stars are white dwarfs (Gentile Fusillo et al. 2021) (Sirius B, Van Maanen’s Star, Wolf 219, CD–38 10980), all continuum non-detections with nominal EIRP_{5 σ} trigger thresholds of 2.3×10^{13} – $8.4 \times 10^{14} \text{ W}$ – among the few mm/submm SETI constraints for white dwarfs (Huang et al. 2026), a channel motivated by post-main-sequence survival and the compact debris discs of metal-polluted ones.

6.3. ALMA as a technosignature instrument: strengths and limits

ALMA’s advantages for this programme are four: aperture and receiver temperature give mm-wave sensitivities no single dish reaches; the tuning range, 125–495 GHz here, is essentially unsearched territory; interferometric spatial discrimination – the control-ring statistic of §4.1 – tests a feature’s position against 512 alternatives, which single-dish and beamformed searches cannot; and the archive itself supplies 102 public execution blocks re-processable without new allocations. Each strength has its mirror. Native channelisation reaches 15.3 kHz at best, orders coarser than

the Hz resolution of dedicated narrowband surveys, so the search tests unresolved spectral excess, not fine structure; coverage is fragmented into 33 frequency islands weighted by other science; the molecular-line mask deliberately excludes 5.3 per cent of the union (§5.3); configurations are heterogeneous, so no completeness transfers between them (§6.2); and targeting is interest-driven, not uniform (§3). The largest unused advantage is the visibilities themselves: ALMA’s fundamental advantage for this programme is not only mm-wave sensitivity but the ability to test whether an apparent spectral feature carries the interferometric phase signature of a celestial point source at a known position. For each surviving candidate, construct competing likelihoods – H_* : a source at the propagated stellar position, $V_{ij} = A(t, \nu) e^{-2\pi i(u_{ij}l_* + v_{ij}m_*)}$; H_{sky} : a source elsewhere on the sky; H_{common} : correlated or common-mode interference; H_{noise} : thermal and calibration residual – and evaluate $\Lambda = \mathcal{L}(V|H_*)/\mathcal{L}(V|H_{\text{noise}}+H_{\text{common}})$. This uses information that single-dish and beamformed searches do not possess, is strictly more powerful than ranking a peak against a control ring, and repositions the control ring as a calibration layer rather than the primary authentication statistic. We regard this visibility-domain likelihood analysis as potentially the programme’s most important methodological contribution; the extracted-spectrum search of this release is its amplitude-only prologue.

6.4. A next-generation ALMA technosignature search

Three changes would enlarge what this experiment can say, ranked by how much signal space they buy – the natural extensions of §6.3’s ledger of strengths and limits.

Drift discrimination on coarse windows. Within the searched ceiling every in-grid drift on a 15.625 MHz window is sub-channel over a track, so 313 of 431 windows measure amplitude but cannot distinguish a drifting carrier from a stationary one (§4.5). Finer channelisation on re-observation, or a longer-baseline drift model spanning multiple execution blocks, would recover that axis.

Enlarge the control ensemble for interesting windows. The 1/513 rank floor (§5.3) cannot certify significance across a survey of thousands of windows. A two-stage scheme – 512 controls routinely, then 10^4 – 10^5 local null evaluations for any window that passes – would settle candidates statistically rather than hand each to astrophysical vetting.

Extend the injection campaign and the polarisation axis. The completeness curve should be measured per band, channel width, integration time, antenna count and primary-beam position rather than transferred by construction, and pushed past 10σ until it asymptotes, with EIRP₉₀ unlocated. That campaign is pre-registered here as the completed survey’s completeness design: a stratified grid over those configuration axes; injections at continuously randomised drift rates and channel phases rather than preferentially on grid points; injected morphologies extended to sinusoidal acceleration tracks following representative sample orbits, whose linear-chirp fitting loss the present campaign cannot measure; trial-level logging from the first run; and an independent re-implementation cross-check of the injection code, closing the open item of §4.6. The polarisation axis is as noted in §4: discrimination against weakly polarised astrophysical backgrounds, at no sensitivity cost.

A boundary on drift linearity. The search assumes drift linear over a track. Curvature matters when $\frac{1}{2}\ddot{\nu}T^2$ exceeds half a native channel, reached only by the shortest-period orbits: $P \lesssim 1.6$ d for a typical fine-channel window (230 GHz, 488 kHz, $T \approx 2000$ s), about an hour for a coarse one. TRAPPIST-1 b, at 1.51 d, sits just inside the boundary and is also the case exceeding the drift ceiling; elsewhere the constant-drift model is adequate, and the chirp-search extension (Appendix F) covers the remainder.

7. CONCLUSIONS

This first processed release of an ALMA-archival technosignature survey searched, within 40 pc, every star with public usable coverage: 107 of 208 planned target/bands, 88 stars in 82 independent systems, complete within the archive and audited for position and identity (§5.2), but covering $\sim 1\%$ of the catalogued stars there and inheriting the archive’s non-uniform targeting. The archive’s channelisation splits what was searched: 118 fine-resolution windows were searched for drift-resolved spectral carriers, 313 coarse windows are sensitive only to unresolved spectral excess (§4.1). Four windows raised a stage-1 statistical screening flag: three are consistent with, and very likely caused by, CO emission – two circumstellar components towards β Pictoris across two transitions and three epochs, one previously reported foreground cloud towards HD 48370 (Cataldi et al. 2023) – and one, towards CP–72 2713, is unclassified, marginal and non-significant at the survey level, held under pinned promote/retire criteria (§5.3). None survives vetting: this release reports 0 credible technosignatures. Among the searched stars, Barnard’s Star and Wolf 359 carry their first published mm/submm technosignature limits (§5.1); forty ancillary continuum measurements (median 5σ limit 0.52 mJy) are non-detections of unquantified statistical power.

Its EIRP_{5 σ} values are nominal trigger thresholds (boxed rule, §4): 1.6×10^{13} – 2.8×10^{17} W, deep enough, toward the nearest systems, for an unresolved carrier matching the total power of an Arecibo-like planetary radar; the occurrence demonstration stays illustrative (Appendix Q). Searched is not constrained: the *Scope of the constraints* box of §4 fixes the frequency, epoch, linewidth, drift and threshold box inside which the non-detection binds – everything outside it, from the masked velocity space to any transmitter-frequency prior spanning ALMA’s tuning range, is unbounded, and every EIRP statement concerns continuous or near-continuous emitters (item iii there).

This release is developmental by construction: the detection statistic and line mask were established on these data, so the false-alarm calibration is retrospective, and the remaining 208–107 planned datasets form the frozen, pre-registered confirmatory sample – the completed survey will test the frozen method on data it has never shaped and supersede this release row by row without invalidating it. First-priority growth (§6.4): drift discrimination on coarse windows, a visibility-domain likelihood ratio, control ensembles past the 1/513 rank floor, and a stratified injection campaign

over band, channel width, integration time and beam offset. By-products ship with the release: a serendipitous molecular-line catalogue, exoplanet-host and white-dwarf subsample limits, and a first technosignature-survey use of a closure-phase diagnostic – demonstrated, with no discriminating power at these flux densities.

ACKNOWLEDGEMENTS

This paper uses ALMA data from the public ALMA Science Archive. The 102 execution blocks searched (project codes and epochs in Appendix A) should be cited from that list in the standard JAO.ALMA#XXXX.X.NNNNN.S form. ALMA is a partnership of ESO (representing its member states), NSF (USA) and NINS (Japan), together with NRC (Canada), MOST and ASIAA (Taiwan), and KASI (Republic of Korea), in cooperation with the Republic of Chile. The Joint ALMA Observatory is operated by ESO, AUI/NRAO and NAOJ. This work has made use of data from the European Space Agency (ESA) mission *Gaia*, processed by the *Gaia* Data Processing and Analysis Consortium (DPAC). This research has made use of the SIMBAD database, operated at CDS, Strasbourg, France.

COMPETING INTERESTS

G.J.W. declares no competing interests. R.D. is an employee of VBRL Holdings Inc. (a private software company with no astronomy or SETI-related product line), which provided no funding for and took no role in this work. The authors declare no competing interests.

AUTHOR CONTRIBUTIONS (CREDIT)

Glenn J. White: Conceptualisation, Methodology, Supervision, Writing – review & editing.

Robin Dey: Software, Investigation, Formal analysis, Data curation, Visualisation, Writing – original draft.

DATA AVAILABILITY

The reduced data products underlying this paper, with the full target-selection and analysis pipeline, are at <https://github.com/Tilanthi/SETI>, live at submission. All numerical values here come from the *submitted-analysis snapshot*: the repository commit tagged at submission freezes pipeline code and input data as run, is retained unchanged, and reproduces every number, table and figure via a single `make paper` invocation of the regeneration scripts shipped with it (the short commit hash the tag resolves to is printed in the build log). The *public release* is built from that snapshot with the two pipeline defects of §5.2 corrected and the affected datasets reprocessed *before* freezing, so it carries one set of products, not a submitted/corrected pair. Those datasets are exactly those excluded from this paper’s results by §5.2 (the Giclas 9–38 pair and the two defective Band 6 windows); the Sirius B and ϵ Eri holography-beam recomputation committed in §4 joins the same freeze, its outcome bounded by the systematic quoted there. No value printed here is silently altered: a change-log names every affected product, table and row with its reason. The machine-readable per-window table carries archive-consistent execution times: where the printed integration-time audit differs (the CD–38 10980 row is optimistic by 7%, Appendix O), it gives the archive’s. Public-release per-window crossing lists are uncapped (the interim product capped them at 50 frequencies; Appendix G).

The complete release will also be deposited on Zenodo, with a DOI minted on final acceptance and inserted here; it comprises (i) per-window search products for all 431 windows (spectral extracts, per-drift-trial peak statistics, the machine-readable line-exclusion mask of §5.3), each row keyed by a machine-readable `system_id` – the independent stellar system that is the statistical unit of §6.2, six bound pairs sharing one identifier and one observational trial – and carrying, per window: archive UID; epoch-propagated target coordinates and field centre; primary-beam gain; integration times; channel frequencies and widths; rms; the drift grid; the target statistic with all 512 control statistics, from which the rank and *p*-value are recomputable; line-mask intervals; disposition; injection-completeness class; pipeline version and commit hash; (ii) control-ring calibration products at all 512 positions; (iii) closure-phase and population-anomaly outputs; (iv) the full injection-recovery campaign products (all 1200 trials of Appendix D, including the 24-tone pilot); (v) the frozen pipeline code and input data snapshot; (vi) the audit tables of Appendix O, machine-readable; (vii) the molecular-line catalogue of Appendix L; (viii) the proper-motion-matched execution-block availability census of §6 with its per-star ledger. Data products carry the Creative Commons Attribution 4.0 International licence (CC BY 4.0), pipeline code the MIT licence. Raw ALMA visibility data are public from the ALMA Science Archive at <https://almascience.org>.

SOFTWARE

The pipeline depends on CASA 6.7.6 (casatools/casatasks 6.7.6-14; calibration, imaging and the local reprocessing of §§5.3–5.4), Python 3 with astropy, numpy and scipy (spectral extraction and statistics), and the ALMA Science Archive and SIMBAD TAP interfaces; all are free and open-source, with exact versions recorded in the frozen release.

APPENDIX

A. EXECUTION-BLOCK METADATA

Execution-block metadata were recovered by block identifier after freezing and are not part of the frozen products: 57 of the 102 searched blocks have archive metadata recovered to date, the remainder being retrieved. The machine-readable release carries the full per-block listing – identifier, proposal code, epoch, MJD – and the per-window block-to-window linkage with antenna configurations, synthesised beam sizes and polarisation setup (Data Availability). The epochs span 2013 October 6 to 2025 June 11 (MJD 56571–60837) across 36 unique proposal codes; the AU Mic crossing

TABLE 13
SURVEY VOCABULARY.

Term	Meaning in this paper
Archive-complete sample	every star within 40 pc with public ALMA archival coverage (168 entries; §3) – complete within the archive, not a census of the stellar population
Bycatch	sample stars observed by ALMA for unrelated science goals
Target/band dataset	one star in one ALMA band; the unit of the 208-dataset plan (107 complete)
Spectral window	one correlator spw of one execution block, after deduplication; the unit of search and bookkeeping (431)
Execution block (EB)	one contiguous ALMA observing session (102 searched; Appendix A)
MOUS	member observation unit set: the ALMA data hierarchy level at which pipeline products are delivered (one calibration recipe per MOUS)
QA2	the ALMA observatory pipeline’s second-stage quality-assurance products: per-MOUS calibrated measurement sets with quality scoring, consumed by this survey’s extraction without further observatory calibration
obscore	the IVOA table protocol of the ALMA Science Archive metadata service from which coverage and EB metadata are queried
Native channel	the correlator channel width of the window, 15.3 kHz–31.25 MHz here
Threshold crossing	one channel×drift cell at the stellar position with $T \geq 5$ (state 1, Table 1)
Candidate	a stage-1 statistical flag surviving every veto (0 in this release; state 3, Table 1)
Control ring	512 off-source test positions per window, used to estimate the local false-alarm distribution empirically
Star-exceeds-ring	the stage-1 statistical flag itself: a window whose star statistic exceeds all 512 control statistics (5/431; state 2, Table 1)
SP / SR statistic	the single-position statistic, operative over all 431 windows, and its fully symmetric region-max variant, a validation check on seven reprocessed windows (§5.3)
Drift trial	one trial Doppler drift rate of the stack grid; 2–1944 per window
Nominal threshold	noise-derived 5σ detection threshold, <i>not</i> a calibrated completeness limit (boxed rule, §4)
EIRP _{5σ}	equivalent isotropic radiated power at the nominal threshold; $4\pi d^2 S_{\min} \Delta\nu_{\text{ch}}$
Molecular-line exclusion mask	the $\pm 50 \text{ km s}^{-1}$ stellar-frame region around catalogued transitions where crossings are excluded (§4.2)
Closure phase	interferometric phase triplet test discriminating point-source structure (Appendix E)
Stellar frame	topocentric $\rightarrow$ barycentric $\rightarrow$ systemic-RV frame chain (§4.2)
Funnel	the candidate-narrowing cascade from channels searched to vetted candidates (Table 6)

window of §5.3 comes from the 2014 August 18 execution block of project 2012.1.00198.S. One further public execution block of that project (uid://A002/X7d80c7/X15c6, executed 2014 June 5) also covers the candidate 230.83 GHz window but was not processed; it is the second epoch of the recurrence test of §5.3.

B. GLOSSARY OF SURVEY VOCABULARY

Quantitative definitions are in the sections cited.

C. FREQUENCY CONVENTIONS, LINEWIDTH READING, AND SMEARING

REFERENCE FRAMES

The line-exclusion mask’s chain – topocentric $\rightarrow$ barycentric (Earth ephemeris at the execution-block mid-time) $\rightarrow$ stellar rest frame (catalogued systemic radial velocity; JPL catalogue rest frequencies, Appendix L) – has as residual error the catalogue radial-velocity uncertainty alone (at most a few km s^{-1} within 40 pc); second-order relativistic terms at the 50 km s^{-1} scale are $\lesssim 5 \text{ kHz}$ at 350 GHz, below the finest native channel searched (15.3 kHz). The β Pictoris and AU Mic kinematic attributions (§5.3, Fig. 8) use the same chain, so their offsets are comparable to the exclusion tolerance. Enlarging the drift-rate grid needs no further frame machinery: the searched rates are topocentric and depend only on elapsed time and channel width, not sky position.

CHANNEL WIDTHS AND THEIR VERIFICATION

Channel widths come from the measurement sets’ own spectral window tables (CHAN_WIDTH). For the local reprocessing set we verified these are the *effective* resolution: the tables carry no EFFECTIVE_BANDWIDTH column (no online spectral averaging) and the full-resolution windows are named FULL_RES, so the 15.625-MHz coarse and 488.28125-kHz fine channelisations are the noise bandwidths the thresholds mean.

LINEWIDTH CONVENTION, AND THE RUNNING-MEDIAN TRANSFER FUNCTION

The detection statistic is the flux density in a single channel after drift correction, so the quoted EIRP_{5 σ} applies to any transmitter narrower than the native channel, flat in the assumed linewidth. Two sub-channel qualifications: flatness is exact for a tone centred in a channel (a boundary-straddling tone splits its power between two, at worst a factor of two; measured in Appendix D); and the channel-to-channel transition follows the correlator’s passband shape, not retained in the frozen products, making the flatness a channel-centred idealisation accurate at the tens-of-per-cent level. Analytically, a tone placed uniformly at random within a channel keeps a fraction $\max(f, 1-f)$ of its power in

TABLE 14

THE ONLY WINDOWS WITH NON-NEGLIGIBLE INTRA-INTEGRATION SMEARING AT THE MAXIMUM SEARCHED DRIFT RATE ($\eta_{\text{smeared}} < 0.99$; THE OTHER 431 MINUS THREE HAVE $\eta_{\text{smeared}} \geq 0.99$, NOMINAL AND EFFECTIVE THRESHOLDS AGREEING TO BETTER THAN 1%). Δ IS THE EXCURSION IN CHANNELS ACROSS ONE INTEGRATION, t_{int} THE PER-INTEGRATION ON-SOURCE TIME, AND $\text{EIRP}_{5\sigma, \text{eff}}$ THE SMEARED THRESHOLD $\text{EIRP}_{5\sigma}/\eta_{\text{smeared}}$.

Window	$\Delta\nu_{\text{ch}}$ (kHz)	t_{int} (s)	Δ (ch)	η_{smeared}	$\text{EIRP}_{5\sigma, \text{eff}}$ (10^{13} W)
β Pic B6	15.3	6.05	1.096	0.574	4.08
η Crv B8	61.0	6.05	0.585	0.865	365.8
η Crv B7	61.0	6.05	0.411	0.932	4.68

TABLE 15

SCOPE OF THE ANALYSES IN THIS RELEASE.

Analysis	Applied to	Validation status	In headline limits?
Primary spectral-carrier search	431 windows / 88 stars (82 systems)	Injection-calibrated for the searched class on one configuration (1200 trials; transfer unmeasured, Appendix D); false-alarm rate empirical (§5.3)	Yes
Closure-phase vetting	4 flagged windows	Simulation-validated; sensitivity-limited on sky (Appendix E)	Disposition only
Injection-recovery validation	6 configurations / 3 stars	This release; scope limits stated (Appendix D)	Completeness only
Chirp/periodicity screen	122 retained spectra	Demonstration-grade (Appendix F)	No
Frequency-occupancy check	366 overlapping star pairs	Executed (Appendix G)	Disposition only

the stronger channel, f uniform over the channel, so the amplitude cost is uniform between 0.5 and 1 (mean 0.75): individual thresholds carry a sub-channel-position systematic of order tens of per cent, and EIRP values are therefore quoted to two significant figures (a randomised through-pipeline robustness test remains pending; §6).

Transmitters wider than one channel. The baseline filter is a running median along frequency, whose transfer function is flat for features narrower than roughly half the median width W and falls to zero for structure broader than W : the survey’s effective sensitivity is therefore to channel-confined morphologies, and W (in channels, fixed per setup in the released code) is the linewidth ceiling. Within that ceiling the sensitivity to a top-hat transmitter spanning N_{ch} native channels follows from the statistic’s single-channel definition: the per-channel flux is the total divided by N_{ch} at unchanged per-channel noise, so to first order

$$\text{EIRP}_{5\sigma}(N_{\text{ch}}) \simeq N_{\text{ch}} \times \text{EIRP}_{5\sigma}(1), \quad N_{\text{ch}} \lesssim W/2, \quad (\text{C1})$$

the same rescaling the main text applies to signals broader than one channel. Engineered signals need not be intrinsically Hz-wide; this function is how the quoted thresholds answer for them. For the deepest window (UV Ceti Band 3), 1.6×10^{13} W at a 15.625-MHz channel corresponds to a minimum detectable spectral power of 1.0×10^6 W Hz $^{-1}$: channelised at 1 Hz from the start the same observation would reach 4×10^9 W, and the factor ~ 4000 between them is the unavoidable price of archival channelisation (§4.2).

Worked example: from flux to EIRP. Take UV Ceti (G 272-61B) Band 3. The searched window’s per-channel rms is 0.244 mJy, so $S_{\text{min}} = 5 \times 0.244 = 1.22$ mJy in one 15.625-MHz native channel. At 2.67 pc that is 1.22×10^{-29} W m $^{-2}$ Hz $^{-1}$, and $\text{EIRP}_{5\sigma} = 4\pi d^2 S_{\text{min}} \Delta\nu = 4\pi (2.67 \text{ pc})^2 \times 1.22 \times 10^{-29} \text{ W m}^{-2} \text{ Hz}^{-1} \times 15.625 \text{ MHz} = 1.6 \times 10^{13}$ W. A wider transmitter needs S_{min} per channel.

INTRA-INTEGRATION SMEARING

The frequency excursion at the maximum searched drift rate across one integration, $\dot{\nu}_{\text{max}} t_{\text{int}}$, is evaluated for every window: at most 1.10 channels across the 431 windows (median 0.001, 90th percentile 0.06). A linearly drifting signal smeared by Δ channels within an integration retains a fraction $\eta_{\text{smeared}} = \text{sinc}(\pi\Delta/2)$ of its peak-channel amplitude, so a worst-drift transmitter’s effective threshold is $\text{EIRP}_{5\sigma}/\eta_{\text{smeared}}$; the three affected windows of 431 are those of Table 14, the median window essentially unsmeared. A signal drifting midway between adjacent trial rates accumulates at most 0.03 channels of residual drift across the grid as realised, so no significant sensitivity is lost between trial rates.

D. INJECTION-RECOVERY SENSITIVITY VALIDATION

The quoted limits assume a 5σ threshold behaves as idealised Gaussian noise predicts. We test that by injecting synthetic narrowband signals of known amplitude into real calibrated visibilities via the extraction code’s own mechanism, as far upstream of the statistic as retained products allow, and measuring what the unmodified pipeline recovers. Three results elsewhere in the paper rest on this campaign and inherit its single-configuration limitation:

TABLE 16

SELECTION FUNCTION OF THE SEARCHED SAMPLE AGAINST A VOLUME-LIMITED REFERENCE CENSUS. REFERENCE CENSUS: GAIA DR3 SOURCES WITH $\varpi > 0$, $\sigma_\varpi/\varpi \leq 0.1$ AND $1/\varpi \leq 40$ pc (17566 OBJECTS; 8594, 49 PER CENT, CARRY A **TEFF_GSPPHOT** ESTIMATE). SAMPLE: THE 88 STARS WITH AT LEAST ONE SEARCHED WINDOW IN THIS RELEASE. DISTANCES FOR THE SAMPLE ARE THE VALUES USED BY THE SEARCH ITSELF AND ARE COMPLETE; EFFECTIVE TEMPERATURES ARE AVAILABLE FOR 52 OF THE 88 STARS (ALL 88 MATCHED TO THE 168-ENTRY ARCHIVE-COMPLETE SAMPLE, 58 BY EXACT NAME), SO THE TEMPERATURE PANEL IS NORMALISED WITHIN EACH COLUMN OVER CLASSIFIED OBJECTS ONLY AND T_{eff} IS USED AS A PROXY FOR SPECTRAL CLASS. “RATIO” IS THE SAMPLE COLUMN FRACTION DIVIDED BY THE CENSUS COLUMN FRACTION, SO 1 TRACKS THE REFERENCE POPULATION AND > 1 IS OVER-REPRESENTATION.

Property	Census	Searched	Ratio
Distance (all 88 searched stars; 17566 census objects)			
$d \leq 5$ pc	48 (0.3%)	12 (13.6%)	49.9
5–10 pc	240 (1.4%)	13 (14.8%)	10.8
10–20 pc	2069 (11.8%)	18 (20.5%)	1.7
20–30 pc	5391 (30.7%)	18 (20.5%)	0.7
30–40 pc	9818 (55.9%)	27 (30.7%)	0.5
T_{eff} class ^a (52 searched; 8594 census)			
A (≥ 7500 K)	24 (0.3%)	1 (1.9%)	6.9
F (6000–7500 K)	402 (4.7%)	9 (17.3%)	3.7
G (5300–6000 K)	860 (10.0%)	16 (30.8%)	3.1
K (3900–5300 K)	1400 (16.3%)	5 (9.6%)	0.6
M (< 3900 K)	5908 (68.7%)	21 (40.4%)	0.6
Other axes			
Known planet hosts ^b	20 (11.9%)	18 (20.5%)	1.7
Searched stars / systems	—	88 / 82	—

^aBins are temperature bins, not MK types: A ≥ 7500 , F 6000–7500, G 5300–6000, K 3900–5300, M < 3900 K. Gaia **teff_gspphot** is unavailable for about half the reference census, preferentially for the coolest and faintest objects, so the census M fraction is a lower limit and the earlier-type ratios are correspondingly upper limits. White dwarfs are not separable in this parameterisation and fall wherever their (unreliable) photometric temperature places them.

^bThe Gaia reference census carries no planet information; the census column here is instead the 168-entry archive-complete sample, of which 20 entries are known planet hosts. The 88 searched stars include 18 hosts of 41 known planets.

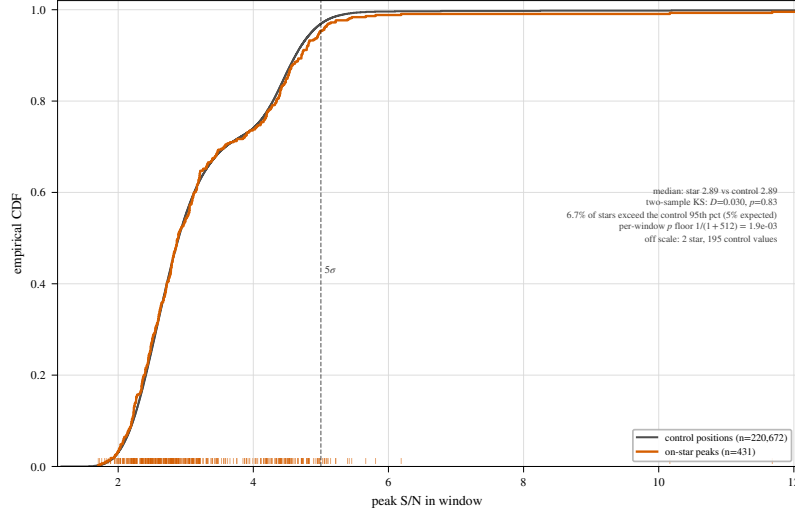


FIG. 11.— Amplitude-level view of the null calibration: the empirical CDF of per-window peak S/N at the 431 on-star positions against all 220 672 pooled control-position peaks. Medians coincide (2.89 vs 2.89), the two-sample Kolmogorov–Smirnov distance is $D = 0.030$ (in-panel annotation), and 6.7 per cent of on-star windows exceed the pooled control 95th percentile against 5 per cent expected. The two off-scale on-star peaks are the HD 48370 Band-6 and β Pictoris Band-3 windows of Table 7.

the 17 $\rightarrow$ 83 per cent drifting-class recovery entering the measured completeness of §6.2 and Table 18; the machinery and boundary-placement amplitude factors quoted in §4.6; and the coarse-window dwell-completeness reading of §4.5.

Six configurations span three bands and both resolution classes: three coarse Band 7 spws of AT Mic, one coarse spw each in Bands 3 and 6 of AU Mic, and AU Mic’s 488-kHz flagband window (49-trial drift grid). Into each goes a complex tone at the star position, at amplitudes 4–10 $\times$ the statistic’s own noise (the five levels of Fig. 12), drift fractions $f_{\text{drift}} = 0, 0.25, 0.5, 0.75, 1.0$ of the window’s searched ceiling, on- and half-grid rates, and channel-centred and boundary placements: 200 trials per configuration, 1200 in all, every marginal in the tables below pooling the grid-phase and placement draws of these 200, under a frozen recovery criterion ($T \geq 5$, peak channel within 3 of the injection, peak drift within one grid step; Fig. 12).

The coarse-window variant of this campaign recovered 0 of 500 injections at every amplitude from 4σ to 10σ under

TABLE 17
CONSOLIDATED FAMILY-WISE TRIALS ACCOUNTING FOR THIS RELEASE; THE RETIRED BOOKKEEPING LADDER AND THE
ASYMMETRIC-DERIVED BRACKETS ARE HISTORICAL (APPENDICES N AND P).

Trials factor	Unit	Count
Effectively independent channel cells ^b	release	$8.5 \times 10^4 - 1.1 \times 10^5$
Windows (star-level tests)	release	431
Windows containing an on-star crossing	release	20
Gaussian-expected chance crossings	release	11.4
Cross-target frequency pairs tested	release	946
Star-exceeds-ring rate $\hat{p}_{sbr}$ (symmetric)	window/release	5/431 (1.2%)
Execution blocks spanned	release	102
Expected flags under the symmetric null ($N/513$)	snapshot	0.84
Family-level Monte-Carlo expectation	release	0.20

^bFrom the measured lag-one channel autocorrelation of the drift-tracked statistic, $\rho_{lag1} = 0.989-0.991$.

TABLE 18

WHAT COMPLETENESS IS MEASURED, BY WINDOW CLASS AND CARRIER MORPHOLOGY. “DETECTION” IS AMPLITUDE RECOVERY AT THE NOMINAL TRIGGER; “DRIFT” IS WHETHER A RECOVERED DRIFT RATE IS INFORMATIVE ABOUT THE TRANSMITTER. N IS THE NUMBER OF SEARCHED WINDOWS OF THAT CLASS. EVERY MEASURED ENTRY IN THIS TABLE DERIVES FROM THE SINGLE CALIBRATION CONFIGURATION (ONE STAR, ONE BAND 6 FINE WINDOW); ITS TRANSFER TO THE OTHER 118 FINE WINDOWS IS AN ASSUMPTION WHOSE ERROR IS NOT BOUNDED (§6.2).

Class	Morphology	Detection	Drift	N
Class A (fine)	drifting	measured: $C_{5\sigma} = 0.42$	measured	118
Class A (fine)	near-static	post-baseline only ^a	n/a ^b	118
Class B (coarse, 15.625 MHz)	any	measured: 83–100% ^c	none ^d	313

^aDwell campaign (§4.5) recovers 91–100 per cent at $f_{\text{dwell}} \geq 0.25$ injecting after the baseline step; through-pipeline recovery is unmeasured (Appendix D). ^bSub-channel drift over the track by definition. ^cPost-baseline (Table 4) and end-to-end (§4.5). ^dEvery in-grid drift is sub-channel over the track, so the drift axis is degenerate and no drift-rate claim is made for coarse windows.

the frozen criterion – a recovery-criterion artefact, not a pipeline suppression; the full history and its withdrawal are audited in Appendix P. In brief: the criterion’s drift-matching clause cannot be satisfied on coarse windows, where every in-grid drift is sub-channel over the track, so the reported peak drift is arbitrary; scoring the same windows on detection alone recovers essentially everything, and an end-to-end injection through the released pipeline returns T_* of 20.5, 65.3 and 217.7 at 1, 3 and 10 times the per-integration noise. The coarse-window limits therefore bound carriers of any dwell fraction, with the completeness of §4.5, and what those windows genuinely lack is drift discrimination. The fine window measures the same boundary in drift space: carriers with $f_{\text{drift}} \leq 0.25$ dwell ≥ 40 per cent of the track per channel and fail the criterion, those with $f_{\text{drift}} \geq 0.5$ surviving as the searched class.

The second result is the completeness curve. On the fine window, drifting carriers ($f_{\text{drift}} \geq 0.5$, pooled over grid phase and sub-channel placement) are recovered 17, 42, 58, 75 and 83 per cent of the time at 4, 5, 6, 8 and 10σ , saturating by twice threshold power (Fig. 4). The machinery alone (tone injected after the baseline step) is amplitude-faithful to within $-9/+14$ per cent for channel-centred tones at every drift fraction; boundary-straddling placement costs a factor 0.6–0.8 in the statistic and recovers 0 per cent at 5σ against 83 per cent for centred ones (67 and 100 per cent at 10σ), crossing 50 per cent near $1.4\times$ threshold amplitude and recovering fully at $\sim 1.6\times$ threshold power. Survey completeness uses the placement-pooled curve, conservative for the boundary-straddling case. For that class $\text{EIRP}_{50} = 1.10 \text{EIRP}_{5\sigma}$ (the interpolated half-recovery point, 5.5σ), while $\text{EIRP}_{90} > 2.0 \text{EIRP}_{5\sigma}$ is a bound only, the curve reaching 83 per cent at the largest amplitude injected. Coarse windows carry no searched-class EIRP_{50} by construction; Table 18 collects the per-class reading. Band 8 windows share the coarse-channel construction, so the static-carrier result holds there by the same mechanism, but their drifting-class completeness is transferred rather than measured (§6). Not spanned: other target environments, primary-beam-offset positions, integration durations and edge-phased drift grids; campaigns on Band 8 data and on the β Pictoris windows are infeasible with the measurement sets retained locally.

Transfer to other windows. Survey completeness is measured on one configuration – AU Mic’s 488-kHz Band 6 window – and applied to all 118 fine windows; the transfer assumption and its unbounded error are stated in §6, and bounding it needs the multi-configuration campaign of §6.4.

E. CLOSURE-PHASE POINT-SOURCE VETTING

A threshold crossing faces one further automated test, available only to an interferometer: the closure phase $\arg(V_{ij}) + \arg(V_{jk}) - \arg(V_{ik})$ around a triangle vanishes for an unresolved point source anywhere on sky and is immune to per-station calibration errors – it asks only whether the signal is a coherent point source at all, so a coherent far-sidelobe interferer would pass. Consistency with zero is scored by a circular (unit-phaser, phases wrapping at $\pm\pi$) z-score of the mean drift-tracked closure phase, its null calibrated empirically on each observation’s own integrations, which also absorbs the shared-baseline covariance that makes one observation’s triangles non-independent. Validated on synthetic visibilities and driven end to end on continuum detections in four local reprocessing fields (including the

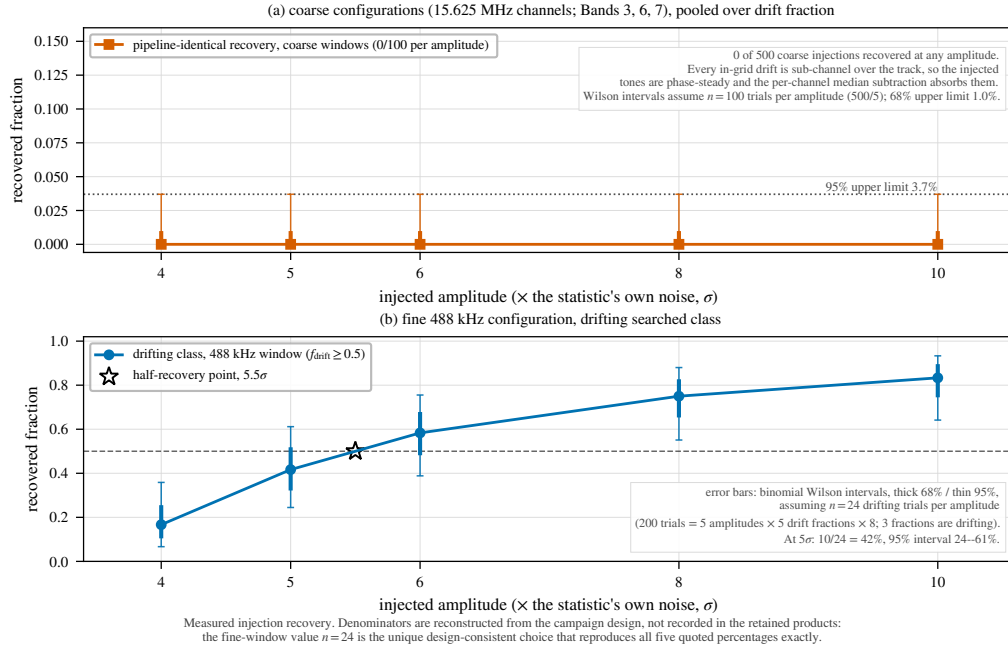


FIG. 12.— Measured injection recovery on real ALMA data (1200 trials, this appendix). Top: the five coarse-channel configurations, pooled over drift fraction and grid phase – the pipeline-identical variant (solid) recovers nothing by construction, leaving only the machinery’s detection fraction (dashed) meaningful. Bottom: the fine-channel configuration, where the drift dimension is real and pipeline-identical recovery $C(A, f_{\text{drift}})$ rises with amplitude for drifting carriers, static and near-static ones failing the frozen criterion. Boundary-straddling tones can fail the frozen criterion’s peak-drift clause at high drift while still detected – hence the non-monotonic drift dependence near the ceiling.

resolved AT Mic Band 7 binary as a resolving control and the AU Mic Band 6 crossing frequency), the capability is banked but not yet useful: per-baseline S/N is 0.5–0.7 at the median and never exceeds 5.1 (0.02 per cent of triangles have all three baselines above S/N = 5), so every field’s closure phases are indistinguishable from noise, including the binary – a flagged window’s formal point-source consistency carries no evidential weight at these flux densities (§4.4). The test triggers only on threshold crossing, at negligible cost, and becomes a discriminator only for deeper releases.

F. CHIRPED-DRIFT AND PERIODICITY SCREEN: SUMMARY AND FULL SPECIFICATION

Stacking at constant linear drift rate makes the primary search blind by design to a transmitter whose drift rate changes during the observation, and to pulsed or periodically-modulated carriers. We implemented a chirped (quadratic-drift) extension of the same stacking on a coarser grid of 21 linear-drift $\times$ 5 chirp-rate trials, and a per-channel periodicity search – a per-position FFT along each channel’s time axis, its peak power against a robust per-channel noise estimate. Both vet the star’s dynamic spectrum against 8 retained control positions, a strict subset of that window’s 512-position ring, not an independent ensemble (§4). On the 122 retained spectra this is a demonstration rather than survey-grade, establishing only that the extension runs and behaves correctly under the null: the flag rates (10.7% chirp, 13.1% periodicity, against the $\sim 11\%$ chance rate) are as expected, no instance is treated as a candidate, and no limit derives from it. Extending both to the full 512-position ring – the spectra being retained – would promote the screen to survey grade.

G. CROSS-TARGET FREQUENCY-OCCUPANCY CHECK: SUMMARY, KERNEL, COORDINATES, AND SCOPE

With no ON/OFF cadence against radio-frequency interference, this single-pointing archival survey substitutes the many unrelated stars sharing similar correlator setups: a channel-confined feature recurring at the same observed *sky frequency* towards different targets is almost certainly RFI or instrumental. Over every target/spw result the check pairs both the best-fit-channel frequency and the formal $\geq 5\sigma$ crossings, matching ν_1, ν_2 towards different targets when $|\nu_1 - \nu_2| < \max(k \Delta\nu_1, k \Delta\nu_2)$ with $k = 3$ (~ 46 kHz; k between ~ 0.2 and ~ 330 in native channels spans the original 5 MHz tolerance). Across the 431 windows and 107 tabulated crossings, no pair falls within the kernel, or within 0.5 MHz; the kernel is narrower than the β Pictoris line offsets of 10.6 and 21.9 MHz, so those windows were dispositioned by line-catalogue inspection instead.

The three frequency coordinates. A crossing’s *sky frequency*, reconstructed from archive metadata in the observation’s recorded frame; its topocentric frequency; and its *correlator channel index*. We probe the first, so a frame-convention inconsistency, or an artefact fixed in the second, would defeat it; correlator-frame artefacts recur at fixed intermediate frequency or channel number, not fixed sky frequency. Matching on the third coordinate (same channel, identical correlator setup) also returns zero pairs; a full intermediate-frequency reconstruction across tunings is not recoverable from the retained metadata.

Exposure and caveats. The null’s power is the count of target pairs sharing frequency coverage: on the ≤ 20 pc

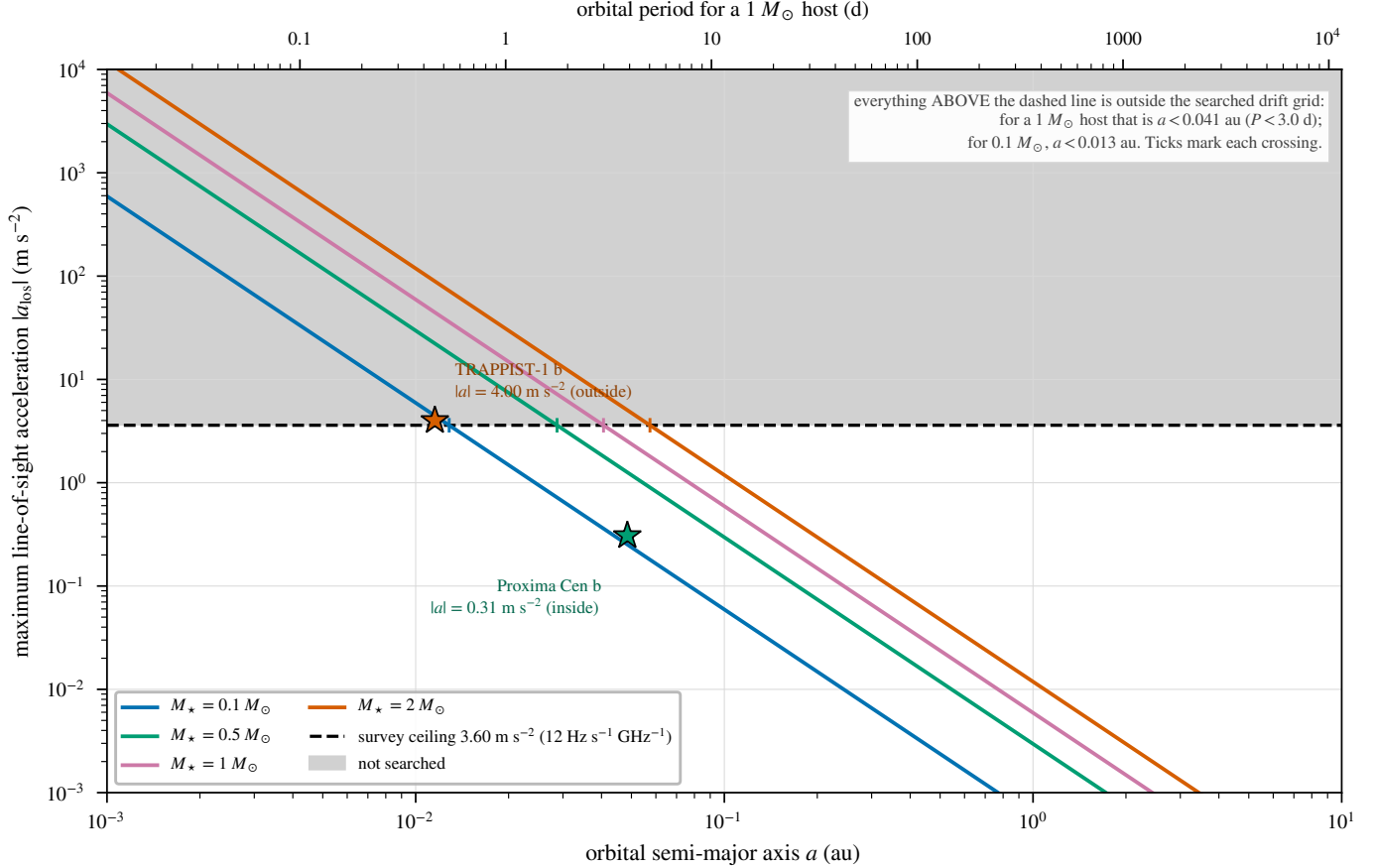


FIG. 13.— The drift-rate ceiling as a physical selection function. Maximum star–planet relative acceleration, edge-on and maximised over orbital phase, as the carrier drift rate it induces (Hz s^{-1} per GHz of carrier frequency; equivalently line-of-sight acceleration, $\dot{\nu}/\nu = \dot{v}/c$), against orbital period, for host-star masses $0.09\text{--}1.8 M_{\odot}$ (lines) and the known planets of the planet-hosting targets (points). Shaded band: the survey’s drift-search ceiling; above it a transmitter lies outside the searched grid. Only TRAPPIST-1 b reaches it (§6).

subset run so far, 366 of 946 star pairs (39%) overlap by at least 0.01 GHz (deepest 14.25 GHz, γ Lep– γ Vir), and 109 pairs involve one of the five stars carrying formal crossings, so every crossing frequency was tested against genuinely overlapping independent targets. The UV Ceti and AT Mic shared execution blocks (§3) are excluded, their intra-pair “coincidences” being identities by construction. The interim product’s per-window crossing lists are capped at 50 frequencies, so the true count is 171 crossings against 107 tabulated – the difference is one window (β Pic Band 6 ultra-fine, 114 formal crossings, the 50 strongest tabulated) whose untabulated 64 could not be tested: the null is demonstrated for the tabulated set, unresolved for the rest, a limitation of the retained product that the public release removes by shipping uncapped crossing lists (Data Availability).

H. PER-TARGET RESULTS

At this scope a typeset table would run to many pages, so these results – data rather than argument – ship with the manuscript source as `per_target_results_v3.32.csv`, summarised here. One row per extracted spectral window (462 at the snapshot epoch, each with its quality-control status) carries star name and distance, ALMA band and execution block, searched frequency range and native channel width, nominal $\text{EIRP}_{5\sigma}$ threshold, line-excluded continuum rms, stellar and control-ring peak S/N, the count of controls at or above the star, the empirical p -value of §5.3, drift ceiling, on-source time, and nearest catalogued transition with velocity offset.

$\text{EIRP}_{5\sigma}$, the nominal trigger threshold, is the instrumental power at a 5σ spectral amplitude before morphology-dependent recovery is folded in, $\text{EIRP}_{5\sigma} = 4\pi d^2 (5\sigma_{\text{comb}}) \Delta\nu$, on the same threshold and primary-beam correction as the continuum column, so the row’s two limits are comparable. The per-channel flux-density threshold follows as $S_{\text{min}} = \text{EIRP}_{5\sigma} / (4\pi d^2 \Delta\nu_{\text{ch}})$, the carrier lane’s per-channel noise and *not* the tabulated rms, that target’s line-excluded full-bandwidth continuum noise. Bands 6 and 7 supply 83 per cent of the windows, an archive imprint rather than a design choice; Band 8 windows are an order of magnitude less sensitive in EIRP at fixed distance and Bands 4 and 5 have one star each, so neither has weight in the aggregate limits.

I. THE VELOCITY-SPACE LINE MASK: JUSTIFICATION AND RETROSPECTIVE APPLICATION

The one-channel line check failed the β Pictoris windows (§5.3): a native channel spans a fraction of a km s^{-1} , while circumstellar kinematics shift real emission by many channels. The replacement mask takes its tolerance from the kinematic budget of circumstellar gas – Keplerian motion where these molecules survive ($\gtrsim 10$ au) $\lesssim 13 \text{ km s}^{-1}$; cold-gas

TABLE 19

PER-BAND SURVEY CHARACTERISATION OF THE 431 SEARCHED WINDOWS. “Sys.” COUNTS UNIQUE STELLAR SYSTEMS SEARCHED IN THE BAND, SO A SYSTEM OBSERVED IN TWO BANDS IS COUNTED IN BOTH ROWS (82 SYSTEMS, 88 STARS IN TOTAL); “EBs” THE EXECUTION BLOCKS CONTRIBUTING, ATTRIBUTED BY THE PROVENANCE OF THE DE-DUPLICATED WINDOWS; “Win.” THE SEARCHED WINDOWS; $\Delta\nu_{\text{ch}}$ THE NATIVE CHANNEL WIDTH IN MHz (MEDIAN, WITH THE RANGE IN BRACKETS); “Union” THE UNIQUE SKY FREQUENCY SEARCHED IN THE BAND, AFTER MERGING OVERLAPPING WINDOWS; $\text{EIRP}_{5\sigma}$ THE MEDIAN NOMINAL THRESHOLD ACROSS THE BAND’S WINDOWS; $\dot{\nu}_{\text{ceil}}$ THE RANGE OF DRIFT-RATE CEILINGS IN Hz s^{-1} PER GHz OF CARRIER FREQUENCY. THE “All” ROW IS COMPUTED OVER THE WHOLE SAMPLE, NOT SUMMED DOWN THE COLUMNS: EBs AND SYSTEMS RECUR ACROSS BANDS AND THE FREQUENCY UNIONS ARE DISJOINT BY CONSTRUCTION.

Band	Sys.	EBs	Win.	$\Delta\nu_{\text{ch}}$ (MHz)	Union (GHz)	Med. $\text{EIRP}_{5\sigma}$ (W)	$\dot{\nu}_{\text{ceil}}$
3	8	9	40	15.63 (0.244–15.63)	20.6	1.1×10^{14}	12.0–13.3
4	1	1	4	15.63	6.9	2.6×10^{13}	12.0
5	1	1	4	15.63	6.9	4.4×10^{13}	12.0
6	54	54	216	15.63 (0.015–31.25)	29.4	1.3×10^{15}	12.0–13.3
7	34	34	155	15.63 (0.061–15.63)	22.6	2.9×10^{15}	12.0–13.3
8	3	3	12	15.63 (0.061–15.63)	6.6	6.1×10^{16}	12.0
All	82	102	431	15.63 (0.015–31.25)	93.1	1.4×10^{15}	12.0–13.3

TABLE 20

MASKED BANDWIDTH (GHz) BY SPECIES AND ALMA BAND ON THE OPERATIVE MASK. COLUMNS B3–B8 AND ‘Total’ ARE WINDOW-SUMMED (DENOMINATOR: THE GROSS SEARCHED BANDWIDTH); ‘Union’ MERGES THE SAME INTERVALS ACROSS WINDOWS AND STARS (DENOMINATOR: THE 93.1 GHz UNIQUE-FREQUENCY UNION). SPECIES ROWS OVERLAP WHERE TWO MASKS SHARE A CHANNEL, SO THEY EXCEED THE ALL-SPECIES TOTALS. PER-STAR INTERVALS ARE IN THE RELEASED VETO-MASK EXPORT.

Species	B3	B4	B5	B6	B7	B8	Total	Union
CO	0.12	0.00	0.00	1.51	0.86	0.00	2.49	0.33
^{13}CO	0.35	0.05	0.07	1.71	0.51	0.00	2.69	0.74
C^{18}O	0.04	0.00	0.00	0.00	0.00	0.00	0.04	0.04
HCN	0.00	0.00	0.00	0.00	0.12	0.00	0.12	0.12
HCO^+	0.10	0.00	0.07	0.00	0.36	0.00	0.53	0.50
CS	0.00	0.00	0.07	0.57	0.11	0.00	0.75	0.29
CN	0.87	0.00	0.00	11.04	2.59	0.00	14.50	2.69
SiO	0.00	0.00	0.00	0.27	0.81	0.00	1.08	0.26
All species (merged)	1.48	0.05	0.21	15.10	5.36	0.00	22.20	4.96

linewidths $\lesssim 1 \text{ km s}^{-1}$; the eight masked species cold dense-gas tracers, no bulk-wind term for the dwarfs dominating the sample; the uncorrected barycentric term bounded by $\pm 30 \text{ km s}^{-1}$ – and is empirically conservative: the largest attributed circumstellar component (β Pic, -2.7 to -3.6 km s^{-1}) sits at 7 per cent of it, and *zero* crossing frequencies lie $13\text{--}50 \text{ km s}^{-1}$ from their nearest transition, so a $\pm 13 \text{ km s}^{-1}$ Keplerian-only mask would have suppressed and released the same windows: the wider tolerance costs no yield and buys robustness against radial-velocity catalogue error.

Two of the six remaining crossing windows fall inside the mask (the unflagged β Pic coarse window at -3.6 km s^{-1} ; HD 285968 at $+8.7 \text{ km s}^{-1}$, already rejected by its control ring, maximum 10.9 against a star peak of 6.5); the rest lie far outside any circumstellar tolerance (AU Mic $+379$, LHS 1140 $+115 \text{ km s}^{-1}$). Of the 20 windows with an on-star crossing, 4 also beat their control ring, three accounted for by the mask. The other 16 are unremarkable the way ordinary noise crossings are: in every one of them the window’s own control ring reaches or exceeds the stellar value (ring-maximum median 5.84 against a crossing median 5.18), the expected Gaussian-tailed population of moderate $5\text{--}5.5\sigma$ crossings with comparably strong peaks among 512 controls. But the transitions that make astrophysical false positives cheap are also natural reference frequencies for a deliberate communicator (Cocconi & Morrison 1959), so the released products keep a known-line-coincident list – every on-source 5σ crossing inside the masked intervals, with its offset from each nearby transition – for a future search with independent discrimination.

J. AU MIC: THE THREE TESTS IN FULL

Localisation. Reprocessing the first-epoch block from the raw science data model with the observatory pipeline’s own calibration recipe (310 integrations against the release pipeline’s 249, on a finer drift grid) and running the symmetric star-versus-control statistic of §5.3 on the crossing window, the star does not beat its control ring. The frozen products agree only asymmetrically: the source-region peak exceeds the 512-control maximum by 0.12, below that statistic’s window-to-window scatter, and the crossing fails the single-position variant too.

Recurrence. The one further public Band 6 block covering 230.83 GHz, the same programme executed 2014 June 5, was searched identically: no channel within ± 10 of the first-epoch flagged channel exceeds 3.1, and nothing is localised to the stellar position. Its band rms (1.48 mJy per 488-kHz channel) matches the first epoch’s 1.9 mJy, so a comparable excess would have been recovered, and the frequency recurs at no other target. Intermittent or precisely-scheduled transmissions outside the sampled windows stay unconstrained, a caveat quantified by the dwell-fraction and epoch-activity analysis of §6.

Statistical scale. The on-source peak (5.97) sits at the 92.1th percentile of the control-maxima distribution over all 431 windows (median 4.56, 90th percentile 5.85; Fig. 14): 34 (7.9 per cent) match or exceed it, add-one empirical

$p = 0.081$, before any multiple-testing correction – a release-wide window-level rate, distinct from the within-window rank quoted in §5.4. It matches no catalogued molecular transition (287 MHz from CO $J=2\rightarrow1$); closure-phase vetting finds it point-source consistent, one-sided evidence only (Appendix E).

Two astrophysical contexts are recorded for future re-examination; neither grounds the rejection. AU Mic is an exceptionally active flare star, but the block’s own quiescent continuum (0.15 mJy) rules the later-epoch GHz-wide flares of MacGregor et al. (2020) (16.8 and 6.1 mJy) out as origin: an 11 mJy channel-confined excess would sit a factor ~ 70 above a continuum that shows nothing, and the same-tuning second epoch shows no burst. The controls share the primary-beam footprint, so smooth structure such as the debris belt raises star and controls alike and cannot alone produce a localised excess.

K. THE ANCILLARY CONTINUUM LANE: SEARCH, DETECTIONS AND POPULATION SCREEN

The continuum lane is an exploratory by-product, not a co-equal search: 57 measurements (17 detections, 40 limits; comparison bins of 4–7 stars), a millimetre excess being at best an indirect, model-dependent observable of waste-heat technology. An “upper limit” is $5\times$ the primary-beam-corrected map rms at the star’s position, the per-target rms of Appendix H rescaling it to any confidence level; the deepest non-detection is towards TRAPPIST-1 in Band 3. Under the combined error model sixteen of the seventeen detections stay above 5σ , the exception (η Corvi Band 7, 4.7σ combined) being marginal. Bright far-from-target peaks (ϵ Eri Band 6 at 10.2σ , 18 arcsec off the phase centre; GJ 674 and CD–23 14742 at 10.7 and 9.9σ) fall outside the 3-arcsec tolerance and are recorded as non-detections, though the retained products cannot say whether the ϵ Eri peak is real emission or a primary-beam artefact (§4).

Every detection is astrophysically unsurprising: the unresolved UV Ceti binary; six young active systems hosting discs or chromospheres (τ Cet, β Pic, AU Mic, AT Mic); six stars at photospheric flux (γ Lep, γ Vir, χ^1 Ori, η Cru); and two late-M dwarfs of documented magnetic activity, LP 349–25 and the auroral emitter LSR J1835+3259 (Hallinan et al. 2007), which merits deeper follow-up. Closure phase cannot discriminate structure for any of the seventeen at these flux densities (Appendix E).

The anomaly screen. Each star’s flux is tested against its own photospheric expectation (Planck function with Gaia DR3 parameters, Pecaut & Mamajek fallback where GSP-Phot is unavailable – close to half the sample; rerunning with fallback parameters imposed on every star leaves the sole flag and the bin scales essentially unchanged) and against same- T_{eff} stars in coarse bins of at least five, through a robust median/MAD statistic with the bin width a fixed pipeline constant. The error model, evaluated before any flag, is $\sigma_{\text{tot}}^2 = \sigma_{\text{map}}^2 + (f_{\text{cal}} S_{\nu})^2 + \sigma_{\text{model}}^2$: f_{cal} 5 per cent in Bands 3–6 and 10 per cent in Bands 7–8; σ_{model} propagated from Gaia temperature and radius errors and the fallback relation’s scatter, dominating for fallback stars as the calibration term does for the brightest detections. The per-target file’s σ_{tot} column carries map and calibration terms only. At the five-star minimum the bin median rests on two or three values, so the screen flags for inspection, not per-star hypothesis tests. One outlier results: χ^1 Ori at Band 3 (excess ratio 2.52, largest of the seventeen; resampled tail probability 0.53 in its six-star bin – a ranked outlier of a small sample, not a significant departure), consistent with a spectroscopic G0V binary’s unresolved companion or a chromosphere. Per-bin medians and log-scatters ship in the machine-readable release; two robustness reruns are pending.

L. A SERENDIPITOUS MOLECULAR-LINE CATALOGUE

The continuum search’s line-exclusion step (§4) flags every $\geq 5\sigma$ outlier channel as a by-product; no existing mm/submm line survey of the solar neighbourhood rests on a distance-limited sample of archival coverage. Twelve of the outlier groups scanned to date are single- or few-channel features with no known transition within 300 MHz, most plausibly statistical; six are plausibly real: CO($J=1\rightarrow0$) and CO($J=2\rightarrow1$) towards β Pictoris (the stellar-frame component of Table 9); a marginal CO($J=2\rightarrow1$)-consistent feature recurring across two epochs towards HD 285968, tentative only; and a single-channel feature 195 MHz from SiO($J=5\rightarrow4$) towards HD 53143, more likely statistical than real. The machine-readable catalogue ships with the tagged public release.

M. LESSONS FOR ARCHIVAL PIPELINES

Three of the four audit items of §5.2 are failure modes generic to automated archive mining. First, every lane needs its own geometric guard: item (ii)’s pointing mismatch aborts the carrier step but not continuum imaging. Second, window-level bookkeeping anomalies yield to cross-target pattern recognition, not per-target sanity checks: item (iii) recurred with matching correlator parameters across two unrelated targets. Third, name resolution is not position resolution: the γ Lupi entry (item iv below) was misidentified through the SIMBAD identifier list and caught only by recovering the Gaia DR3 parallax digit-for-digit; that check should be standard.

N. FALSE-ALARM ACCOUNTING UNDER THE SYMMETRIC CRITERION

Earlier drafts carried a four-model bookkeeping ladder (permutation, hypergeometric, block bootstrap and a Poisson rate heuristic, $P(F\geq 3) = 6.1\text{--}9.2$ per cent) calibrating a false-alarm rate for the *asymmetric* statistic; the symmetric statistic of §5.3 removes that need, giving a per-window rate $1/513$ and survey expectation $431/513 = 0.84$ flags, so the ladder is retired, not updated.

The $1/513$ rank floor of §5.3 falls only if the control ring grows, which the primary beam bounds; and windows are not fully independent, grouped into 102 EBs sharing weather, calibration and correlator setup, so $N/513$ is an

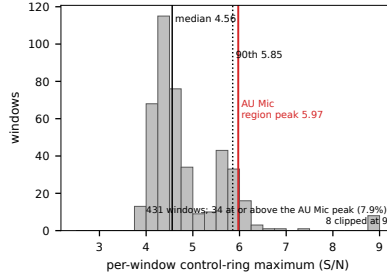


FIG. 14.— Diagnostic distribution for the marginal AU Mic crossing: control-ring peak S/N of all 431 windows, the maxima expected from noise plus background structure alone. Dashed: the formal 5σ threshold. Solid: the on-source peak; dotted: its own control maximum. Three windows lie beyond the plotted range, marked at upper right (HD 285968 Band 6 at 10.9; β Pictoris Band 6 at 9.1 and 15.2, the latter the CO-filled control ring).

expectation over a mildly correlated sample, not N independent trials. Both limits are conservative for the conclusion drawn – that no window warrants candidate status – widening, not narrowing, the null range.

THE SYMMETRIC STAR-VERSUS-CONTROL NULL IN FULL

The per-window p -value of the test of §5.3 is $p = (1 + \#\{T_i \geq T_\star\})/(N + 1)$, and its seven measured windows are the six validation windows of the local reprocessing set plus the AU Mic crossing’s reprocessed first-epoch window (§5.4).

Table 17 consolidates the family-wise trials factors: every per-window factor applies identically at the 512 controls, making the control ring and its empirical star-exceeds-ring rate the operative calibration, not any analytic correction.

ROBUSTNESS OF THE EMPIRICAL CALIBRATION

The release evaluates 4.60×10^7 channel $\times$ drift cells; a Gaussian one-sided 5σ tail (2.87×10^{-7}) over that count predicts 11.4 chance crossings against the 20 windows with an on-star crossing. Raw counts run higher because the drift-tracked statistic is strongly correlated between adjacent channels ($\rho_{\text{lag1}} = 0.989\text{--}0.991$, i.e. $8.5 \times 10^4\text{--}1.1 \times 10^5$ effectively independent cells), so crossings arrive in contiguous runs; collapsed to clusters (kernel of 3 native channels) they leave far fewer independent trials, and that count agrees with the trials expectation.

No single stratum drives the star-exceeds-ring rate. The stratification was run on the ≤ 20 pc subset under the withdrawn asymmetric statistic, whose counts it quotes – the operative symmetric rate is too small (5 windows) to stratify. Across band, channelisation and primary-beam offset that rate moved only between 11 and 20 per cent, and the disc-hosting caveat inverts empirically: disc- or planet-hosting stars contribute every crossing window and every stage-1 statistical flag but a *lower* star-exceeds-ring rate (9.8 against 18.3 per cent for non-hosts), so circumstellar emission shared by star and ring does not drive the statistic. Excluding the 15 β Pictoris windows leaves one unattributed crossing among 416, against 0.78 expected ($P(\geq 1) = 54$ per cent).

Beyond the shared primary-beam gain, neighbouring control positions are correlated below the synthesised-beam scale, so independent spatial trials fall from 512 to $N_{\text{eff}} \simeq 30\text{--}43$, as the symmetric reprocessing runs measure directly – a floor to which the empirical 1.2 per cent rate, measured on the correlated ensembles themselves, is immune by construction. A continuum source in the ring raises the carrier statistic only if it also emits a narrow line in-window, and Appendix G covers recurring frequencies.

O. INTEGRATION-TIME AUDIT: METHOD AND EXCEPTIONS

The archive cross-check of §5.5 uses the ALMA Science Archive’s `ivoa.obscore` service (queried 2026 September 8), whose `t_exptime` rows give the science-intent exposure per observing unit set and pointed science source, summed over its executions, covering every window-level record of the subset by star and EB. For each of the 57 searched EBs the pointed source came from the archive’s field list (aliases resolved by hand: Sirius B through the Sirius A pointings; the UV Ceti/G 272-61 and AT Mic pairs share fields), its on-source time compared with the archive exposure. `obscore` counts science-intent scans only, hiding target time in calibrator-intent scans, so the archive figure is a lower bound and any pipeline over-count conservative; per-execution durations are not exposed, so multi-execution sets are compared through their aggregate. The audit is conservative for both flagged hosts (§5.3): β Pictoris’s 1824–1845 s sit inside an archive exposure of 47 444 s, AU Mic’s inside 109 334 s, no disposition changing, and the archive values lower the exposure-weighted coverage total by under 1 per cent.

Two of the 58 star-execution-block groups over-count: AT Mic’s wide windows in block A002_Xcd97a1_X3f1e (pipeline 1258 s against the archive’s 877 s, ratio 1.43) and CD–38 10980 in A002_Xe5808e_X80a1 (617 against 575 s, 1.07); the 53 remaining groups are all at or below the archive exposure, three more (Wolf 28, the narrow AT Mic window, G 272-61) agreeing to within 1 per cent. The AT Mic block shows the signature of a dump-counting defect: three windows report 208 integrations where 145 exist, and more on-source time than the block’s own 1140 s elapsed span. The duty table uses the archive-consistent value for both AT Mic entries; the over-counted values survive only in per-window machine-readable fields. The CD–38 10980 excess is within the intent-filter’s conservative margin and its limit is unaffected, thresholds deriving from measured rms rather than integration time.

TABLE 21
PRE-FREEZE VALIDATION CHANGES: ISSUE, CORRECTION, EFFECT.

Item	Correction	Effect
Threshold convention	per-channel-width $\text{W Hz}^{-1} \rightarrow$ total power (W; §2)	all EIRP values restated; no flux or rms changes
Search statistic	asymmetric region-max $\rightarrow$ symmetric single-position at star and controls (§5.4); release re-run, criterion frozen	flagged windows 65/431 (15.1%) $\rightarrow$ 5/431 (1.2%); the AU Mic crossing becomes unflagged
Baseline-operator description	withdrawal of the “temporal-median subtraction” claim; operative operator restated (running median along frequency, §4.1)	wording only; no numerical product changes; static-carrier recovery measured instead (§4.5)
Line exclusion	one-channel frequency match $\rightarrow \pm 50 \text{ km s}^{-1}$ stellar-frame velocity mask, applied retrospectively to every crossing (§5.3)	rejects the two β Pic windows; no other disposition changes
Occurrence framing	completeness-corrected constraint $\rightarrow$ illustrative conditional limit with the $F_i=D_i=1$ conditioning stated wherever quoted (§6)	labels only; the numbers are unchanged

P. DEVELOPMENT AUDIT AND FROZEN-ANALYSIS HISTORY: VALIDATION CHANGES, STATISTIC REVISION, EXCLUSIONS AND DEFECTS

PRE-FREEZE VALIDATION CHANGES

The pipeline as released differs from earlier working versions in five substantive ways, each settled before the freeze and logged here once (Table 21); none changes a released data product except where stated. The inference-status box of §1 governs all of it: performance estimated on this release is retrospective validation.

HISTORICAL FALSE-ALARM DIAGNOSTICS OF THE RETIRED STATISTIC

Kept so the main-text account (§5.3) can be symmetric-only; nothing here enters the operative false-alarm probability. Under a pure-noise null the region maximum over n_{src} positions beats single controls with probability $n_{\text{src}}/(n_{\text{src}} + n_{\text{ctrl}}) - 21$ per cent for a typical window – the order of the measured region-max star-exceeds-ring rate (15.1 against the symmetric 1.2 per cent, an order of magnitude apart, never to be interchanged); bookkeeping built on the asymmetric rate expected 65 flagged windows against the symmetric 0.84, with secondary models spanning $P(F \geq 3) = 1.0\text{--}9.2$ per cent. That bracket was the sensitivity of the pre-freeze accounting to the statistic choice – the deposit the freeze protected against. This history mixes three failure kinds, kept distinct: a *design error* (the asymmetric statistic), a *data/bookkeeping defect* (the noise-defect windows below), and *analysis choices* frozen with their alternatives stated.

THE FOUR CLASSES OF EXCLUDED ENTRY, ITEMISED

The process-level failures of §5.2 are item (i); the six noise-defect windows lie across five stars (51 Eri, HD 10647, HD 139664, TWA 7, HD 92945 twice). The supporting audit is clean across all 431 windows bar the four χ^1 Ori Band 3 windows it flags as elevated: their radiometric quantity $\sigma\sqrt{t_{\text{on}}}\Delta\nu_{\text{ch}}$ lies $0.9\text{--}1.1\times$ the Band 3 median, weakening their thresholds by at most $\sim 10\text{--}20$ per cent in amplitude, and all 168 sample identifiers re-resolve to 168 distinct sources inside 40 pc.

(i) **Narrowband process failures.** Four target/bands (Barnard’s Star [B7], Wolf 359 [B7], SCR J1845–6357, Wolf 358) have no carrier result; each keeps a valid continuum non-detection and no EIRP threshold.

(ii) **Giclas 9–38 pair.** Giclas 9–38A and Giclas 9–38B [B3] are excluded entirely: the pair’s only matched MOUS points ~ 52 arcmin from either star, $35\times$ its own primary-beam FWHM. The carrier step correctly aborts; continuum imaging lacks that guard, so the target/band waits on one (Appendix M).

(iii) **Window-level noise defect.** One spectral window in each of two Band 6 targets (HD 10647, HD 139664) reports 12.1 s on source against 393–520 s for siblings in the same block, yet a noise estimate $\sim 1000\times$ deeper – the opposite of radiometer scaling for a $\sim 40\times$ shorter integration; both figures come from one machine-written product, defective together, and the window’s carrier result is excluded, the combined continuum withheld, their other windows kept. Table 22 gives, for the four defect windows the released products carry, every quantity those products hold: all are 15.62-MHz coarse windows whose reported rms is inconsistent with their reported integration by three to four orders of magnitude, the anomalous normalisation entering the noise estimate, not the channelisation or window placement. Localised against their own blocks, the defect rms sits a factor 388–669 *below* the shallowest sibling of the same block and channel width – a normalisation signature, not weather or configuration, which would depress the siblings alike. The root cause sits in the upstream extractor, outside the frozen tree, and naming it needs quantities (antenna and baseline counts, calibration weights, visibility units, flagged fraction) that are properties of the raw execution blocks; pulling those blocks is deferred to the completed survey’s pipeline release.

(iv) **The “ γ Lupi” resolution.** The object observed under that entry is the F4V debris-disc host HD 139664 at 17.40 pc, not the naked-eye γ Lupi (§5.2); all such entries are relabelled, no measured value changing.

Q. POPULATION-INFERENCE FRAMEWORK: DERIVATION AND DEMONSTRATION TABLES

This appendix consolidates the numerical demonstrations condensed out of §6.2, labelled *illustrative calculation only* throughout. Everything here is conditional on Eq. 4 with $F_i=D_i=1$, and f is an *effective conditional in-band transmitter fraction given occupancy of each target’s own observed frequencies* – defined on the 82 systems processed

TABLE 22

THE FOUR NOISE-DEFECT WINDOWS CARRIED BY THE RELEASED PER-WINDOW PRODUCTS (THE TWO HD 92945 WINDOWS FAIL THE SAME TEST UPSTREAM AND ARE EXCLUDED BEFORE THE EXPORT): ALL THE QUANTITIES THOSE PRODUCTS CARRY. $q_a = \sigma \sqrt{t_{\text{on}} \Delta \nu_{\text{ch}}}$, IN UNITS OF THE SAMPLE MEDIAN.

Star	ν (GHz)	$\Delta \nu_{\text{ch}}$ (MHz)	t_{on} (s)	rms (μJy)	q_a/med
51 Eri	230.5251	15.62	24.2	0.6	1.7×10^{-4}
HD 139664	230.5548	15.62	12.1	1.0	2.0×10^{-4}
HD 10647	230.5357	15.62	12.1	1.0	2.1×10^{-4}
TWA 7	346.3790	15.62	48.4	0.5	2.2×10^{-4}

TABLE 23

OCCURRENCE-DEMONSTRATION LIMITS (CONDITIONAL IN-BAND TRANSMITTER FRACTION, $F_i=D_i=1$ THROUGHOUT; EQ. 4). (A) THE 95 PER CENT UPPER LIMIT $f_{95}(P)$ ABOVE EIRP P : n SYSTEMS WITH NON-ZERO COMPLETENESS AT THAT P ; THE *measured* COLUMN CARRIES THE INJECTION-MEASURED RECOVERY, WHOSE SINGLE-CONFIGURATION TRANSFER ERROR IS UNQUANTIFIED. (B) DUTY-CYCLE DEPENDENCE AT $P \geq 10^{16}$ W WITH $D_i = 1 - (1 - p_{\text{epoch}})^{n_i}$: THRESHOLDED (82 SYSTEMS, UNIT RECOVERY) AND MEASURED-COMPLETENESS (59 FINE-CALIBRATED) FORMS. ‘UNCONSTRAINED’: NO BOUND EXISTS AT THAT COMBINATION.

P (W)	thresholded (unit C)		measured C	
	n	f_{95}	n	f_{95}
3×10^{13}	4	52.7%	2	unconstrained
10^{14}	16	17.1%	11	47.1%
10^{15}	57	5.1%	45	10.1%
10^{16}	81	3.6%	59	6.2%
10^{17}	82	3.7%	57	6.2%

(b)

Epoch activity	p_{epoch}	f_{95} thresholded	f_{95} measured
1 (continuous)		3.6%	6.2%
0.5		6.4%	10.8%
0.1		29.0%	48%
0.01		unconstrained	unconstrained

here, not the 168-star archive-complete sample and still less the 40 pc stellar population.

Illustrative calculation only. The demonstration number and its sensitivity analysis follow; none of it is a survey result. At EIRP $\geq 10^{16}$ W the measured form gives $f_{95} \approx 6$ per cent over the 59 systems with a fine-channel window – conditional on in-band, always-on emission of the searched morphology ($F_i=D_i=1$ by conditioning, not measurement); on a recovery curve measured on one fine-channel configuration and transferred by assumption to all 118 fine windows ($0.5\text{--}2\times$ scaling, the plausible range once unmeasured channel-count and beam-offset terms are admitted, moves the limit to $5.0\text{--}12.3$ per cent – a bracketing range, not a confidence interval; and on the archivally observed subsample, in which F and G stars are over-represented ($\times 3.1\text{--}3.4$) and M dwarfs under-represented ($\times 0.6\text{--}0.7$) relative to the 40 pc census (§3), so no demographic statement about the solar neighbourhood follows. Under any transmitter-frequency prior spread across ALMA’s tuning range the bound vanishes entirely: the per-system fine-channel union has median 1.73 GHz against a ~ 75 GHz range, so the conditioning supplies the entire content of the number. The full derivation, the thresholded and bracketing forms (idealised brackets 3.6 and 8.6 per cent at 10^{16} W), the epoch-activity degradation (p_{epoch} and the intra-track dwell fraction are distinct axes; panel b of Table 23), the transfer-error analysis, a Bayesian cross-check (6.1 per cent) and Table 23 are consolidated in this appendix.

Independence. The C_i are not independent: 431 windows group into 102 execution blocks sharing weather, calibration and correlator setup. Across the 103 execution blocks of the frozen export – 102 of them contributing searched windows (Table 5), the withheld ϵ Eri Band 6 block A002_X99c183_X4f1e accounting for the difference – the intraclass correlation of the window noise floor is 0.55 (shared weather and configuration), of the control-normalised stellar rank 0.24, and of the raw stellar statistic 0.05. Residual dependence at this level is one-sided – positive correlation between detections inflates the true f_{95} above the independent product. Quantified for the rank correlation: over the 74 execution blocks carrying fine-channel windows (mean 1.59 windows each), 0.24 implies a design effect of 1.14, moving the demonstration number only from 6 to 7.0 per cent – an order below the completeness-transfer term that dominates the bracket above.

Two forms. Eq. 5 is evaluated on the 82 independently observed systems (88 carrier-searched stars; six bound pairs share observations and count once each). The *thresholded* form is the step function the retained products support, $R_i(P) = 1$ when system i ’s deepest threshold lies below P , else 0. The *measured* form integrates the injection-recovery curve of Appendix D over each system’s searched fine windows (the transferred curve existing only there), folding in the $17 \rightarrow 83$ per cent recovery of the searched class measured between four and ten times the noise; it is defined on the 59 systems with a fine-channel window.

Transfer error, and why this is a pilot number. The drifting-class recovery curve is measured on one fine-channel configuration, then applied to all 118 fine windows, which span Bands 3–8, on-source times of 21–5274 s and a range of primary-beam offsets. Achieved noise, integration time and drift-grid density shift the curve mainly in *position*; the channelisation and median-subtraction dwell fraction that control its *shape* are common to every fine window by construction. None of this has been measured, so the transfer is an assumption, and its error is not bounded by the ${}^{+16}_{-14}$ percentage-point binomial interval, which describes only sampling noise within the one configuration run. One term is bounded analytically: coherently averaging N_{int} on-source integrations suppresses noise as $\sqrt{N_{\text{int}}}$, so the amplitude anchor can shift by at most $\sqrt{5274/21} \approx 16$ between the shortest and longest windows, and the calibration configuration (1506 s on source) sits a factor 1.2 in $\sqrt{t_{\text{on}}}$ from the median fine window (2177 s). Propagated as a bracket: shifting the amplitude anchor over that factor moves the 95-per-cent limit by only 0.1 points (most fine windows sit on the plateau); scaling the transferred curve itself by 0.5–2 $\times$ moves it from 6.2 to 5.0–12.3 per cent. The retained products preserve the campaign’s pooled recovery fractions but not its trial-level records, so their denominators come from the design, not a log: the coarse-window searched class has no calibrated completeness at all, making this a *pilot, configuration-limited* estimate, secondary to the non-detection and threshold range. Trial-level logging is the corresponding first change of §6.4.

Which number is which. The measured form carries the only measured recovery fraction; the thresholded form (3.6 per cent) assumes unit recovery, a uniform $C = 0.42$ gives 8.6 per cent, and a Bayesian treatment with a uniform prior on f , $p(f | \text{data}) \propto \prod_i (1 - f C_i)$, gives 6.1 per cent – bracketing calculations, none a prevalence measurement, and each applying to transmitters above its P , not one well-defined class. Under a uniform transmitter-frequency prior across ALMA’s tuning range, $F_i = \int_{\Delta\nu_i} p(\nu) d\nu$ rather than one: F_i averages 4.5×10^{-3} and $\sum_i C_i = 0.27$ over Bands 3–8, so one detection would need $f \approx 3.8$ transmitters per system – outside the conditional regime the release constrains nothing (Scope box, §4).

Epoch-activity ladder. Writing p_{epoch} for the chance a transmitter is on in any one epoch, a system observed in n_i independent execution blocks is sampled n_i times, so $D_i = 1 - (1 - p_{\text{epoch}})^{n_i}$; 18 of the 82 systems have $n_i > 1$ (13 with two epochs, five with three). The Bernoulli form assumes independent epochs; the release spans 2013–2025 at year-scale separations (β Pictoris: 2013, 2019, 2022), so orbit-tied beacons enter at effectively random phase and, with at most three epochs, the ladder is a phase-averaged bound. An intermittent transmitter is at least as plausible *a priori*, so the $p_{\text{epoch}} = 1$ numbers should never be quoted without this dependence.

Series-level error control. The prespecified family-level false-alarm procedure for this release is the Monte-Carlo pseudo-target calibration of §5.3 (expected 0.20 flags against a rank-only budget of 0.84); the Bonferroni, Wilson and Poisson calculations of Appendix N are robustness checks against it, not independent claims. Across releases the cumulative look-elsewhere effect of component (v) applies: each release’s unattributed flags will be reported against the *cumulative* expected count (a Poisson tail on the running total), the series-level false-discovery rate controlled by Benjamini–Hochberg on per-window empirical spatial rank probabilities (the 1/513 floor limits their resolution) and promoted flags keeping the per-release pre-commitment of §5.3. The rule is provisional and will be fixed, with power simulations, before the second release.

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